

Constrained Consensus-Based Dynamics for Decentralized Path Planning in UAV Swarms*

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Abstract

Safe navigation of a UAV swarm in a constrained environment can be posed rigorously as a distributed constrained optimization problem. Each vehicle plans its own trajectory from local information and exchanges with nearby neighbours, while the swarm as a whole must satisfy shared safety requirements. Consensus-based coordination is effective for such systems, but most existing formulations either relax the hard constraints imposed by obstacles and vehicle limits, or replace them by heuristic potential fields that carry no convergence or safety guarantee. This report develops a decentralized path-planning formulation built on constrained consensus-based dynamics, and reports what has been established for its first, fully deterministic instance.

Multi-UAV planning is written as the minimization of a sum of local objectives under two families of constraints: local feasibility sets encoding the workspace, vehicle speed limits and obstacle clearance; and pairwise coupling constraints enforcing separation between vehicles. Coordination is obtained through agreement constraints resolved by local interaction. A central modelling point is made precise here: agreement is imposed between computational copies of the same source trajectory, never between the physical trajectories of distinct vehicles, since forcing distinct vehicles onto identical paths would contradict separation. This is the Trajectory-Copy Consensus reformulation, and it is proved to preserve the feasible set, the optimal value and the set of global minimizers of the original problem.

For the deterministic instance the report establishes: exact inter-sample safety for the continuous piecewise-linear trajectory, rather than safety at sampled instants only; existence of a global minimizer by a constructive argument; a complete characterization of strict convexity of the objective, together with an explicit admissible instance carrying two distinct global minimizers, which shows that uniqueness cannot be assumed; set-valued stability of the solution under perturbation of the problem data; a first-order analysis identifying precisely where standard constraint qualifications can fail, and a correspondingly conditional Karush–Kuhn–Tucker theorem; and the mapping of the reformulated problem onto a two-level distributed algorithm whose assumptions are stated and checked rather than assumed. A numerical study on a two-vehicle crossing instance attains the specified agreement and stopping tolerances, is reproducible to the bit, and its saved plan is certified safe in exact rational arithmetic rather than by a floating-point tolerance. An independent centralized solve of the same instance reproduces the distributed objective value to seven significant figures and the trajectory to thirty-three micrometres, and exhibits two distinct global minimizers,

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confirming the non-uniqueness result numerically. The limits of the result are stated with the same precision as the result itself.

Because obstacle-induced feasible sets are nonconvex, the analysis is stated in terms of stationarity, feasibility and safety margins, not global optimality. Dynamic environments and uncertainty in sensing and communication are treated as the next stage of the work and are scoped here, not claimed. A physics-based ROS 2 and Gazebo environment with two vehicles is presented as the simulation track on which later versions will be validated. The contribution is an integration: existing literature supplies strong but separate components, and this work assembles them into a single constrained consensus formulation for UAV swarm path planning with an explicit and defensible boundary between what is proved and what is not.

Keywords: UAV swarm; decentralized optimization; constrained consensus; inter-sample collision avoidance; alternating direction method of multipliers; constraint qualification; ROS 2 and Gazebo.

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Notation and Abbreviations

Every symbol used in this report is listed here once, in the order in which it first becomes active in the text. Each symbol is also defined again, in words, at the point where it is first used, so that the report can be read linearly without consulting this table. Index ranges are written explicitly because several families are indexed over samples $k = 0, \dots, K$ while others are indexed over segments $k = 0, \dots, K - 1$, and the distinction matters throughout.

Two reservations hold everywhere and are stated here once. First, k is the physical sample or segment index and nothing else. Second, r carries two roles that never appear in the same expression: unsubscripted and in superscript position, r is the outer iteration index of the distributed algorithm; subscripted by a vehicle index, r_i is that vehicle's geometric radius.

Table N.1 Every symbol used in this report appears here once, in the order in which it first becomes active in the text.

Symbol	Meaning
$V = \{1, \dots, N\}$	Set of vehicles, also the node set of the communication graph; $N \geq 2$.
$G = (V, E)$	Generic undirected graph on the vehicle set: the nodes are the vehicles and an edge records that the two vehicles it joins can exchange information directly. Every graph used in this report is a graph of this form, obtained by specifying which pairs are edges; the graph of the solve launched at t_0 is G_0^{comm} .
E	Edge set of a generic graph $G = (V, E)$: a set of unordered pairs $\{i, j\}$ of distinct vehicles. Decorated versions denote specific edge sets, and their meanings are separate: E^{all} , E_k^{comm} , E_k^{coll} and E_0^{comm} below.
i	Vehicle index, $i \in V$, used for a physical source, that is, for a vehicle whose own trajectory is being planned.
j	A second vehicle index, $j \in V$ with $j \neq i$, used whenever an unordered pair $\{i, j\}$ of distinct physical vehicles is in play.
a	Vehicle index used for a <i>computational</i> role: the holder of a trajectory copy, or one endpoint of an agreement edge. In the closest-point derivations of Section 4.5, a is instead a local temporary vector, defined at the point of use; it is never a node index there.
b	A second vehicle index in the same computational sense: the other endpoint of an agreement edge. In the closest-point derivations of Sections 4.5 and 4.6, b is likewise a local temporary vector and not a node index.
$q \in \{2, 3\}$	Spatial dimension of the workspace.
t_0	Planning instant at which the problem is posed.
$\Delta t > 0$	Sampling period of the planning grid.
$K \geq 1$	Number of prediction intervals, hence $K + 1$ samples.
$t_k = t_0 + k\Delta t$	Sample instant, $k = 0, \dots, K$.
k	Physical sample index ($k = 0, \dots, K$) or segment index ($k = 0, \dots, K - 1$). It is never an algorithm iteration index.

Symbol	Meaning
$p_i^k \in \mathbb{R}^q$	Planned position of vehicle i at sample t_k , $k = 0, \dots, K$.
$\hat{p}_i(t_0)$	Measured or estimated position of vehicle i at t_0 ; deterministic input data.
$p_i^0 := \hat{p}_i(t_0)$	Initial position; fixed data, not a decision variable.
x_i	Complete sampled trajectory record $(p_i^0, \dots, p_i^K)$ of vehicle i .
$x_{i,\text{free}}$	Free block $(p_i^1, \dots, p_i^K) \in \mathbb{R}^{qK}$ of vehicle i : the part the optimizer chooses.
x_{free}	Stack $(x_{1,\text{free}}, \dots, x_{N,\text{free}}) \in \mathbb{R}^{NqK}$ of all free blocks.
$n = NqK$	Number of free scalar decision variables.
$X_i(t_0)$	Trajectory domain of vehicle i with its initial position held fixed.
Full_{p^0}	Map attaching the fixed initial positions to a free stack, producing full records.
π_{free}	Map removing the fixed initial positions from a full record.
$\lambda \in [0, 1]$	Interpolation parameter along one segment.
$\lambda(t)$	Value $(t - t_k)/\Delta t$ for $t \in [t_k, t_{k+1}]$.
λ^*	Unconstrained minimizer of a segment distance in λ .
$\hat{\lambda}$	Constrained closest-point parameter, i.e. λ^* clipped to $[0, 1]$.
v_i^k	Segment velocity $(p_i^{k+1} - p_i^k)/\Delta t$, $k = 0, \dots, K - 1$.
$v_i^{\max} > 0$	Maximum speed of vehicle i .
$g_i \in \mathbb{R}^q$	Goal position of vehicle i .
$\alpha_i \geq 0$	Weight of the terminal goal term of vehicle i .
$\beta_i \geq 0$	Weight of the squared-step term of vehicle i ; distinct from the algorithm penalty β^r .
$f_i(x_i)$	Local objective of vehicle i .
F	Swarm objective $\sum_i f_i$. Written $F(x_{\text{free}})$ in physical variables and $F(Z)$ in the computational variables of Section 8; it is the same function.
$\mathcal{W}$	Workspace: a nonempty, compact, convex, full-dimensional polytope of admissible positions.
$A_{\mathcal{W}}$	Coefficient matrix of the halfspace representation of the workspace $\mathcal{W} = \{p \in \mathbb{R}^q : A_{\mathcal{W}}p \leq b_{\mathcal{W}}\}$. Row m of $A_{\mathcal{W}}$ is the outward normal of the m -th bounding hyperplane of the polytope. It is fixed problem data describing the flyable region, and it is unrelated to the matrix A of the agreement system.
$b_{\mathcal{W}}$	Right-hand side of that same halfspace representation $\mathcal{W} = \{p \in \mathbb{R}^q : A_{\mathcal{W}}p \leq b_{\mathcal{W}}\}$. Entry m of $b_{\mathcal{W}}$ is the offset of the m -th bounding hyperplane. It is fixed problem data, in the same length units as a position.
$a_m^\top$	Row m of $A_{\mathcal{W}}$, written as a row vector: the outward normal of the m -th bounding hyperplane of the workspace. It carries the single scalar inequality $a_m^\top p \leq b_m$, and every retained row satisfies $a_m \neq 0$.

Symbol	Meaning
b_m	Entry m of $b_{\mathcal{W}}$: the offset of the m -th bounding hyperplane of the workspace, appearing in the same single scalar inequality $a_m^\top p \leq b_m$.
$M_{\mathcal{W}}$	Number of retained workspace inequalities.
$\mathcal{O}$	Index set of obstacles; finite, possibly empty.
o	Obstacle index.
$c_o \in \mathbb{R}^q$	Centre of obstacle o ; constant over the horizon.
$R_o > 0$	Radius of obstacle o .
$r_i > 0$	Geometric radius of vehicle i .
$\delta_i^{\text{obs}} > 0$	Additional vehicle-to-obstacle safety margin.
$\delta_{ij}^{\text{uav}} > 0$	Additional vehicle-to-vehicle safety margin.
κ_{io}^{obs}	Required centre clearance $R_o + r_i + \delta_i^{\text{obs}}$ to obstacle o .
κ_{ij}^{uav}	Required centre clearance $r_i + r_j + \delta_{ij}^{\text{uav}}$ between vehicles i and j .
E^{all}	All unordered vehicle pairs $\{\{i, j\} : 1 \leq i < j \leq N\}$.
E_k^{comm}	Direct communication pairs available at sample t_k .
E_k^{coll}	Pairs exchanging collision-relevant information at sample t_k .
R_{ij}^{comm}	Communication range between vehicles i and j .
$R_{ij}^{\text{coll,info}}$	Range within which collision-relevant information is exchanged.
$\zeta(e, t_k)$	Communication latency on edge e ; reserved notation, not used by the present model.
$C_i(t_0)$	Local feasible set of vehicle i : workspace, speed and obstacle clearance.
$\mathcal{F}_{0\text{-D}}(t_0)$	Complete feasible set of the swarm, including all pairwise separation constraints.
$\mathcal{F}_{0\text{-D}}^{\text{full}}(t_0)$	The complete feasible set written in full trajectory records, that is, with the fixed initial positions included in each record.
$\mathcal{F}_{0\text{-D}}^{\text{free}}(t_0)$	The same complete feasible set written in free coordinates, that is, after the fixed initial positions have been removed by π_{free} .
(P)	The planning problem, stated in full in (P): minimize the swarm objective over all free trajectory stacks whose full records lie in $\mathcal{F}_{0\text{-D}}(t_0)$.
(P _c)	The compact restatement (P _c) of the same problem, used whenever the argument needs the problem as a whole rather than its individual constraint rows. It is the same problem as (P), written more briefly.
d_i^k	Segment displacement $p_i^{k+1} - p_i^k$, $k = 0, \dots, K - 1$.
s_i^k	Squared displacement $\ d_i^k\ ^2$; also written s when a single segment is in play.
a_{io}^k	Vector $p_i^k - c_o$ from obstacle centre to the segment start.
$D_{i,o,k}^{\text{obs}}$	Exact minimum Euclidean distance from segment k of vehicle i to c_o .
$D_{i,j,k}$	Exact minimum Euclidean distance between vehicles i and j over segment k .
$M_{i,o,k}^{\text{obs}}$	Exact minimum <i>squared</i> vehicle-to-obstacle segment distance.

Symbol	Meaning
$M_{i,j,k}^{\text{uav}}$	Exact minimum <i>squared</i> vehicle-to-vehicle relative segment distance.
$z_{i,o,k}$	Vector from c_o to the closest point of the segment.
$z_{i j,k}$	Oriented closest relative vector from vehicle j to vehicle i on segment k .
$h_{i,j,k}^{\text{coll}}$	Separation function evaluated at sample k only.
$h_{i,j,k}^{\text{inter,coll}}$	Exact inter-sample separation function, unsquared form.
$h_{i,j,k}^{\text{inter,coll,sq}}$	Exact inter-sample separation function, squared form; the one used by the solver.
$g_{i,m,k}^{\mathcal{W}}$	Workspace inequality $a_m^\top p_i^k - b_m \leq 0, k = 1, \dots, K$.
$g_{i,k}^{\text{spd}}$	Squared speed inequality, $k = 0, \dots, K - 1$.
$g_{i,o,k}^{\text{obs,seg}}$	Exact squared segment-to-obstacle inequality, $k = 0, \dots, K - 1$.
φ_ℓ	Generic scalar inequality of the reduced problem, used in the first-order analysis.
$\mathcal{A}(x_{\text{free}}^*)$	Index set of inequalities active at a feasible point.
$J_{\mathcal{A}}$	Jacobian of the active inequalities.
$m_{\mathcal{A}}$	Number of active inequalities.
h	Direction certifying the Mangasarian–Fromovitz condition; distinct from the separation functions $h_{i,j,k}^*$, which always carry indices.
τ	Uniform margin in the linear-programming test for that condition.
$\mu_{i,m,k}^{\mathcal{W}}$	Multiplier of a workspace inequality.
$\nu_{i,k}^{\text{spd}}$	Multiplier of a speed inequality.
$\xi_{i,o,k}^{\text{obs}}$	Multiplier of an obstacle-segment inequality; distinct from the algorithm slack ξ .
$\eta_{i,j,k}^{\text{uav}}$	Multiplier of a pairwise separation inequality.
$\mathcal{L}$	Lagrangian of the reduced physical problem.
θ	Vector collecting the numerical problem data in the stability analysis.
Θ_{adm}	Set of admissible data vectors, with N, K, q held constant.
$v(\theta)$	Optimal value as a function of the data.
$S(\theta)$	Set of global minimizers as a function of the data.
G_0^{comm}	Communication graph used by the solve launched at t_0 ; fixed and connected.
$\tilde{x}_{a \leftarrow i}$	Copy of the free trajectory of source i held at vehicle a .
$\tilde{x}_{a \leftarrow i}^{\text{full}}$	Same copy with the initial position attached.
V_i^{copy}	Set of vehicles holding a copy of source i .
$n_i = V_i^{\text{copy}} $	Number of holders of source i .
G_i^{copy}	Connected topology over the holders of source i ; may contain cycles.
T_i^{eq}	Spanning tree of G_i^{copy} carrying the agreement constraints.
γ	Guardian map assigning each pair to one of its two endpoints.
L_a^{guard}	Number of pairs guarded by vehicle a .

Symbol	Meaning
V_i^{req}	Holders of source i required by the guardian assignment.
V_i^{relay}	Holders present only to keep the copy topology connected.
n_i^{relay}	Number of relay holders used for source i .
L_a^{relay}	Number of relay copies stored at vehicle a .
I_a^{copy}	Set of sources represented at vehicle a .
$Q = qK$	Dimension of one free source trajectory.
z_a	Computational block held by vehicle a : its own free trajectory and its foreign copies.
$d_a = Q I_a^{\text{copy}} $	Dimension of z_a .
$S_{a \leftarrow i}$	Selector extracting the source- i representation from z_a .
$\chi_{a \leftarrow i}^{\text{full}}(z_a)$	Full source- i trajectory represented inside z_a .
$Z = \text{col}(z_1, \dots, z_N)$	Stack of all computational blocks.
M_{eq}	Number of agreement-tree edges, $\sum_i (n_i - 1)$.
C_i^{eq}	Agreement incidence matrix of source i ; distinct from the local feasible set $C_i(t_0)$.
$\mathcal{F}_{\text{copy}}(t_0)$	Feasible set of the reformulated problem, in the computational variables.
(P')	The reformulated problem (P'): the same planning problem expressed in the computational variables, with agreement between copies imposed explicitly.
$F_{0\text{-D}}^*$	Optimal value of the physical planning problem (P), that is, the smallest value the swarm objective attains on $\mathcal{F}_{0\text{-D}}(t_0)$.
J_{copy}^*	Optimal value of the reformulated problem (P'), that is, the smallest value attained in the computational variables. Section 8 proves $J_{\text{copy}}^* = F_{0\text{-D}}^*$.
$e = (a, b; i)$	Agreement edge carrying copies of source i at endpoints a and b .
$u_e \in \mathbb{R}^Q$	Auxiliary variable introduced on edge e by the algorithmic split.
$U = \text{col}(u_e)$	Stack of all edge auxiliaries.
$\mathcal{B}$	Compact convex box with $\mathcal{W} \subset \text{int}(\mathcal{B})$, used to compactify foreign copies.
$\mathcal{B}_K = \mathcal{B}^K$	Horizon-wide version of that box.
X_a	Local set of admissible blocks z_a for vehicle a ; compact, in general nonconvex.
$X = \prod_a X_a$	Product of the local sets.
$\bar{X} = \mathcal{B}_K^{M_{\text{eq}}}$	Compact convex domain of the edge auxiliaries.
A	Matrix acting on Z in the split agreement system $AZ + BU = 0$: for every half-edge (e, a) it extracts the copy $S_{a \leftarrow i} z_a$ that the edge constrains. It is unrelated to the workspace matrix $A_{\mathcal{W}}$.
B	Matrix acting on U in the same split agreement system $AZ + BU = 0$: it subtracts the edge auxiliary u_e from each of the two half-edge values of edge e , so that $B_e = \begin{bmatrix} -I_Q \\ -I_Q \end{bmatrix}$ and $B^\top B = 2I$.

Symbol	Meaning
$m_{\text{split}} = 2QM_{\text{eq}}$	Number of scalar rows of that system.
ξ	Slack variable of the outer method, with $\xi = 0$ at agreement; not a physical multiplier.
y	Multiplier of the split agreement system inside the inner method.
Λ^r	Outer multiplier at outer iteration r .
$\underline{\Lambda}$	Componentwise lower bound used to safeguard the outer multiplier: the update is clipped so that $\Lambda^r \geq \underline{\Lambda}$ in every component.
$\bar{\Lambda}$	Componentwise upper bound used in the same safeguard: the update is clipped so that $\Lambda^r \leq \bar{\Lambda}$ in every component. Together the two bounds keep the outer multiplier sequence bounded, which the convergence theory requires.
r	Outer iteration index.
t	Inner iteration index.
$\beta^r > 0$	Outer penalty at iteration r .
$\rho^r = 2\beta^r$	Inner penalty, tied to β^r by the algorithm.
$\vartheta > 1$	Growth factor of the outer penalty.
$\omega \in [0, 1)$	Progress factor deciding whether the outer penalty grows.
ϵ_1^r	Positive tolerance applied to $R_1^{r,t}$ at outer iteration r ; the inner loop may stop only once $R_1^{r,t} \leq \epsilon_1^r$, and $\epsilon_1^r \rightarrow 0$ as $r \rightarrow \infty$.
ϵ_2^r	Positive tolerance applied to $R_2^{r,t}$ in the same way, with $\epsilon_2^r \rightarrow 0$ as $r \rightarrow \infty$.
ϵ_3^r	Positive tolerance applied to $R_3^{r,t}$ in the same way, with $\epsilon_3^r \rightarrow 0$ as $r \rightarrow \infty$. Because ρ^r scales R_1 and R_2 but not R_3 , the three tolerances are naturally of different magnitudes.
$\mathcal{H}_a$	Set of half-edges incident to vehicle a .
$w_{e,a}^{r,t}$	Value $S_{a \leftarrow i} z_a^{r,t}$ selected on half-edge (e, a) .
$q_{e,a}^{r,t}$	Local residual of half-edge (e, a) entering the local objective.
$\phi_a^{r,t}$	Local objective solved by vehicle a at inner iteration t .
$\psi_{e,a}^{r,t}$	Shifted endpoint quantity used in the edge update.
$R_1^{r,t}$	First inner stopping quantity (81): the block stationarity residual produced by the most recent change in the edge auxiliaries and the slack, seen through $A^\top$ and scaled by the inner penalty ρ^r .
$R_2^{r,t}$	Second inner stopping quantity (82): the edge-auxiliary stationarity residual produced by the most recent change in the slack, seen through $B^\top$ and scaled by ρ^r .
$R_3^{r,t}$	Third inner stopping quantity (83): the primal residual $\ AZ^{r,t} + BU^{r,t} + \xi^{r,t}\ $ of the slack-augmented agreement system. It carries no factor ρ^r .
T^{redux}	Fixed tree used to compute global sums by neighbour-to-neighbour exchange.

Symbol	Meaning
$R_{\text{TCC,eq}}$	Largest disagreement between copies of the same source, in metres, stacked ℓ_2 norm.
R_{target}	Residual $\ AZ + BU\ $ of the split agreement system, in metres, stacked ℓ_2 norm.
R_{id}	Residual of the algebraic identity $\Lambda^r + \beta^r \xi^r + y^r = 0$.
$R_{\rho\beta}$	Residual of the relation $\rho^r = 2\beta^r$.
τ_{eq}	Numerical acceptance threshold for $R_{\text{TCC,eq}}$: the largest copy-versus-source disagreement, in metres, that a completed solve is allowed to leave behind.
τ_{target}	Numerical acceptance threshold for R_{target} : the largest residual $\ AZ + BU\ $ of the split agreement system, in metres, that a completed solve is allowed to leave behind. It is a different quantity from τ_{eq} and is not interchangeable with it.
$\sigma_{i,m,k}^W$	Signed workspace margin of the computed trajectory, $k = 1, \dots, K$.
$\sigma_{i,k}^{\text{spd}}$	Signed speed margin, $k = 0, \dots, K - 1$.
$\sigma_{i,o,k}^{\text{obs}}$	Signed obstacle-clearance margin, $k = 0, \dots, K - 1$.
$\sigma_{i,j,k}^{\text{uav}}$	Signed pairwise separation margin, $k = 0, \dots, K - 1$.
$\hat{\sigma}$	Floating-point evaluation of a margin σ .
Δ_k	Largest copy-versus-source discrepancy over the two endpoints of segment k .
$\hat{N}_C(x)$	Regular (Fréchet) normal cone of a closed set C at x .
$N_C(x)$	Limiting normal cone of C at x ; the notion used by the convergence statements.
$e_Z^{r,t}$	Error witness of the local step, defined in (100): the amount by which the block returned by the local solver fails exact stationarity of its own subproblem, measured with the multiplier $y^{r,t-1}$ that was in force while that subproblem was solved.
$d_Z^{r,t}$	Stationarity witness, defined in (100): the same residual evaluated with the updated multiplier $y^{r,t}$. It is the quantity the convergence theory requires to vanish, and it obeys $\ d_Z^{r,t}\ \leq \ e_Z^{r,t}\ + R_1^{r,t}$.

Abbreviations.

Table N.2 Every abbreviation used in this report is expanded here, and each one is expanded again at its first use in the text.

Abbreviation	Meaning
UAV	Unmanned aerial vehicle.
MAV	Micro aerial vehicle.
CBO	Consensus-based optimization, in the particle/statistical-physics sense.
CCBO	Constrained consensus-based optimization, the problem class used in this work.

Abbreviation	Meaning
TCC	Trajectory-Copy Consensus, the reformulation introduced in Section 8.
NLP	Nonlinear program.
QCQP	Quadratically constrained quadratic program.
KKT	Karush–Kuhn–Tucker.
LICQ	Linear independence constraint qualification.
MFCQ	Mangasarian–Fromovitz constraint qualification.
CRCQ	Constant rank constraint qualification.
CPLD	Constant positive linear dependence condition.
SBCQ	Sequentially bounded constraint qualification.
SOSC	Second-order sufficient condition.
SCC	Strict complementarity condition.
ALM	Augmented Lagrangian method.
ADMM	Alternating direction method of multipliers.
ALADIN	Augmented Lagrangian based alternating direction inexact Newton method.
MPC	Model predictive control.
NMPC	Nonlinear model predictive control.
DMPC	Distributed model predictive control.
RHC	Receding-horizon control.
MILP	Mixed-integer linear program.
CBF	Control barrier function.
QP	Quadratic program.
IPOPT	Interior Point OPTimizer, the nonlinear programming solver used for the local steps.
ROS	Robot Operating System.
DOI	Digital object identifier.

1 Introduction

A swarm of unmanned aerial vehicles operating in a cluttered environment must solve a coordination problem that is genuinely shared. Each vehicle has its own destination, its own speed limit and its own view of the obstacles around it, and would like to plan using only what it can measure and what its neighbours can tell it. Yet the requirement that no two vehicles collide, and that none of them strikes an obstacle, is a property of the swarm as a whole. A plan that is excellent for one vehicle in isolation may be unacceptable once its neighbours' plans are known. This tension between local computation and shared safety is what makes swarm path planning a coordination problem rather than a collection of independent planning problems.

Optimization gives a natural language for this. Let each vehicle carry a decision variable describing its own future trajectory, a local objective expressing where it wants to go and how smoothly it wants to get there, and a local feasible set expressing the workspace it may occupy, the speed it may not exceed and the obstacles it must clear. Then attach, to each pair of vehicles, a coupling constraint requiring them to stay apart. The swarm problem is the minimization of the sum of the local objectives subject to all of these constraints at once. Written this way, the problem inherits a large and mature body of theory: distributed optimization over graphs, constrained consensus, dual decomposition for coupled constraints, and augmented Lagrangian and splitting methods.

That theory does not, however, transfer to the swarm problem without work, and the reasons are specific. Obstacle clearance makes the local feasible sets nonconvex, which removes the convexity assumption on which most constrained consensus results rest. The coupling between vehicles is geometric and pairwise, not an aggregate resource constraint of the kind that dual decomposition handles most directly. Safety is required not merely at the instants at which the trajectory is sampled, but at every instant in between, and enforcing it only at samples is not a small approximation: two vehicles can be comfortably separated at two consecutive samples and pass through each other between them. Finally, and most importantly for the formulation, the word *consensus* cannot be taken at face value. In its usual form, consensus drives the agents' decision variables towards a common value. If the decision variable is the physical trajectory, then exact consensus means that two distinct vehicles fly the same path, which is precisely what collision avoidance forbids.

This report develops a decentralized planning formulation that confronts each of those points explicitly, and reports in full what has been established for its first instance: a completely deterministic model, in which the workspace and the obstacles are static and exactly known, the communication structure is given, and no uncertainty is present. That instance is called Model 0-D throughout, where the suffix stands for *deterministic* and carries no other meaning; in particular it is not a statement about dimension, and the spatial dimension is $q \in \{2, 3\}$.

Working with a deterministic instance first is a deliberate methodological choice, not a limitation of ambition. Uncertainty and dynamic environments change the meaning of almost every object in the formulation — what a feasible set is, what a safety guarantee asserts, what an agreement constraint can be expected to deliver when the information being agreed upon is

stale. Establishing an unambiguous deterministic baseline first makes it possible to say later, precisely, which conclusions survive the extension and which do not.

The contributions reported here are the following.

1. **A planning model with exact inter-sample safety.** The trajectory is represented by finitely many sampled positions joined by straight segments, and safety is imposed as a minimum-distance condition over each complete segment rather than at its endpoints. Section 4 derives the exact minimum-distance expressions in closed form, with no auxiliary optimization variable and no smoothing approximation, and shows by explicit example why the sampled condition is insufficient.
2. **A complete well-posedness analysis of the resulting problem.** Section 6 proves existence of a global minimizer by a constructive argument, characterizes exactly when the objective is strictly convex, and then exhibits a fully admissible instance with a strictly convex objective and two distinct global minimizers, which settles the question of uniqueness in the negative. It also states what stability under perturbation of the data can and cannot mean when the solution is set-valued.
3. **A first-order analysis that locates its own failure modes.** Section 7 identifies three concrete geometric mechanisms by which the standard constraint qualifications fail for this problem, and states the Karush–Kuhn–Tucker theorem in the conditional form those mechanisms force. The conditions under which first-order necessity may be invoked are therefore explicit rather than assumed.
4. **Trajectory-Copy Consensus.** Section 8 resolves the difficulty with the word consensus. Agreement is imposed between computational *copies of the same source trajectory* held at different vehicles, never between the trajectories of distinct vehicles. The physical trajectories remain distinct variables and are optimized jointly. Theorem 8.9 proves that this reformulation preserves the feasible set, the optimal value and the set of global minimizers exactly.
5. **A distributed solver whose assumptions are checked, not assumed.** Section 9 maps the reformulated problem onto a two-level method combining an outer augmented Lagrangian with an inner three-block alternating direction method, derives each of its update steps from the local optimality conditions, and states the source assumptions that the mapping requires.
6. **A numerical study with an exact safety certificate.** Section 10 reports a two-vehicle crossing instance solved to the specified agreement and stopping tolerances, reproducible to the bit across independent executions. The safety of the computed plan is certified in exact rational arithmetic, not by a floating-point tolerance, and the relationship between that certificate and the finite agreement error is proved as a lemma rather than asserted. An independent centralized solve of the same instance reproduces the objective to seven significant figures and the trajectory to thirty-three micrometres, and independently exhibits the non-uniqueness proved in Section 6.

It is equally important to state what this report does not claim. The value of a formulation of this kind depends as much on the sharpness of its boundary as on the strength of its results, and that boundary is stated here at the outset rather than buried at the end. This report does not claim global optimality for the nonconvex problem it poses; it does not claim that any constraint qualification holds universally for this problem, and indeed proves that none of the standard ones does; it does not claim a convergence theorem for the distributed method on this problem class; it does not claim probabilistic safety, since no uncertainty model is part of the deterministic instance; and it does not claim that a certified planned trajectory is a certified flown trajectory, since no tracking controller or vehicle dynamics are modelled. Section 14 states the full boundary in one place, and each individual result is accompanied by the conditions under which it holds.

The contribution claimed is an integration. The literature reviewed in Section 2 provides strong components — constrained consensus over graphs, distributed optimization with coupled inequality constraints, chance-constrained and robust planning, distributed predictive control, and safety-filtering methods — but provides them separately. What is assembled here is a single constrained consensus formulation for UAV swarm path planning that carries local feasibility, exact pairwise geometric coupling, an agreement mechanism compatible with physical separation, and an explicit account of where its own guarantees end.

One phrase in common use needs to be separated at the outset from what is done here, because the two are easily confused. The phrase *Consensus-Based Optimization (CBO)* is used in the literature for a family of particle methods in which a population of candidate solutions is driven towards a weighted average and analysed through its mean-field limit. The present work is not of that kind, and should not be read as an instance of it. Here, consensus is a hard algebraic agreement constraint between computational copies of a trajectory, imposed inside a constrained optimization problem and resolved by a deterministic distributed algorithm. The connection is one of vocabulary and of general spirit — coordination through agreement rather than through a central coordinator — not of method. Where the distinction matters for a particular statement, it is repeated locally.

The remainder of the report is organized as follows. Section 2 surveys the literature and states the gap this work addresses. Section 3 defines the deterministic planning model. Section 4 derives exact inter-sample safety. Section 5 states the optimization problem and its solver representation. Section 6 treats existence, uniqueness and stability. Section 7 develops the first-order analysis. Section 8 introduces and justifies Trajectory-Copy Consensus. Section 9 specifies the distributed algorithm. Section 10 reports the numerical study. Section 11 presents the simulation environment. Sections 12 and 13 report two scoping studies, on uncertainty and on higher-order trajectory representations. Section 14 states what is established and what is not. Section 15 sets out the continuation of the work, and Section 16 concludes. Supporting material is collected in the Appendix.

Earlier exploratory work on this topic, carried out before the model presented here was settled, is retained separately and is not part of the results reported in this document; it is mentioned only where it explains why a particular modelling choice was made.

2 State of the Art

The problem posed in Section 1 sits at the meeting point of several mature literatures. None of them is empty, and it would be both false and unnecessary to claim that decentralized planning, collision avoidance or distributed optimization are unexplored. The purpose of this section is the opposite: to set out what each literature establishes solidly, so that the contribution of the present work can be stated narrowly and defended. The review is organized by mathematical proximity to the formulation of Section 5 rather than by application area, and closes with an explicit statement of the gap.

2.1 Consensus and constrained consensus over graphs

The coordination grammar used here descends from the consensus literature. Olfati-Saber and Murray [1] established convergence of consensus dynamics under switching topology and communication delays, and the subsequent survey of Olfati-Saber, Fax and Murray [2] consolidated the algebraic graph-theoretic conditions — connectivity, balancedness, spectral properties of the Laplacian — under which agreement is reached. Ren, Beard and Atkins [3] gave the multi-vehicle reading of the same theory, and Cortés [4] extended agreement beyond the arithmetic mean to general consensus functions, which matters as soon as the quantity being agreed upon is not a plain average.

The bridge from agreement to optimization was made by Nedić and Ozdaglar [5], who analysed distributed subgradient methods minimizing a sum of local objectives over a time-varying network, and quantified the trade-off between solution accuracy and iteration count. The closest single antecedent of the present formulation is Nedić, Ozdaglar and Parrilo [6], which adds *local constraint sets* to that picture: each agent carries its own feasible set, and the algorithm alternates averaging with projection. That paper is the correct starting point for the phrase constrained consensus, and it supplies the vocabulary used throughout Section 8: projection after averaging, the distinction between the consensus value and the optimizer, and the connectivity assumptions under which the two coincide. Nedić, Olshevsky and Rabbat [7] later mapped the communication–computation trade-offs that govern how sparse the graph may be made.

What this family establishes is strong and directly reusable. What it does not provide is equally clear. The local constraint sets are, with few exceptions, assumed convex, and the convergence arguments rest on that convexity through the non-expansiveness of the projection. Obstacle clearance destroys it: the set of positions at distance at least κ from a point is the complement of an open ball, and is not convex. Nor does this family address trajectory-valued decision variables with time structure, or geometric coupling between agents.

2.2 Distributed optimization with coupled inequality constraints

The second family addresses exactly the object that the first one lacks: constraints that tie agents together. Falsone, Margellos, Garatti and Prandini [8] give a dual-decomposition scheme for local objectives with local constraints and coupling constraints over a time-varying

network, and their treatment of the coupled constraint by dual exchange is the most direct available template for translating a pairwise safety requirement into distributed optimization terms. Li, Feng and Xie [9] provide a distributed proximal primal–dual algorithm for coupled inequality constraints over directed, unbalanced and time-varying networks, and Li, Yi and Xie [10] add the online dimension, tracking both dynamic regret and cumulative constraint violation, which is the natural language for a feasible set that moves.

The splitting methods underlying these schemes are surveyed comprehensively by Boyd, Parikh, Chu, Peleato and Eckstein [11], whose treatment of the alternating direction method of multipliers, of consensus reformulations and of augmented Lagrangian decomposition is the standard reference for the algorithmic layer of the present work.

The limitation, for present purposes, is structural. The coupling treated in this family is typically of resource-sharing type: a sum of agent contributions constrained in aggregate. Pairwise geometric separation is not of that type. It is not aggregate, it is not additively separable across agents, and — decisively — it is not convex. The translation from $\sum_i g_i(x_i) \leq 0$ to $h_{ij}(x_i, x_j) \leq 0$ with h_{ij} encoding a minimum distance is not a matter of notation.

2.3 Planning under uncertainty

The third family supplies the language in which safety under uncertainty is expressed. Blackmore, Ono and Williams [12] formulate chance-constrained optimal path planning with obstacles, allocating a total risk budget across individual constraints and converting the probabilistic requirement into a deterministic tightening. Ono and Williams [13] extend the construction to decentralized chance-constrained finite-horizon control for multi-agent systems, which is the closest available bridge between probabilistic safety and decentralized coordination. The underlying theory is classical: Prékopa [14] and Shapiro, Dentcheva and Ruszczyński [15] for chance constraints and stochastic programming, and Ben-Tal, El Ghaoui and Nemirovski [16] for the robust counterpart when a bounded uncertainty set is more appropriate than a distribution.

This family is not used in the deterministic model reported here, by construction. It is reviewed because it defines the shape of the extension scoped in Section 12, and because its central caution applies already: a probabilistic guarantee is only as meaningful as the probability model behind it, and inventing a distribution in order to be able to write a chance constraint is not a modelling step but an evasion of one.

2.4 Distributed and receding-horizon predictive control

The fourth family is the closest to realistic vehicle behaviour. Dunbar and Murray [17] established distributed receding-horizon control for multi-vehicle formation stabilization with trajectory exchange between coupled subsystems, and Keviczky, Borrelli and Balas [18] treated large-scale dynamically decoupled systems whose costs and constraints are coupled through a graph — mathematically very close to vehicles whose dynamics are independent but whose safety constraints are not. Dai, Cao, Xia and Gao [19] combine distributed predictive control

with both formation and obstacle avoidance. Wang and Ding [20] and Ravanshadi, Amini Boroujeni and Pourgholi [21] extend the same line to synthesis and to time-varying obstacles. The architectural landscape is surveyed by Negenborn and Maestre [22] and treated at length in [23]; the underlying theory of predictive control, terminal ingredients and recursive feasibility is in Rawlings, Mayne and Diehl [24].

This family supplies dynamics, prediction horizons and replanning, and it is the natural home of the receding-horizon extension discussed in Section 12. It is not, however, a constrained consensus formulation. Coordination in distributed predictive control is typically achieved by exchanging predicted trajectories and assuming compatibility, rather than by imposing agreement as a hard constraint inside a single optimization problem. The distinction is not cosmetic: it determines whether the coupling is a constraint of the problem or a protocol around it.

2.5 Vehicle-specific navigation and safety filtering

The fifth family is where the mathematics meets flying hardware. Lindqvist, Sopasakis and Nikolakopoulos [25] present a scalable distributed collision-avoidance scheme for multi-agent UAV systems based on nonlinear predictive control with predicted trajectory broadcasting, evaluated up to ten aerial agents. Mansouri et al. [26] give a unified predictive scheme for micro aerial vehicle navigation with three-dimensional collision avoidance under position uncertainty, validated in Gazebo, and Papadimitriou, Jafari, Mansouri and Nikolakopoulos [27] address varying communication delays in the same setting.

A distinct and rigorous line within this family enforces safety as a last-layer filter rather than through trajectory optimization. Ames, Grizzle and Tabuada [28] introduced control barrier function quadratic programs, and Wang, Ames and Egerstedt [29] developed safety barrier certificates for collision-free multirobot systems, together with a careful discussion of conservativeness, feasibility and deadlock — vocabulary that this report borrows in Section 14. Earlier optimization-based trajectory planning with collision avoidance, formulated as a mixed-integer linear program, is due to Richards and How [30], and Mastellone, Stipanović, Graunke, Intlekofer and Spong [31] treated formation control and collision avoidance for nonholonomic multi-agent systems both theoretically and experimentally.

These methods provide realistic validation scenarios and a mature failure vocabulary. They are not, however, trajectory-level decentralized planning formulations: a safety filter modifies a nominal command to keep a forward-invariant safe set, which is a different object from a planner that optimizes a horizon of positions subject to explicit constraints.

2.6 Potential fields as contrast

Artificial potential fields, introduced by Khatib [32], remain the most widely used heuristic for reactive obstacle avoidance, and they are cited here deliberately as a contrast rather than as a baseline to reproduce. They are computationally light and intuitive, but they offer no convergence guarantee, no feasibility certificate and no principled way to state what safety margin

has been achieved. The motivation for posing swarm planning as a constrained optimization problem at all is precisely to replace such guarantees-by-construction-absent with statements that can be proved or disproved.

2.7 Consensus-Based Optimization (CBO) in the particle sense

A separate body of work uses the term Consensus-Based Optimization (CBO) for derivative-free particle methods in which a population of candidate solutions is driven towards a weighted average and analysed through a mean-field limit [33, 34]. That literature is mathematically elegant and is a legitimate line of research, but it solves a different problem with different tools. It is recorded here to prevent a terminological confusion, not because the present work builds on it. The formulation developed in this report is a deterministic constrained optimization problem with hard agreement constraints; no particle ensemble, no weighted average and no mean-field limit appears.

2.8 What is established, and what remains difficult

Collecting the above, the following can be taken as established by the literature and is used here without reproof: consensus over graphs, including under switching topology and delay; distributed optimization for sums of local convex objectives over networks; constrained consensus with convex local sets and projection-based algorithms; handling of coupled inequality constraints by dual decomposition and proximal primal–dual schemes; chance-constrained and robust formulations of probabilistic safety; distributed predictive control for dynamics, prediction and replanning; and barrier-certificate safety filtering with its associated failure vocabulary.

What remains difficult, and is not resolved by importing any one of these families, is the following.

1. **Nonconvex local feasibility.** Obstacle clearance makes the local sets nonconvex, so the projection-based guarantees of constrained consensus weaken or fail.
2. **Pairwise geometric coupling.** Separation constraints are structured differently from aggregate resource coupling, and are themselves nonconvex.
3. **Continuous-time safety.** Enforcing separation at sampled instants does not enforce it between them, and the gap is not small.
4. **The meaning of agreement.** Exact consensus on physical trajectories contradicts separation, so the agreement variable must be chosen with care rather than inherited.
5. **Assumption transfer.** Distributed algorithms for nonconvex problems carry assumptions — exact local stationarity, constraint qualifications, dual boundedness — that are easy to state and hard to verify, and that are frequently assumed silently.

2.9 The gap addressed in this work

The literature does not leave this problem empty; it leaves it *separated*. Each of the five families supplies a mature component, and no one of them supplies the combination. Constrained consensus gives the coordination grammar but under convexity; coupled-constraint optimization gives the dual machinery but for aggregate coupling; chance-constrained planning gives probabilistic safety but usually centralized; predictive control gives dynamics and replanning but is not a consensus formulation; safety filtering gives rigorous last-layer avoidance but not trajectory-level planning.

Gap addressed. To formulate decentralized UAV swarm path planning as a single Constrained Consensus-Based Optimization (CCBO) problem that simultaneously carries local feasibility with nonconvex obstacle clearance, exact pairwise geometric coupling enforced between samples and not only at them, an agreement mechanism compatible with physical separation, sparse communication, and an explicit account of which guarantees hold under which conditions.

This is an integration and formulation gap, and it is claimed as such. The remainder of this report addresses it for the deterministic instance, and states precisely how far that instance goes.

3 The Deterministic Planning Model

This section fixes the planning model. Every quantity introduced here is deterministic and known at the planning instant; nothing in this section is random, estimated online, or time-varying beyond the explicit dependence on the sample index. The choices made are recorded together with the reason for making them, because several of them are simplifications that a later version of the model is expected to revisit, and it is important to distinguish a simplification adopted deliberately from an omission.

3.1 Vehicles, horizon and data

Let

$$V := \{1, \dots, N\}, \quad N \geq 2,$$

be the set of vehicles, and let $q \in \{2, 3\}$ be the spatial dimension. The vehicles are also the nodes of a graph, and every graph used in this report is written in the standard form

$$G = (V, E), \quad E \subseteq \{\{i, j\} : i, j \in V, i \neq j\},$$

in which the node set V is the set of vehicles and the edge set E is a set of unordered pairs of distinct vehicles, an edge $\{i, j\} \in E$ recording that vehicles i and j can exchange information directly. Different graphs on the swarm differ only in which pairs they admit as edges. Section 3

below distinguishes the three edge sets that must not be confused with one another, and the graph on which the decentralized solve is actually carried out is $G_0^{\text{comm}} = (V, E_0^{\text{comm}})$.

Planning is performed at an instant t_0 over a uniform grid

$$t_k = t_0 + k \Delta t, \quad k = 0, \dots, K, \quad (1)$$

with sampling period $\Delta t > 0$ and $K \geq 1$ prediction intervals. The grid has $K + 1$ samples and K segments; throughout, a quantity indexed by a *sample* ranges over $k = 0, \dots, K$ and a quantity indexed by a *segment* over $k = 0, \dots, K - 1$.

Each vehicle i carries deterministic data: a goal position $g_i \in \mathbb{R}^q$, a geometric radius $r_i > 0$, a maximum speed $v_i^{\text{max}} > 0$, and objective weights $\alpha_i, \beta_i \geq 0$. For each pair $\{i, j\}$ a separation margin $\delta_{ij}^{\text{uav}} > 0$ is prescribed, and for obstacle clearance each vehicle carries a margin $\delta_i^{\text{obs}} > 0$.

Obstacles are indexed by a finite, possibly empty set $\mathcal{O}$. Obstacle o has centre $c_o \in \mathbb{R}^q$ and radius $R_o > 0$, and is static over the planning horizon:

$$c_o(t) = c_o, \quad t \in [t_0, t_K]. \quad (2)$$

The radius-based description models a disc in $q = 2$ and a ball in $q = 3$. It applies to another shape only when that shape admits a rigorously equivalent fixed-centre, constant-radius clearance test; it must not be extended to an arbitrary geometry by analogy. Static obstacles are a deliberate simplification of this first model: determinism alone would not force obstacles to be static, and moving obstacles would require their own inter-sample safety analysis rather than a reuse of the formulas derived in Section 4.

3.2 Workspace

The workspace is the set of admissible vehicle centre positions. It is *not* the obstacle-free region: obstacles are separate constraints. It is taken to be a polytope

$$\mathcal{W} = \{p \in \mathbb{R}^q : A_{\mathcal{W}} p \leq b_{\mathcal{W}}\}, \quad \text{int}(\mathcal{W}) \neq \emptyset, \quad (3)$$

with the represented polyhedron bounded, so that $\mathcal{W}$ is nonempty, compact, convex and full-dimensional. Row m of the representation is written $a_m^\top p \leq b_m$, and every retained row satisfies $a_m \neq 0$; Section 5.4 states how a representation is validated before it is used. An axis-aligned box is an admissible special case but not the only one.

Convexity of $\mathcal{W}$ is used, and is not decorative: Proposition 4.1 below shows that it is exactly what makes a condition imposed at the samples sufficient for the whole continuous trajectory.

3.3 Communication, information and physical safety are three different things

It is convenient, and wrong, to identify the set of vehicles that can talk to each other with the set of vehicles that must avoid each other. The model keeps the three notions apart. Write

$$E^{\text{all}} := \{\{i, j\} : 1 \leq i < j \leq N\}$$

for all unordered pairs. At sample t_k , let E_k^{comm} be the pairs able to communicate directly, and E_k^{coll} the pairs actually exchanging collision-relevant information. These satisfy

$$E_k^{\text{coll}} \subseteq E_k^{\text{comm}} \subseteq E^{\text{all}}, \quad k = 0, \dots, K, \quad (4)$$

with the corresponding range convention

$$r_i + r_j + \delta_{ij}^{\text{uav}} < R_{ij}^{\text{coll,info}} \leq R_{ij}^{\text{comm}}.$$

The hard separation requirement, however, is imposed on *every* pair in E^{all} , regardless of whether the pair can communicate:

$$\text{physical safety} \neq \text{collision-related information} \neq \text{communication availability}. \quad (5)$$

Sparse communication changes how a constraint is *evaluated and enforced*; it never deletes the constraint. Making this explicit at the model level is what later allows Section 8 to reduce communication without reducing safety.

The edge sets are exogenous data for the solve considered here. State-dependent link activation, packet loss and latency dynamics are outside this model; the symbol $\zeta(e, t_k)$ is reserved for latency and is not used. For the decentralized solve launched at t_0 , the operative structure is the fixed connected graph $G_0^{\text{comm}} = (V, E_0^{\text{comm}})$. This choice affects only how the problem is computed, never what the problem is.

3.4 Decision variable and the fixed initial state

For vehicle i the complete sampled trajectory record is

$$x_i = (p_i^0, p_i^1, \dots, p_i^K), \quad p_i^k \in \mathbb{R}^q, \quad (6)$$

where the superscript k in p_i^k is an *index* and not an exponent: it names the sample t_k at which the position is taken, so p_i^k is a point of $\mathbb{R}^q$ and never a power of p_i . The same reading applies to every superscripted sample or segment quantity in this report, among them the segment displacement d_i^k , the segment velocity v_i^k and the squared displacement s_i^k . Where a genuine power is intended, the quantity raised to it is parenthesized, as in $(\kappa_{io}^{\text{obs}})^2$. The first entry

$$p_i^0 := \hat{p}_i(t_0) \quad (7)$$

is the measured or estimated position at the planning instant. It is input data. It is not a decision variable, and it is not imposed through a redundant equality constraint; it is simply not optimized. The optimizer chooses the free block

$$x_{i,\text{free}} = (p_i^1, \dots, p_i^K) \in \mathbb{R}^{qK}, \quad (8)$$

and the full decision vector $x_{\text{free}} = (x_{1,\text{free}}, \dots, x_{N,\text{free}})$ lies in $\mathbb{R}^n$ with $n = NqK$.

Because some statements are simplest in full records and others in free coordinates, the translation between them is named once. For fixed initial data $p^0 = (p_1^0, \dots, p_N^0)$ define

$$\text{Full}_{p^0}(x_{\text{free}}) := ((p_i^0, x_{i,\text{free}}))_{i=1}^N, \quad (9)$$

and let π_{free} remove those initial entries. Then

$$\pi_{\text{free}}(\text{Full}_{p^0}(x_{\text{free}})) = x_{\text{free}}. \quad (10)$$

This introduces neither a variable nor a constraint; it records which representation is in use. We write $\mathcal{F}_{0-D}^{\text{full}}(t_0)$ for the feasible full records and $\mathcal{F}_{0-D}^{\text{free}}(t_0) = \pi_{\text{free}}(\mathcal{F}_{0-D}^{\text{full}}(t_0))$ for their free-coordinate image, and use the shorter $\mathcal{F}_{0-D}(t_0)$ when no ambiguity arises.

Fixing the initial position has three consequences used later: initial safety becomes a check on the input data rather than a constraint carrying a multiplier (Section 3.6); the movement penalty is anchored, which is what makes the strict-convexity characterization of Theorem 6.3 come out as it does; and the constant trajectory used in the existence proof is well defined.

3.5 Objective

For vehicle i ,

$$f_i(x_i) = \alpha_i \|p_i^K - g_i\|^2 + \beta_i \sum_{k=0}^{K-1} \|p_i^{k+1} - p_i^k\|^2, \quad (11)$$

and the swarm objective is

$$F(x_{\text{free}}) = \sum_{i=1}^N f_i(x_i). \quad (12)$$

The first term rewards ending near the goal. The second penalizes large successive displacements; it is a sum of squared steps and is *not* the geometric path length, which would be $\sum_k \|p_i^{k+1} - p_i^k\|$. The squared form is chosen because it is smooth everywhere, including where a step vanishes, whereas the norm is not differentiable there.

The goal is soft: no equality $p_i^K = g_i$ is imposed. This is essential and not incidental. It is what allows the constant trajectory to be feasible in Theorem 6.1, and therefore what makes the feasible set nonempty without an assumption.

3.6 Admissibility of the initial data

Since p_i^0 is not optimized, the planner cannot repair an inadmissible initial state. The input data must therefore satisfy, before optimization begins,

$$A_{\mathcal{W}} p_i^0 \leq b_{\mathcal{W}} \quad \forall i \in V, \quad (13)$$

$$R_o + r_i + \delta_i^{\text{obs}} \leq \|p_i^0 - c_o\| \quad \forall i \in V, \forall o \in \mathcal{O}, \quad (14)$$

$$r_i + r_j + \delta_{ij}^{\text{uav}} \leq \|p_i^0 - p_j^0\| \quad \forall \{i, j\} \in E^{\text{all}}, \quad (15)$$

where the left-hand sides of (14) and (15) are the required centre clearances, written out in full here so that each condition can be read without a substitution:

$$\kappa_{io}^{\text{obs}} := R_o + r_i + \delta_i^{\text{obs}}, \quad \kappa_{ij}^{\text{uav}} := r_i + r_j + \delta_{ij}^{\text{uav}}. \quad (16)$$

All three conditions are written in the same direction, with $\leq$, so that none of them has to be read backwards against the others. Every constraint of the planning problem is stated this

way; the reduced representation of Section 5.2, which the solver uses, then moves each of them to the single normalized form *left-hand side at most zero*.

Conditions (13)–(15) are admissibility checks on the data. They carry no multipliers and appear nowhere in the first-order conditions of Section 7. If they fail, the problem as posed has no meaning, and the correct response is to reject the input rather than to solve a relaxed problem.

4 Continuous Trajectory Reconstruction and Exact Safety

The decision variable is a finite list of positions, but a vehicle flies a continuous path. This section fixes how the continuous path is reconstructed from the samples, and then derives safety conditions that hold at every instant of the horizon rather than at the sampled instants alone. The derivations are given in full because the resulting expressions are used directly by the solver and because the difference between sampled and inter-sample safety is the difference between a plan that is safe and a plan that merely looks safe.

4.1 Piecewise-linear reconstruction

For segment $k \in \{0, \dots, K-1\}$ and $t \in [t_k, t_{k+1}]$ set

$$\lambda(t) := \frac{t - t_k}{\Delta t} \in [0, 1], \quad (17)$$

and define the reconstructed position by linear interpolation,

$$p_i(t) = (1 - \lambda(t)) p_i^k + \lambda(t) p_i^{k+1}. \quad (18)$$

The reconstructed trajectory is a continuous polyline determined entirely by the finite decision variable. On the interior of each segment its velocity is constant,

$$\dot{p}_i(t) = \frac{p_i^{k+1} - p_i^k}{\Delta t} =: v_i^k. \quad (19)$$

The velocity may jump at the knots, so the model asserts no global C^1 or C^2 regularity and makes no claim about acceleration, curvature, jerk, thrust or vehicle dynamics. Section 13 discusses what a smoother representation would require.

4.2 Workspace membership transfers exactly from samples to the continuum

Proposition 4.1 (Exact workspace transfer). *Under the reconstruction (18) and with $\mathcal{W}$ convex,*

$$p_i^k \in \mathcal{W} \quad \forall k = 0, \dots, K \quad \Longleftrightarrow \quad p_i(t) \in \mathcal{W} \quad \forall t \in [t_0, t_K].$$

Proof. ($\Leftarrow$) The samples are values of $p_i(t)$ at $t = t_k$. ($\Rightarrow$) Fix $t \in [t_k, t_{k+1}]$. By (18), $p_i(t)$ is a convex combination of p_i^k and p_i^{k+1} , both in $\mathcal{W}$ by hypothesis, and $\mathcal{W}$ is convex. $\square$

No semi-infinite constraint is therefore needed for the workspace. The implication fails for a nonconvex workspace: the property that the segment joining any two points of a set remains in the set is exactly convexity, so this is the strongest statement available and it is available only because of the modelling choice made in (3).

4.3 The speed bound is exact on each segment

Proposition 4.2 (Exact speed transfer). *The sampled step condition*

$$\|p_i^{k+1} - p_i^k\| \leq v_i^{\max} \Delta t, \quad k = 0, \dots, K-1, \quad (20)$$

holds if and only if $\|\dot{p}_i(t)\| \leq v_i^{\max}$ on the interior of every segment, hence almost everywhere on $[t_0, t_K]$.

Proof. Immediate from (19): the velocity is constant on the interior of segment k and equals $(p_i^{k+1} - p_i^k)/\Delta t$, so the two conditions are the same inequality divided by $\Delta t > 0$. The knots form a finite set and carry no measure. $\square$

4.4 Why sampled separation is not enough

The two propositions above might suggest that sampled conditions always suffice. They do not, and the failure is not marginal.

Example 4.3 (Sampled safety hides a crossing). Take $q = 2$ and consider the relative position of two vehicles at the two ends of one segment,

$$a = (2, 0), \quad b = (-2, 0).$$

Both have norm 2, so any required separation $\kappa \leq 2$ is satisfied at both samples. The reconstructed relative position is $(1 - \lambda)a + \lambda b = (2 - 4\lambda, 0)$, which vanishes at $\lambda = \frac{1}{2}$: the two vehicles occupy the same point at the midpoint of the interval. The same construction works against an obstacle: a vehicle travelling from $(-2, 0)$ to $(2, 0)$ is at distance 2 from an obstacle centred at the origin at both samples, while the straight segment passes exactly through the centre.

Example 4.3 is the reason the rest of this section exists. Safety is therefore imposed as a condition on the minimum distance attained over each complete segment.

4.5 Exact vehicle–obstacle clearance

Fix a vehicle i , an obstacle o and a segment $k \in \{0, \dots, K-1\}$. Write

$$a := a_{io}^k = p_i^k - c_o, \quad d := d_i^k = p_i^{k+1} - p_i^k, \quad s := s_i^k = \|d\|^2, \quad (21)$$

so that the reconstructed position relative to the obstacle centre is $a + \lambda d$ for $\lambda \in [0, 1]$. The exact minimum distance over the segment is

$$D_{i,o,k}^{\text{obs}} := \min_{\lambda \in [0,1]} \|a + \lambda d\|, \quad (22)$$

and the requirement, written with $\leq$ like every other constraint in this report, is $R_o + r_i + \delta_i^{\text{obs}} \leq D_{i,o,k}^{\text{obs}}$, that is $\kappa_{io}^{\text{obs}} \leq D_{i,o,k}^{\text{obs}}$.

It is convenient to work with the square. Let

$$\psi(\lambda) := \|a + \lambda d\|^2 = \|a\|^2 + 2\lambda a^\top d + \lambda^2 s. \quad (23)$$

This is a one-dimensional quadratic with $\psi'' = 2s \geq 0$, so it is convex, and its behaviour splits into cases according to s .

Degenerate segment, $s = 0$. Then $d = 0$, the vehicle does not move on this segment, $\psi(\lambda) = \|a\|^2$ for every λ , and the minimum is $\|a\|^2$.

Nondegenerate segment, $s > 0$. Then ψ is strictly convex and

$$\psi'(\lambda) = 2a^\top d + 2\lambda s,$$

which vanishes at the unique unconstrained minimizer

$$\lambda^* = -\frac{a^\top d}{s}. \quad (24)$$

Since ψ is strictly convex, its minimum over $[0, 1]$ is attained at $\hat{\lambda} = \min\{1, \max\{0, \lambda^*\}\}$, and the three cases are distinguished by the sign of $a^\top d$ relative to $-s$:

- if $a^\top d \geq 0$ then $\lambda^* \leq 0$, the minimum is at $\hat{\lambda} = 0$ and equals $\|a\|^2$;
- if $a^\top d \leq -s$ then $\lambda^* \geq 1$, the minimum is at $\hat{\lambda} = 1$ and equals $\|a + d\|^2$;
- if $-s < a^\top d < 0$ then $\lambda^* \in (0, 1)$ and, substituting (24) into (23),

$$\psi(\lambda^*) = \|a\|^2 - 2\frac{(a^\top d)^2}{s} + \frac{(a^\top d)^2}{s} = \|a\|^2 - \frac{(a^\top d)^2}{s}.$$

Collecting the cases gives the exact squared minimum distance in closed form, with no auxiliary variable and no approximation:

$$M_{i,o,k}^{\text{obs}} = \begin{cases} \|a\|^2, & s = 0, \\ \|a\|^2, & s > 0, a^\top d \geq 0, \\ \|a\|^2 - \frac{(a^\top d)^2}{s}, & s > 0, -s < a^\top d < 0, \\ \|a + d\|^2, & s > 0, a^\top d \leq -s. \end{cases} \quad (25)$$

The clearance requirement is then, equivalently and in a form with no square root,

$$g_{i,o,k}^{\text{obs,seg}} := (\kappa_{io}^{\text{obs}})^2 - M_{i,o,k}^{\text{obs}} \leq 0. \quad (26)$$

The case boundaries in (25) are exact. If an implementation later guards the division by a numerical threshold, that is a safeguard of the implementation and not part of this definition.

4.6 Exact vehicle–vehicle clearance

The pairwise case reduces to the same computation performed on the *relative* trajectory. Fix a pair $\{i, j\}$ and a segment k , and set

$$a := p_i^k - p_j^k, \quad b := p_i^{k+1} - p_j^{k+1}. \quad (27)$$

Both vehicles are reconstructed by (18) on the same segment with the same parameter λ , so their difference is

$$p_i(t) - p_j(t) = (1 - \lambda)a + \lambda b = a + \lambda d, \quad d := b - a,$$

which is exactly the object treated in Section 4.5. Hence the exact minimum separation over the segment is

$$D_{i,j,k} := \min_{\lambda \in [0,1]} \|(1 - \lambda)a + \lambda b\|, \quad (28)$$

and, with $s = \|d\|^2 = \|b - a\|^2$, the same case analysis yields the squared value $M_{i,j,k}^{\text{uav}}$ by (25). When $a \neq b$, the unconstrained minimizer can be written directly in terms of the endpoints:

$$\lambda^* = -\frac{a^\top (b - a)}{\|b - a\|^2} = \frac{\|a\|^2 - a^\top b}{\|b - a\|^2}, \quad (29)$$

again clipped to $[0, 1]$. If $a = b$ the relative position is constant over the segment and the minimum is $\|a\|$.

The separation requirement admits two equivalent forms. The unsquared form,

$$h_{i,j,k}^{\text{inter,coll}}(x_i, x_j) := \kappa_{ij}^{\text{uav}} - D_{i,j,k}(x_i, x_j) \leq 0, \quad (30)$$

is the one to state physically. The squared form,

$$h_{i,j,k}^{\text{inter,coll,sq}}(x_i, x_j) := (\kappa_{ij}^{\text{uav}})^2 - M_{i,j,k}^{\text{uav}}(x_i, x_j) \leq 0, \quad (31)$$

is the one to give a solver. Since the required clearances are strictly positive, the two define the same feasible set; they are nevertheless different functions, with different gradients and different conditioning, and the distinction is maintained throughout. Constraint (31) is imposed for every pair in E^{all} and every segment $k = 0, \dots, K - 1$.

Remark 4.4 (Why the inequalities are not strict). Safety is written with $\leq$ rather than $<$. Because $\delta_i^{\text{obs}} > 0$ and $\delta_{ij}^{\text{uav}} > 0$, equality means that the prescribed positive margin is exactly attained, not that bodies touch. The non-strict form also makes the safety sets closed, which is used in the compactness argument of Theorem 6.1; a strict inequality would define an open set on which a minimum need not be attained.

Remark 4.5 (Sampled separation is retained only for exposition). The sampled function $h_{i,j,k}^{\text{coll}}(x_i, x_j) = \kappa_{ij}^{\text{uav}} - \|p_i^k - p_j^k\|$, $k = 0, \dots, K$, remains useful when describing the model in words. It is not part of the solver formulation, because $\lambda = 0$ and $\lambda = 1$ belong to $[0, 1]$ and the segment condition therefore already contains both endpoints.

5 The Planning Problem and Its Solver Representation

5.1 Local and complete feasible sets

For each vehicle i , collect the constraints that involve only that vehicle:

$$C_i(t_0) := \left\{ \begin{array}{l} A_W p_i^k \leq b_W, \quad k = 1, \dots, K, \\ x_i \in X_i(t_0) : \left\| p_i^{k+1} - p_i^k \right\| \leq v_i^{\max} \Delta t, \quad k = 0, \dots, K-1, \\ R_o + r_i + \delta_i^{\text{obs}} \leq D_{i,o,k}^{\text{obs}}, \quad o \in \mathcal{O}, \quad k = 0, \dots, K-1 \end{array} \right\}, \quad (32)$$

where $X_i(t_0) = \{\hat{p}_i(t_0)\} \times (\mathbb{R}^q)^K$ records that the first entry is fixed. All three rows are written with $\leq$, in the same direction, so that none of them has to be read against the others.

Two of the symbols in (32) are recalled here so that the set can be read without turning back. The left-hand side of the third row is the required centre clearance $\kappa_{io}^{\text{obs}} = R_o + r_i + \delta_i^{\text{obs}}$, written out in full: obstacle radius, plus vehicle radius, plus the vehicle's obstacle safety margin. Its right-hand side $D_{i,o,k}^{\text{obs}}$ is the quantity defined in (22): the *exact* minimum Euclidean distance between the centre c_o of obstacle o and the reconstructed position of vehicle i , taken over the whole of segment k and not merely at its two endpoints. The third row therefore says that the required clearance must not exceed the distance actually achieved anywhere along the segment.

The workspace condition is imposed for $k = 1, \dots, K$ because $k = 0$ was already checked on the input data in (13).

The complete feasible set adds the pairwise separation constraints, which are the only constraints that couple distinct vehicles:

$$\mathcal{F}_{0\text{-D}}(t_0) := \left\{ (x_1, \dots, x_N) : \begin{array}{l} x_i \in C_i(t_0), \quad \forall i \in V, \\ h_{i,j,k}^{\text{inter,coll}}(x_i, x_j) \leq 0, \quad \forall \{i, j\} \in E^{\text{all}}, \quad k = 0, \dots, K-1 \end{array} \right\}. \quad (33)$$

Two readings of (33) are worth fixing here. First, the subscript 0-D names *Model 0, Deterministic*: it identifies the deterministic instance of the model, the one in which the workspace and the obstacles are static and exactly known and no uncertainty is present. It says nothing about the spatial dimension, which is $q \in \{2, 3\}$ and is carried by q alone. Second, the function in the second row is the exact inter-sample separation function (30), whose value is

$$h_{i,j,k}^{\text{inter,coll}}(x_i, x_j) = \kappa_{ij}^{\text{uav}} - D_{i,j,k}(x_i, x_j) = (r_i + r_j + \delta_{ij}^{\text{uav}}) - D_{i,j,k}(x_i, x_j), \quad (34)$$

in which $D_{i,j,k}(x_i, x_j)$ is the exact minimum Euclidean distance between vehicles i and j over the whole of segment k . The row $h_{i,j,k}^{\text{inter,coll}} \leq 0$ is thus the statement $r_i + r_j + \delta_{ij}^{\text{uav}} \leq D_{i,j,k}(x_i, x_j)$: the two radii plus the pairwise safety margin must not exceed the separation actually achieved anywhere along the segment.

The planning problem can now be stated in full. Written out with every component expanded — the objective of each vehicle, the fixed initial state, the local rows of (32) and

the pairwise rows of (33) — it is

$$\begin{array}{ll}
\min_{\{x_{i,\text{free}}\}_{i=1}^N} & \sum_{i=1}^N \left[\alpha_i \|p_i^K - g_i\|^2 + \beta_i \sum_{k=0}^{K-1} \|p_i^{k+1} - p_i^k\|^2 \right] \\
\text{subject to} & p_i^0 = \hat{p}_i(t_0), \quad i \in V, \\
& x_{i,\text{free}} = (p_i^1, \dots, p_i^K) \in \mathbb{R}^{qK}, \quad i \in V, \\
& A_{\mathcal{W}} p_i^k \leq b_{\mathcal{W}}, \quad i \in V, k = 1, \dots, K, \\
& \|p_i^{k+1} - p_i^k\| \leq v_i^{\max} \Delta t, \quad i \in V, k = 0, \dots, K-1, \\
& R_o + r_i + \delta_{i,o}^{\text{obs}} \leq D_{i,o,k}^{\text{obs}}, \quad i \in V, o \in \mathcal{O}, k = 0, \dots, K-1, \\
& r_i + r_j + \delta_{ij}^{\text{uav}} \leq D_{i,j,k}, \quad \{i, j\} \in E^{\text{all}}, k = 0, \dots, K-1.
\end{array} \tag{P}$$

Every line of (P) has been met already: the objective is (11) summed over the swarm, the first two rows are the decision variable of (8) together with the fixed initial state (7), the next three rows are the local set (32), and the last row is the pairwise coupling of (33) in the equivalent form (34). The two exact distances $D_{i,o,k}^{\text{obs}}$ and $D_{i,j,k}$ are the closed-form segment minima of Section 4; they are not extra optimization variables.

The same problem is written more briefly whenever an argument needs it as a whole rather than row by row. In that compact form it is

$$\boxed{\min_{\{x_{i,\text{free}}\}_{i=1}^N} \sum_{i=1}^N f_i(x_i) \quad \text{subject to} \quad (x_1, \dots, x_N) \in \mathcal{F}_{0\text{-D}}(t_0)} \tag{P_c}$$

with f_i as in (11) and $\mathcal{F}_{0\text{-D}}(t_0)$ as in (33). Problems (P) and (P_c) are the same problem; only the level of detail differs, and references to either may be read as references to the planning problem itself.

Problem (P) is written as a single optimization problem over all vehicles. That is a statement about the mathematics, not about the computing architecture: it says what is to be optimized, not who computes it. Section 8 determines how the coupling in (33) is represented so that the problem can be solved without a central coordinator, and proves that the representation changes nothing.

5.2 Reduced representation used for analysis and computation

For differential analysis and for the solver, it is convenient to write (P) in the free variable x_{free} with the constraints as explicit scalar inequalities. Four families suffice:

$$g_{i,m,k}^{\mathcal{W}} = a_m^\top p_i^k - b_m \leq 0, \quad i \in V, m = 1, \dots, M_{\mathcal{W}}, k = 1, \dots, K, \tag{35}$$

$$g_{i,k}^{\text{spd}} = \|p_i^{k+1} - p_i^k\|^2 - (v_i^{\max} \Delta t)^2 \leq 0, \quad i \in V, k = 0, \dots, K-1, \tag{36}$$

$$g_{i,o,k}^{\text{obs,seg}} = (\kappa_{io}^{\text{obs}})^2 - M_{i,o,k}^{\text{obs}} \leq 0, \quad i \in V, o \in \mathcal{O}, k = 0, \dots, K-1, \tag{37}$$

$$h_{i,j,k}^{\text{inter,coll,sq}} = (\kappa_{ij}^{\text{uav}})^2 - M_{i,j,k}^{\text{uav}} \leq 0, \quad 1 \leq i < j \leq N, k = 0, \dots, K-1. \tag{38}$$

The squared speed form (36) is used rather than the norm form (20) for the same reason as in the objective: it is smooth where the step vanishes.

Three properties of this representation are worth stating explicitly, because each of them is a decision that could have been made differently.

- **No closest-point variable is introduced.** The parameter $\hat{\lambda}$ is not an optimization variable. It is eliminated in closed form by (25). An alternative would be to keep it and impose complementarity conditions, which turns the problem into a mathematical program with complementarity constraints; that route is not taken here, and the reason is recorded in Section 8.2.
- **No smoothing is applied.** The minimum in (22) is exact. Replacing it by a soft minimum would change the feasible set, and therefore the problem, in exchange for regularity.
- **Sampled safety constraints are omitted.** They are implied by the segment constraints, as noted in Remark 4.4, so including them would add rows without changing the feasible set — while worsening the constraint qualification analysis of Section 7 by introducing redundancy.

5.3 Classification of the constraint families

The four families behave very differently, and the differences drive the analysis that follows. Table 5.1 sets them side by side.

Table 5.1 The four constraint families differ in what they couple, in the geometry they induce and in the regularity they carry, and those differences drive the analysis that follows.

Family		Couples	Geometry	Regularity and role
Initial admissibility		data only	convex workspace; nonconvex safety regions	outside the optimization; no multipliers
Workspace g^w		one sample	convex, affine	C^∞ ; feasible set convex
Speed g^{spd}		two consecutive samples	convex quadratic sublevel set	C^∞ ; feasible set convex
Obstacle $g^{\text{obs,seg}}$	segment	two consecutive samples	nonconvex, exterior of a ball	value function: locally Lipschitz; C^1 where $s > 0$; not C^2 at regime switches; nonsmooth at $s = 0$
Pairwise $h^{\text{inter,coll,sq}}$	segment	two vehicles, two samples	nonconvex	as above; the only family coupling distinct vehicles

Two observations follow, and both are used later. First, *quadratic does not mean convex*: the sampled squared separation function is a concave quadratic whose sublevel set is the exterior of a ball, and it is exactly this that makes $\mathcal{F}_{0-D}(t_0)$ nonconvex in general. Second, the two segment families are value functions of a parametric minimization, so their regularity must be established rather than assumed; this is done in Section 6.5.

5.4 Validating the workspace representation

The halfspace representation (3) is data, and data can be malformed. Before it is used — in particular before any normalized margin is computed by dividing by $\|a_m\|$ — it is validated:

$$a_m \neq 0 \quad \text{for every retained row } m = 1, \dots, M_{\mathcal{W}}. \quad (39)$$

A row with $a_m = 0$ and $b_m \geq 0$ is satisfied by every point; it is removed, which does not change $\mathcal{W}$. A row with $a_m = 0$ and $b_m < 0$ is satisfied by no point; the representation is rejected as inconsistent rather than silently repaired. Exact duplicate rows and other plainly redundant rows should be stripped where practicable, since redundancy degrades the constraint qualification analysis without changing the set. Complete irredundancy is not required. Note that (39) is an exact algebraic condition: a small but nonzero normal is a legitimate row, and discarding it because it is small would change the workspace.

6 Existence, Uniqueness, Stability and Regularity

Before an algorithm is chosen, four questions must be answered about (P): does a solution exist, is it unique, does it depend reasonably on the data, and are the functions involved regular enough for first-order methods. This section answers all four. Two of the answers are negative, and those are as useful as the positive ones, because they determine what may legitimately be claimed later.

6.1 Nonemptiness and existence

Theorem 6.1 (Existence of a global minimizer). *Assume the initial data satisfy (13)–(15). Then $\mathcal{F}_{0-D}(t_0)$ is nonempty and compact, and problem (P) attains its minimum.*

Proof. Nonemptiness. Choose the constant trajectory

$$p_i^k := p_i^0, \quad i = 1, \dots, N, \quad k = 1, \dots, K, \quad (40)$$

so that every vehicle remains at its initial position. We verify each constraint of (33). Workspace membership at every sample holds because $p_i^0 \in \mathcal{W}$ by (13). Every displacement is zero, so $\|p_i^{k+1} - p_i^k\| = 0 \leq v_i^{\max} \Delta t$. For obstacle clearance on segment k , we have $d = 0$, hence $s = 0$ and, by the first case of (25), $M_{i,o,k}^{\text{obs}} = \|p_i^0 - c_o\|^2 \geq (\kappa_{io}^{\text{obs}})^2$ by (14). For pairwise separation, $a = b = p_i^0 - p_j^0$, so the relative position is constant along the segment and the minimum separation equals $\|p_i^0 - p_j^0\| \geq \kappa_{ij}^{\text{uav}}$ by (15). Hence the constant trajectory lies in $\mathcal{F}_{0-D}(t_0)$, which is therefore nonempty. Note where the soft goal is used: the constant trajectory does not reach g_i , and would be infeasible if a terminal equality had been imposed. *Compactness.* Every free sample is constrained to lie in $\mathcal{W}$, which is bounded, so $\mathcal{F}_{0-D}^{\text{free}}(t_0)$ is a bounded subset of $\mathbb{R}^n$. For closedness, observe that each defining condition is of the form $\{x : \phi(x) \leq 0\}$ with ϕ continuous: the workspace and speed functions are polynomials, and the segment functions $M_{i,o,k}^{\text{obs}}$, $M_{i,j,k}^{\text{uav}}$ are minima over the compact set $[0, 1]$ of functions jointly continuous in (λ, x) , hence continuous in x by the elementary form of Berge’s maximum theorem [35]. A finite intersection of closed sets is closed, so $\mathcal{F}_{0-D}^{\text{free}}(t_0)$ is closed and bounded, hence compact.

Attainment. F in (12) is a polynomial, hence continuous, and a continuous function on a nonempty compact set attains its minimum. $\square$

6.2 Convexity of the objective, and a complete strict-convexity criterion

Proposition 6.2 (Convexity). *F is convex on $\mathbb{R}^n$.*

Proof. Each term of (11) is a nonnegative multiple of the squared Euclidean norm of an affine function of x_{free} : $p_i^K - g_i$ is affine, and so is $p_i^{k+1} - p_i^k$ once p_i^0 is treated as a constant. The squared norm is convex and nondecreasing in the norm, the composition of a convex function with an affine map is convex, and a nonnegative combination of convex functions is convex. $\square$

Theorem 6.3 (Strict convexity criterion). *F is strictly convex in x_{free} if and only if*

$$\begin{cases} \alpha_i + \beta_i > 0 & \text{for every } i, \quad K = 1, \\ \beta_i > 0 & \text{for every } i, \quad K \geq 2. \end{cases} \quad (41)$$

Proof. F is a quadratic form plus affine terms and is separable across vehicles, so it is strictly convex if and only if each f_i is strictly convex in $x_{i,\text{free}}$, and that holds if and only if the second directional derivative is positive in every nonzero direction. Fix i and a perturbation $z_i = (z_i^1, \dots, z_i^K)$, adjoining $z_i^0 = 0$ because p_i^0 is fixed. Differentiating (11) twice along $\tau \mapsto x_i + \tau z_i$ gives

$$\frac{d^2}{d\tau^2} f_i(x_i + \tau z_i) = 2\alpha_i \|z_i^K\|^2 + 2\beta_i \sum_{k=0}^{K-1} \|z_i^{k+1} - z_i^k\|^2. \quad (42)$$

If $K = 1$ the sum has the single term $\|z_i^1 - z_i^0\|^2 = \|z_i^1\|^2$, so (42) equals $2(\alpha_i + \beta_i) \|z_i^1\|^2$. This is positive for every $z_i^1 \neq 0$ precisely when $\alpha_i + \beta_i > 0$, which proves both directions.

Let $K \geq 2$. If $\beta_i > 0$ and (42) vanishes, then every successive difference vanishes, so $z_i^1 = z_i^0 = 0$ and inductively $z_i^k = 0$ for all k ; hence the form is positive on every nonzero direction. Conversely, if $\beta_i = 0$ then (42) reduces to $2\alpha_i \|z_i^K\|^2$; choosing $z_i^1 \neq 0$ and $z_i^2 = \dots = z_i^K = 0$, which is possible because $K \geq 2$, gives a nonzero direction with zero curvature. So strict convexity forces $\beta_i > 0$. $\square$

The role of the fixed initial position is visible in the proof: it supplies the anchor $z_i^0 = 0$ that converts “all successive differences vanish” into “the perturbation vanishes”. Without it, the step penalty alone would be blind to a constant shift of the whole trajectory.

6.3 Uniqueness fails, and an explicit instance shows it

Strict convexity of the objective would give uniqueness if the feasible set were convex. It is not: the exterior of an obstacle ball and the set of separated configurations are nonconvex, and $\mathcal{F}_{0-D}(t_0)$ inherits this. The next theorem shows that the failure is real and not an artefact of a pathological instance.

Theorem 6.4 (No general uniqueness). *There exists an instance of (P) satisfying all conditions of Section 3, with F strictly convex, that has at least two distinct global minimizers.*

Proof. Take $q = 2$, $N = 2$, $K = 1$, and a single obstacle centred at the origin, $c_o = (0, 0)$. Write

$$d := R_o + r_1 + \delta_1^{\text{obs}} = \kappa_{1o}^{\text{obs}} > 0.$$

For vehicle 1 set $p_1^0 = (-2d, 0)$, $g_1 = (2d, 0)$ and $\alpha_1 = \beta_1 = 1$. For vehicle 2 set $p_2^0 = g_2 = (0, L)$ and $\alpha_2 = \beta_2 = 1$, with L chosen below. Choose a rectangular workspace containing $[-3d, 3d] \times [-d, L + d]$, and speed limits satisfying $v_1^{\text{max}} \Delta t \geq \sqrt{3}d$ and $v_2^{\text{max}} > 0$ arbitrary. Choose L with

$$L > R_o + r_2 + \delta_2^{\text{obs}}, \quad L > \frac{\sqrt{3}}{2}d + \kappa_{12}^{\text{uav}}. \quad (43)$$

Since $K = 1$ and $\alpha_i + \beta_i = 2 > 0$, Theorem 6.3 gives strict convexity of F .

The objective of vehicle 1 depends only on $\|p\|$. Writing $p := p_1^1$,

$$f_1(p) = \|p - (2d, 0)\|^2 + \|p - (-2d, 0)\|^2.$$

Expanding both squares, the cross terms $-2p^\top(2d, 0)$ and $-2p^\top(-2d, 0)$ cancel, leaving

$$f_1(p) = 2\|p\|^2 + 8d^2. \quad (44)$$

A lower bound. The obstacle constraint on segment 0 requires the minimum distance over the segment to be at least d ; since p is the endpoint of that segment, in particular $\|p\| \geq d$. With (44), $f_1(p) \geq 10d^2$ for every feasible p . Vehicle 2 has $g_2 = p_2^0$, so $f_2(p_2^1) = 2\|p_2^1 - (0, L)\|^2 \geq 0$ with equality exactly at $p_2^1 = (0, L)$. Hence $F \geq 10d^2$ on the feasible set.

Two feasible points attaining the bound. Take $p_2^1 = (0, L)$ and consider

$$p_A = (-d, 0), \quad p_B = \left(-\frac{d}{2}, \frac{\sqrt{3}}{2}d\right),$$

both of norm exactly d , so both give $F = 10d^2$. It remains to verify feasibility of each.

For p_A , the segment from $(-2d, 0)$ to $(-d, 0)$ has points of norm between d and $2d$, so obstacle clearance holds; its length is $d \leq \sqrt{3}d$, so the speed bound holds.

For p_B , parametrize the segment as $P(\lambda) = (-2d, 0) + \lambda(\frac{3}{2}d, \frac{\sqrt{3}}{2}d)$ for $\lambda \in [0, 1]$. Then

$$\|P(\lambda)\|^2 = d^2 \left[\left(-2 + \frac{3}{2}\lambda\right)^2 + \frac{3}{4}\lambda^2 \right] = d^2 [4 - 6\lambda + 3\lambda^2].$$

The bracket has derivative $-6 + 6\lambda$, which vanishes only at $\lambda = 1$; since the bracket is convex, its minimum over $[0, 1]$ is attained at $\lambda = 1$ and equals $4 - 6 + 3 = 1$. Hence $\min_{\lambda \in [0, 1]} \|P(\lambda)\| = d$ and obstacle clearance holds with equality at the endpoint. The segment length is $d\sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3}d$, so the speed bound holds by construction.

For pairwise separation, vehicle 2 stays at $(0, L)$ while vehicle 1 remains in the half-plane $y \leq \frac{\sqrt{3}}{2}d$ on either segment. Therefore the separation is at least $L - \frac{\sqrt{3}}{2}d > \kappa_{12}^{\text{uav}}$ by (43). Vehicle 2 clears the obstacle since $L > R_o + r_2 + \delta_2^{\text{obs}}$, and both points lie in the chosen workspace.

Both configurations are therefore feasible and attain the lower bound $10d^2$, so both are global minimizers, and they are distinct. $\square$

Remark 6.5. The midpoint of p_A and p_B has strictly smaller objective value, as strict convexity demands; it is simply infeasible, lying inside the obstacle clearance disc. This is exactly how

strict convexity and nonuniqueness coexist. Theorem 6.4 does not say that every instance has multiple minimizers: an instance whose feasible set happens to be convex, with a strictly convex objective, has exactly one. It says that uniqueness cannot be assumed in general, and Section 10.8 exhibits the same phenomenon numerically on the instance actually solved in this report.

6.4 Dependence on the problem data

Because uniqueness fails, the object whose stability can be discussed is the *set* of minimizers, not a single minimizer. Let θ collect the numerical data of the problem — goals, radii, margins, speed limits, objective weights, obstacle centres and radii — ranging over an admissible set Θ_{adm} on which the structural quantities N , K and q are held constant. Define

$$v(\theta) := \min_{x_{\text{free}} \in \mathcal{F}_{0-D}^{\text{free}}(t_0)(\theta)} F(x_{\text{free}}; \theta), \quad S(\theta) := \arg \min_{x_{\text{free}} \in \mathcal{F}_{0-D}^{\text{free}}(t_0)(\theta)} F(x_{\text{free}}; \theta). \quad (45)$$

By Theorem 6.1 both are well defined and $S(\theta)$ is nonempty and compact for every admissible θ .

Three statements can be made, and it is important that they are not confused with one another.

Proposition 6.6 (Outer stability). *Under convergence of the data in the Euclidean sense for a finite obstacle register, the feasible-set correspondence $\theta \mapsto \mathcal{F}_{0-D}^{\text{free}}(t_0)(\theta)$ has closed graph and is upper hemicontinuous, and the optimal value v is lower semicontinuous.*

This is the one-sided statement, and it is the one that holds without further hypotheses. It says that feasible points cannot appear discontinuously in the limit, and that the value cannot jump downwards. It does *not* say that the feasible set varies continuously: lower hemicontinuity is a separate property and can fail, because an arbitrarily small outward perturbation of an obstacle can delete a feasible point that was only just feasible.

Proposition 6.7 (Fixed-domain continuity). *If only the objective data (g, α, β) are perturbed, the feasible set is fixed and compact, so by the maximum theorem [35] the value v is continuous and S is upper hemicontinuous. If in addition the nominal minimizer is unique, every sequence of minimizers along data converging to the nominal data converges to it.*

Proposition 6.8 (What constraint perturbation requires). *When constraint data move, continuity of v requires a lower-stability property in addition. A sufficient condition is lower hemicontinuity of the feasible correspondence; a weaker sufficient condition is that one nominal global minimizer can be recovered under the perturbation. For a finite family of inequalities, strict positive slack at a nominal minimizer yields a positive minimum slack and hence one such recovery mechanism. Feasibility at equality is not robust: a minimizer lying exactly on a safety boundary can be destroyed by an arbitrarily small outward perturbation of that boundary.*

The standard references for these notions are [35, 36, 37]. Changing N , K , q , or adding and removing obstacles, changes the structure of the problem and is not treated as a small perturbation. Perturbing the workspace is treated separately: outer stability survives under

set convergence in the Hausdorff sense with local boundedness, but full value continuity again needs its own hypotheses.

The honest summary is therefore

$$\boxed{\text{Problem (P) is not globally well posed in the single-solution sense, but it possesses rigorous set-valued stability properties.}} \quad (46)$$

One consequence deserves emphasis because it is easy to misread as instability of the method: under nonuniqueness, a small change of the data may switch which of two equal-cost routes is selected. That is a property of the problem, not a defect of the solver, and it is exactly why $S(\theta)$ rather than a single-valued map is the right object.

A separate remark: perturbing the communication graph has *exactly zero* effect on v and S . Communication does not enter (P) at all; it enters only the reformulation of Section 8, which is proved there to leave the problem unchanged.

6.5 Regularity of the segment value functions

The two segment families are value functions of the parametric minimization (22), and first-order methods need to know how regular they are. Write

$$M(u, v) := \min_{\lambda \in [0,1]} \|(1 - \lambda)u + \lambda v - c\|^2 \quad (47)$$

for the generic squared value with endpoints u, v and reference point c , and put $s = \|v - u\|^2$.

Proposition 6.9 (Regularity and gradient). *The function M is continuous and locally Lipschitz on $\mathbb{R}^q \times \mathbb{R}^q$. On the region $s > 0$ it is continuously differentiable, including across the surfaces where the minimizing λ switches between the interior and an endpoint, and its gradients are*

$$\nabla_u M = 2(1 - \hat{\lambda})z, \quad \nabla_v M = 2\hat{\lambda}z, \quad z := (1 - \hat{\lambda})u + \hat{\lambda}v - c, \quad (48)$$

where $\hat{\lambda}$ is the constrained minimizer. It is in general not twice continuously differentiable across those surfaces. At $s = 0$ it is in general not differentiable, and its directional derivative at $u = v = p$ is

$$M'(p, p; h_0, h_1) = 2 \min\{(p - c)^\top h_0, (p - c)^\top h_1\}, \quad (49)$$

which is not linear in (h_0, h_1) .

Proof. Continuity follows as in Theorem 6.1. For $s > 0$ the minimizer $\hat{\lambda}$ is the clip to $[0, 1]$ of the smooth function λ^* of (24), and (48) is Danskin's formula applied to the smooth objective ψ , the minimizer being unique by strict convexity.

The only point requiring care is the switching surface, say $\lambda^* = 0$, i.e. $a^\top d = 0$ with $a = u - c$, $d = v - u$. On the interior side, the value is $\|a\|^2 - (a^\top d)^2/s$, whose gradient with respect to u is $2a - 2(a^\top d)d/s + \dots$, and at $a^\top d = 0$ every term carrying the factor $a^\top d$ vanishes, leaving $2a$. On the endpoint side the value is $\|a\|^2$, with gradient $2a$. The two one-sided gradients agree, so M is C^1 across the surface; the same computation at $\lambda^* = 1$ gives the symmetric conclusion. The second derivatives do not match, because the interior branch contributes a term in $dd^\top/s$ that the endpoint branch does not, so M is not C^2 there.

At $s = 0$, put $u = v = p$ and perturb by (h_0, h_1) . To first order the segment endpoints become $p + \tau h_0$ and $p + \tau h_1$, so $\|(1 - \lambda)(p + \tau h_0) + \lambda(p + \tau h_1) - c\|^2 = \|p - c\|^2 + 2\tau(p - c)^\top((1 - \lambda)h_0 + \lambda h_1) + O(\tau^2)$. Minimizing the linear term over $\lambda \in [0, 1]$ selects the smaller of the two endpoint values, which gives (49). Since a minimum of two linear functionals is concave and positively homogeneous but not linear, no gradient exists in general. $\square$

Corollary 6.10 (Unsquared distance). *The unsquared distance $D = \sqrt{M}$ is C^1 wherever $s > 0$ and $D > 0$. Positive required clearance guarantees $D > 0$ at feasible points, but not on arbitrary infeasible trial points, which is one reason the solver formulation (38) uses the squared form.*

Proposition 6.9 is the precise statement behind a claim that is often made loosely. The exact safety model is smooth enough for first-order methods on the nondegenerate region, and it is not smooth enough for methods requiring global C^2 regularity. Both halves matter, and Section 9 selects an algorithm family on exactly this basis.

6.6 Summary of the well-posedness study

The five preceding subsections answer five separate questions, and the answers are not all of the same kind: two are unconditional, one is an exact characterization, one is negative and settled by an explicit instance, and one holds only under a hypothesis that this problem does not supply in general. Table 6.1 states each answer together with what it costs to obtain, so that no result is carried forward under stronger conditions than it actually has.

Table 6.1 Each well-posedness question is answered unconditionally, by an exact criterion, or by an explicit counterexample, and every answer carries the hypotheses it actually needs.

Question	What is established	What it requires
Does a minimizer exist?	Yes, always. $\mathcal{F}_{0-D}(t_0)$ is nonempty and compact and (P) attains its minimum (Theorem 6.1). Nonemptiness is constructive: the constant trajectory is feasible, because the goal is soft.	Only admissibility of the input data, (13)–(15). No convexity, no qualification.
Is the objective convex?	Yes, always (Proposition 6.2). F is a sum of squared affine maps.	$\alpha_i, \beta_i \geq 0$. Nothing else.
Is it strictly convex?	Exactly when $\beta_i > 0$ for every i if $K \geq 2$, and exactly when $\alpha_i + \beta_i > 0$ for every i if $K = 1$ (Theorem 6.3). This is a characterization, not a sufficient condition.	Nothing beyond the weights. The criterion is checkable on the data before solving.
Is the minimizer unique?	No, and not by accident. An instance satisfying every condition of Section 3, with F strictly convex, carries two distinct global minimizers (Theorem 6.4). Strict convexity of the objective cannot rescue uniqueness once the feasible set is nonconvex.	Nothing: the instance is exhibited explicitly, so the negative answer is settled rather than conjectured.
Is the solution stable in the data?	Partly. The feasible correspondence has closed graph and is upper hemicontinuous and the value is lower semicontinuous (Proposition 6.6); with the feasible set held fixed the value is continuous and the solution set upper hemicontinuous (Proposition 6.7). Continuity of the value under <i>constraint</i> perturbation needs a lower-stability property in addition (Proposition 6.8).	Strict slack at a nominal minimizer suffices for the missing half. Feasibility at equality is not robust: a minimizer sitting exactly on a safety boundary is destroyed by an arbitrarily small outward perturbation.
How smooth are the safety functions?	M is locally Lipschitz everywhere, is C^1 on $\{s > 0\}$ including across the surfaces where the closest point switches between the interior and an endpoint, and is in general not C^2 there (Proposition 6.9). The unsquared distance $D = \sqrt{M}$ is C^1 only where $s > 0$ and $D > 0$ (Corollary 6.10).	Nondegeneracy $s > 0$ on the segment concerned. This is what forces the squared form on the solver and rules out any method needing global C^2 regularity.

Read as a chain, the five answers say this. A solution exists without assumptions. The objective is convex, and it is strictly convex under a condition that can be checked on the data. Strict convexity nevertheless fails to deliver uniqueness, because the feasible set is nonconvex, and this is proved rather than suspected. Consequently every later statement is written for a *set* of minimizers and not for a single one, and the stability results are set-valued for the same reason. Finally, the safety functions are smooth enough for first-order analysis on the nondegenerate region and no smoother, which is the property that selects the algorithm family in Section 9.

7 First-Order Analysis: Active Gradients, Constraint Qualifications and Optimality

This section develops the first-order theory of (P) in the reduced representation of Section 5.2. Its main outcome is negative and deliberate: no standard constraint qualification holds for this problem in general, and three explicit geometric mechanisms are exhibited that break them. The Karush–Kuhn–Tucker theorem is then stated in the conditional form those mechanisms force, so that the conditions under which first-order necessity may be invoked are visible rather than implicit.

7.1 The active system

Let $\varphi_\ell(x_{\text{free}}) \leq 0$ denote generically a retained inequality from (35)–(38), and at a feasible point x_{free}^* let

$$\mathcal{A}(x_{\text{free}}^*) := \{\ell : \varphi_\ell(x_{\text{free}}^*) = 0\} \quad (50)$$

be the active index set, of cardinality $m_{\mathcal{A}}$. Where the active constraints are differentiable, the active Jacobian is

$$J_{\mathcal{A}}(x_{\text{free}}^*) = \begin{pmatrix} \nabla \varphi_{\ell_1}^\top \\ \vdots \\ \nabla \varphi_{\ell_{m_{\mathcal{A}}}}^\top \end{pmatrix} \in \mathbb{R}^{m_{\mathcal{A}} \times n}. \quad (51)$$

By Proposition 6.9, differentiability of an active segment row requires $s > 0$ on that segment. A degenerate but *strictly inactive* row may be dropped from a locally equivalent problem, since by continuity it remains inactive in a neighbourhood. A degenerate row that is *active* falls outside the smooth theory entirely and must be treated by nonsmooth means; this case is excluded by hypothesis wherever the smooth theorem is invoked, and never silently assumed away.

7.2 Structural facts about the active gradients

The following facts are elementary but decisive, and each is proved because each is used.

Proposition 7.1 (Speed gradients are independent). *The gradients of the active speed constraints (36) are nonzero and linearly independent, across all vehicles and all segments.*

Proof. Independence across vehicles is immediate: the constraints of different vehicles involve disjoint blocks of variables. Fix a vehicle i and let $A \subseteq \{0, \dots, K-1\}$ be its active segments. Activity means $\|p_i^{k+1} - p_i^k\| = v_i^{\max} \Delta t > 0$, so $d_i^k \neq 0$ and the gradient is nonzero: with respect to p_i^{k+1} it is $2d_i^k$, and with respect to p_i^k it is $-2d_i^k$, where for $k=0$ only the first of these is a gradient with respect to a free variable.

Suppose $\sum_{k \in A} c_k \nabla g_{i,k}^{\text{spd}} = 0$. Let $k^* = \max A$. The variable $p_i^{k^*+1}$ appears in $g_{i,k}^{\text{spd}}$ only for $k = k^*$ and $k = k^* + 1$, and $k^* + 1 \notin A$ by maximality; hence the block of the identity corresponding to $p_i^{k^*+1}$ reads $2c_{k^*} d_i^{k^*} = 0$, giving $c_{k^*} = 0$. Removing k^* from A and repeating the argument downwards gives $c_k = 0$ for every $k \in A$. $\square$

Proposition 7.2 (Three mechanisms of constraint-qualification failure). *Each of the following can occur at an admissible feasible point.*

- (i) **Zero gradient on the first segment.** If an active safety row on segment $k = 0$ attains its closest point at $\hat{\lambda} = 0$, its gradient with respect to all free variables is zero.
- (ii) **Duplicate gradients on adjacent segments.** If two active safety rows on consecutive segments attain their closest points at the shared sample, their gradients coincide exactly.
- (iii) **Positive dependence with nonzero gradients.** Active rows can be positively linearly dependent even though each individual gradient is nonzero.

Proof. (i) By (48) the gradient with respect to the segment's second endpoint is $2\hat{\lambda}z$, which vanishes when $\hat{\lambda} = 0$; the first endpoint of segment 0 is p_i^0 , which is not a free variable. Hence every free-variable partial derivative of that row vanishes.

(ii) Consider segments k and $k + 1$ sharing the sample p_i^{k+1} , with the closest point of segment k at $\hat{\lambda} = 1$ and that of segment $k + 1$ at $\hat{\lambda} = 0$, both equal to p_i^{k+1} , so that the corresponding vectors z coincide. By (48), the gradient of the first row is $2z$ in the block p_i^{k+1} and zero in the block p_i^k ; that of the second is $2z$ in the block p_i^{k+1} and zero in the block p_i^{k+2} . The two gradients are therefore identical.

(iii) Place a vehicle midway between two obstacles positioned symmetrically about it, with both clearance constraints active. The two gradients are nonzero and point in opposite directions, so a positive combination of them vanishes. $\square$

Mechanism (i) is specific to this model and arises from the fixed initial position. Note that it is a consequence of a modelling decision that is otherwise beneficial, which is precisely the kind of interaction that a first-order analysis is supposed to surface.

7.3 Linear independence: not available

Definition 7.3 (LICQ). The linear independence constraint qualification holds at a feasible x_{free}^* with all active rows differentiable if $\text{rank } J_{\mathcal{A}}(x_{\text{free}}^*) = m_{\mathcal{A}}$.

Since (P) has no equality constraints, LICQ is exactly full row rank of the active Jacobian. A necessary counting condition is $m_{\mathcal{A}} \leq n = NqK$, which is not sufficient.

Corollary 7.4. *There is no general LICQ guarantee for problem (P): mechanisms (i) and (ii) of Proposition 7.2 each produce a rank-deficient active Jacobian.*

7.4 The Mangasarian–Fromovitz condition, and where it survives

Definition 7.5 (MFCQ). With inequality constraints only, the Mangasarian–Fromovitz constraint qualification holds at x_{free}^* if there exists $h \in \mathbb{R}^n$ with

$$J_{\mathcal{A}}(x_{\text{free}}^*)h < 0 \quad \text{componentwise,} \tag{52}$$

equivalently if the only $\mu \geq 0$ with $J_{\mathcal{A}}^\top \mu = 0$ is $\mu = 0$.

LICQ implies MFCQ, and the converse fails: duplicated nonzero gradients destroy linear independence while leaving a common strict descent direction. MFCQ is therefore the weaker and more useful condition here. It is nevertheless not universal for this problem.

Corollary 7.6. *MFCQ can fail at admissible feasible points of (P). Mechanism (i) of Proposition 7.2 destroys it immediately, since a zero active gradient admits $\mu \neq 0$ with $J_A^\top \mu = 0$; mechanism (iii) destroys it with all gradients nonzero.*

What does survive is a statement about the workspace subsystem alone, and it is worth isolating because it is the only unconditional positive result available.

Proposition 7.7 (Workspace-only MFCQ). *Let the representation satisfy (39). Then the workspace rows alone satisfy MFCQ at every point of $\mathcal{W}$.*

Proof. Since $\mathcal{W}$ is full-dimensional, choose $\bar{p} \in \text{int}(\mathcal{W})$. If the row $a_m^\top p \leq b_m$ is active at p , then $a_m^\top \bar{p} = b_m$ while $a_m^\top \bar{p} < b_m$, so $a_m^\top (\bar{p} - p) < 0$. The direction $h = \bar{p} - p$, applied in the corresponding sample block, therefore satisfies (52) for all active workspace rows simultaneously. $\square$

This is a statement about a subsystem and establishes neither full-model MFCQ nor LICQ. A Slater-type argument is not available for the full model, because the safety sets are nonconvex. The constant rank and constant positive linear dependence conditions remain available as alternatives but are not established here.

For a verified active set, MFCQ is checkable by a bounded linear program:

$$\max_{h, \tau} \tau \quad \text{subject to} \quad J_A h \leq -\tau \mathbf{1}, \quad \|h\|_\infty \leq 1, \quad (53)$$

whose optimal value satisfies $\tau^* > 0$ if and only if MFCQ holds. If the active set is empty, LICQ and MFCQ hold vacuously and the associated diagnostics are recorded as not applicable rather than replaced by artificial numbers.

7.5 First-order necessary conditions

Theorem 7.8 (Conditional first-order necessity). *Let x_{free}^* be a local minimizer of (P) in the reduced representation. Assume that every active segment row is differentiable at x_{free}^* , that is, the nondegeneracy condition $s > 0$ holds on each active segment, and assume MFCQ at x_{free}^* . Then there exist multipliers*

$$\mu_{i,m,k}^{\mathcal{W}} \geq 0, \quad \nu_{i,k}^{\text{spd}} \geq 0, \quad \xi_{i,o,k}^{\text{obs}} \geq 0, \quad \eta_{i,j,k}^{\text{uav}} \geq 0,$$

one for each retained row, such that primal feasibility, dual feasibility, complementarity and stationarity hold:

$$\varphi_\ell(x_{\text{free}}^*) \leq 0, \quad \mu_\ell^* \geq 0, \quad \mu_\ell^* \varphi_\ell(x_{\text{free}}^*) = 0, \quad (54)$$

$$\nabla F(x_{\text{free}}^*) + \sum_\ell \mu_\ell^* \nabla \varphi_\ell(x_{\text{free}}^*) = 0. \quad (55)$$

If LICQ also holds the multiplier vector is unique. Strict complementarity is not asserted. If the active set is empty, the conditions reduce to $\nabla F(x_{\text{free}}^*) = 0$.

Proof. Under the stated differentiability hypothesis the problem is, in a neighbourhood of x_{free}^* , a smooth inequality-constrained nonlinear program with the same local minimizers. MFCQ is then the classical qualification under which the Karush–Kuhn–Tucker conditions are necessary

at a local minimizer, and under which the set of multipliers is nonempty, convex and compact; see [38, 39]. Uniqueness under LICQ is the standard consequence of the active gradients being a basis. $\square$

The corresponding Lagrangian is

$$\begin{aligned} \mathcal{L} = & F(x_{\text{free}}) + \sum_{i=1}^N \sum_{k=1}^K \sum_{m=1}^{M_{\mathcal{W}}} \mu_{i,m,k}^{\mathcal{W}} g_{i,m,k}^{\mathcal{W}} + \sum_{i=1}^N \sum_{k=0}^{K-1} \nu_{i,k}^{\text{spd}} g_{i,k}^{\text{spd}} \\ & + \sum_{i=1}^N \sum_{o \in \mathcal{O}} \sum_{k=0}^{K-1} \xi_{i,o,k}^{\text{obs}} g_{i,o,k}^{\text{obs,seg}} + \sum_{1 \leq i < j \leq N} \sum_{k=0}^{K-1} \eta_{i,j,k}^{\text{uav}} h_{i,j,k}^{\text{inter,coll,sq}}. \end{aligned} \quad (56)$$

There is one multiplier per unordered pair and segment, not two.

Three cautions accompany Theorem 7.8, and they are the reason it is stated conditionally. First, the initial admissibility conditions (13)–(15) carry no multipliers and appear nowhere in (56). Second, a degenerate row may be omitted from the smooth stationarity system only when it is strictly inactive, and then its multiplier is zero and its physical feasibility is still evaluated; the product of a zero multiplier with an undefined gradient is not a valid term and is never written. Third, and most important, (55) is a *necessary* condition at a local minimizer. Since $\mathcal{F}_{0-D}(t_0)$ is nonconvex, it is not sufficient, and a point satisfying it is not thereby certified optimal — locally or globally.

7.6 Summary of the first-order analysis

The outcome of this section is mostly negative, and the negatives are the useful part: they are what stop a later claim from resting on a qualification that this problem does not have. Table 7.1 collects the six findings and what each one forces on the rest of the report.

Table 7.1 The first-order analysis locates its own failure modes: three geometric mechanisms break the standard qualifications, one family of rows survives, and necessity is therefore stated conditionally.

Question	What is established	What it forces
Which active gradients are well behaved?	The active speed gradients are nonzero and linearly independent, across all vehicles and all segments (Proposition 7.1).	Speed rows are never the cause of a qualification failure. Every failure below originates in a safety row.
Can the qualifications fail?	Yes, by three distinct geometric mechanisms, each of which occurs at an admissible feasible point (Proposition 7.2): a safety row on the first segment whose closest point sits at $\hat{\lambda} = 0$ has zero gradient; two active rows on consecutive segments whose closest points both sit at the shared sample have identical gradients; and active rows can be positively linearly dependent with every individual gradient nonzero.	The failures are structural, not pathological. They arise from the segment geometry itself, so no reparameterization of the same model removes them.

Question	What is established	What it forces
Does LICQ hold?	No. There is no general linear-independence guarantee; mechanisms (i) and (ii) each produce a rank-deficient active Jacobian (Corollary 7.4).	Multiplier uniqueness cannot be assumed anywhere in this report.
Does MFCQ hold?	Not in general. Mechanism (i) destroys it immediately, because a zero active gradient admits a nonzero μ with $J_{\mathcal{A}}^{\top}\mu = 0$; mechanism (iii) destroys it with all gradients nonzero (Corollary 7.6).	The Karush–Kuhn–Tucker theorem cannot be invoked unconditionally, and MFCQ must be carried as an explicit hypothesis wherever it is used.
Does anything survive?	Yes. Under the validity condition on the halfspace representation, the workspace rows alone satisfy MFCQ at every point of $\mathcal{W}$ (Proposition 7.7).	The workspace description can be validated once, offline, and then relied upon; the difficulty is isolated in the safety rows.
Are the first-order conditions necessary?	Conditionally. At a local minimizer at which every active segment row is differentiable ($s > 0$ on each active segment) and MFCQ holds, multipliers exist satisfying primal feasibility, dual feasibility, complementarity and stationarity (Theorem 7.8). Under LICQ the multiplier vector is unique. Strict complementarity is not asserted.	Two hypotheses must be checked at the point in question, not assumed. The initial admissibility conditions carry no multipliers and appear nowhere in the Lagrangian (56).
Are they sufficient?	No. $\mathcal{F}_{0-D}(t_0)$ is nonconvex, so a point satisfying (55) is not thereby certified optimal, locally or globally.	Every later numerical statement is about stationarity, feasibility and margins — never about optimality.

The practical consequence is a rule followed for the remainder of the report. Wherever a first-order argument is used, the two hypotheses of Theorem 7.8 — nondegeneracy of the active segment rows and MFCQ at the point concerned — are stated at that place, and a computed point that satisfies the stationarity system is described as stationary, never as optimal.

8 Trajectory-Copy Consensus

Problem (P) is written as a single optimization over all vehicles, and its only inter-vehicle coupling is the pairwise separation family (38). To solve it without a central coordinator, that coupling must be represented so that each vehicle can evaluate the constraints it is responsible for using information it actually holds. This section constructs such a representation and proves that it changes nothing about the problem.

8.1 Why agreement cannot be imposed on physical trajectories

The obvious move is to give every vehicle a copy of the whole decision vector and require the copies to agree, which is the standard consensus reformulation of distributed optimization. Applied naively here it is not merely inefficient; it is wrong.

Proposition 8.1 (Physical consensus contradicts separation). *Suppose $N \geq 2$ and the agreement constraint $x_i = x_j$ is imposed on the physical trajectories of two distinct vehicles $i \neq j$ for which $\kappa_{ij}^{\text{uav}} > 0$. Then the resulting problem is infeasible.*

Proof. If $x_i = x_j$ then $p_i^k = p_j^k$ for every k , so on any segment $a = b = 0$ in (27) and the minimum separation (28) equals 0. The requirement $D_{i,j,k} \geq \kappa_{ij}^{\text{uav}} > 0$ therefore fails. $\square$

What distributed computation actually needs is not that vehicles fly the same path, but that a vehicle evaluating a shared constraint has consistent access to the trajectory information that constraint depends on. This suggests introducing *copies of a trajectory* as computational objects, and requiring copies of the *same* trajectory to agree, while the physical trajectories of distinct vehicles remain distinct variables. That is Trajectory-Copy Consensus.

8.2 Alternatives considered

Two other routes were examined before the copy architecture was adopted, and both are recorded because the reasons for not taking them are part of the justification.

Full replication. Every vehicle could carry a replica of the entire swarm trajectory and synchronize all replicas. This is correct and conceptually simple, but each vehicle then stores and exchanges $\Theta(NqK)$ scalars regardless of how few constraints it is responsible for. It was not adopted because the replication is largely unnecessary: most of what a vehicle would store plays no role in any constraint it evaluates. No claim of measured performance advantage is made for the alternative adopted; the argument is structural.

Edge-relative variables. One could optimize relative quantities such as $x_{i,\text{free}} - x_{j,\text{free}}$ on communication edges. Pairwise geometry is then compact, but absolute quantities — goal, workspace, obstacle clearance — require an anchor or a reconstruction; a safety pair that is not a communication edge needs relative information composed along a path; and independently stored edge differences must satisfy consistency conditions around every cycle. This direction is not impossible and is not dismissed; it introduces machinery that the deterministic baseline does not need.

A third route, an exact complementarity reformulation of the closest-point minimization that keeps $\hat{\lambda}$ as a variable, was considered and set aside: it converts the problem into a mathematical program with complementarity constraints, whose constraint qualifications fail systematically and whose solver requirements are heavier than those studied here.

8.3 Architecture

Let $G_0^{\text{comm}} = (V, E_0^{\text{comm}})$ be the communication graph available at t_0 , assumed fixed and connected; it may be sparse. Put

$$Q := qK, \tag{57}$$

the dimension of one free source trajectory. For a source vehicle i , a vehicle a may hold a copy

$$\tilde{x}_{a \leftarrow i} \in \mathbb{R}^Q, \quad \text{with the anchor} \quad \tilde{x}_{i \leftarrow i} := x_{i, \text{free}}. \quad (58)$$

The anchor says that a vehicle's copy of *its own* trajectory is that trajectory. The fixed initial position is not a copied quantity; the full trajectory represented by a copy is $(p_i^0, \tilde{x}_{a \leftarrow i})$.

Let $V_i^{\text{copy}} \ni i$ be the set of holders of source i , with $n_i := |V_i^{\text{copy}}|$, and let G_i^{copy} be a connected subgraph of G_0^{comm} on those holders. Select a spanning tree $T_i^{\text{eq}} = (V_i^{\text{copy}}, E_i^{\text{eq}})$ and impose agreement on its edges only:

$$\tilde{x}_{a \leftarrow i} - \tilde{x}_{b \leftarrow i} = 0, \quad \{a, b\} \in E_i^{\text{eq}}. \quad (59)$$

Theorem 8.2 (Multi-hop agreement). *Under the anchor (58) and the tree constraints (59), every holder carries exactly the source trajectory: $\tilde{x}_{a \leftarrow i} = x_{i, \text{free}}$ for all $a \in V_i^{\text{copy}}$.*

Proof. A tree contains a unique path $i = v_0, v_1, \dots, v_p = a$ from the source to any holder a . Each edge of that path carries a constraint from (59), so $\tilde{x}_{v_0 \leftarrow i} = \tilde{x}_{v_1 \leftarrow i} = \dots = \tilde{x}_{a \leftarrow i}$, and the anchor gives $\tilde{x}_{v_0 \leftarrow i} = x_{i, \text{free}}$. $\square$

Theorem 8.2 is what makes sparse communication acceptable: a holder need not talk to the source directly. It also explains why a tree suffices — additional edges would impose constraints implied by those already present.

8.4 Assigning responsibility for pairwise constraints

Each pairwise constraint must be evaluated by exactly one vehicle, or it will be duplicated. Assign to every pair one of its two endpoints,

$$\gamma : E^{\text{all}} \rightarrow V, \quad \gamma(\{i, j\}) \in \{i, j\}, \quad (60)$$

and call $\gamma(\{i, j\})$ the guardian of that pair. The guardian role is computational responsibility only: it confers no authority over either trajectory and does not make the guardian a leader. If $\gamma(\{i, j\}) = j$ then j must hold a copy of source i . The required holders of source i are therefore

$$V_i^{\text{req}} = \{i\} \cup \{j \neq i : \gamma(\{i, j\}) = j\}, \quad (61)$$

and any additional holders, present only to keep G_i^{copy} connected, are relays: $V_i^{\text{relay}} = V_i^{\text{copy}} \setminus V_i^{\text{req}}$, with $n_i^{\text{relay}} = |V_i^{\text{relay}}|$. Write L_a^{guard} for the number of pairs guarded by a , L_a^{relay} for the number of relay copies it stores, and $I_a^{\text{copy}} = \{i : a \in V_i^{\text{copy}}\}$ for the sources it represents.

The following counting results characterize the cost of the architecture exactly.

Theorem 8.3 (Required foreign-copy count). $\sum_{i \in V} (|V_i^{\text{req}}| - 1) = \binom{N}{2}$.

Proof. By (61), $|V_i^{\text{req}}| - 1 = |\{j \neq i : \gamma(\{i, j\}) = j\}|$, so the sum counts the pairs $\{i, j\}$ once for each i that is *not* the guardian. Since γ assigns exactly one of the two endpoints to each pair, each pair contributes exactly 1, and there are $\binom{N}{2}$ pairs. $\square$

Theorem 8.4 (Optimal guardian load). $\min_{\gamma} \max_{a \in V} L_a^{\text{guard}} = \lceil \frac{N-1}{2} \rceil$.

Proof. Since $\sum_a L_a^{\text{guard}} = \binom{N}{2} = \frac{N(N-1)}{2}$ for any γ , the average load is $(N-1)/2$, so $\max_a L_a^{\text{guard}} \geq \lceil (N-1)/2 \rceil$ for every γ . For achievability, identify γ with an orientation of the complete graph K_N , each pair oriented towards its guardian; then L_a^{guard} is the in-degree of a . For N odd, the circulant orientation joining a to $a+1, \dots, a + \frac{N-1}{2}$ modulo N gives every vertex in-degree exactly $(N-1)/2$. For N even, a near-regular tournament has in-degrees in $\{N/2-1, N/2\}$, whose maximum is $N/2 = \lceil (N-1)/2 \rceil$. $\square$

Proposition 8.5 (Storage). *For every a , $|I_a^{\text{copy}}| = 1 + L_a^{\text{guard}} + L_a^{\text{relay}}$, and*

$$\frac{1}{N} \sum_{a \in V} |I_a^{\text{copy}}| = \frac{N+1}{2} + \frac{1}{N} \sum_{i \in V} n_i^{\text{relay}}. \quad (62)$$

With q and K fixed, total foreign-copy storage scales as $\Theta(N^2 q K)$.

Proof. A vehicle represents its own source, one source per pair it guards, and one per relay copy, which gives the first identity. Summing over a and using $\sum_a L_a^{\text{guard}} = \binom{N}{2}$ and $\sum_a L_a^{\text{relay}} = \sum_i n_i^{\text{relay}}$ gives $\sum_a |I_a^{\text{copy}}| = N + \frac{N(N-1)}{2} + \sum_i n_i^{\text{relay}}$; dividing by N yields (62). The scaling follows from Theorem 8.3: $\binom{N}{2}$ foreign copies of $Q = qK$ scalars are required. $\square$

The quadratic term in Proposition 8.5 deserves a comment, because it is sometimes mistaken for a defect of the architecture. It is inherited from the modelling decision that separation is imposed on *all* pairs, made in (5). Any exact representation of an all-pairs requirement carries it. Removing it would require a certified screening argument proving that certain pairs can never come into conflict, which is a modelling change and not an implementation optimization. Two structural facts complete the picture of when the architecture is realizable.

Theorem 8.6 (Connectivity is exactly what is needed). *All guardian copy demands are realizable for every source, with relays permitted, if and only if G_0^{comm} is connected.*

Proof. ($\Leftarrow$) If G_0^{comm} is connected then for each i there is a connected subgraph containing V_i^{req} , namely any spanning connected subgraph of the whole graph restricted appropriately; take G_i^{copy} to be such a subgraph.

($\Rightarrow$) Suppose G_0^{comm} is disconnected, and pick i and j in different components. The pair $\{i, j\}$ has a guardian, say j , so j must hold a copy of source i , and by (59) the holders of source i must form a connected subgraph of G_0^{comm} containing both i and j . No such subgraph exists. $\square$

Proposition 8.7 (Agreement Jacobian). *For each source i , the agreement system (59) has Jacobian equal to the incidence matrix of T_i^{eq} tensored with I_Q , hence of full row rank $(n_i-1)Q$. Stacking over sources preserves full row rank.*

Proof. The incidence matrix of a tree on n_i nodes has rank $n_i - 1$, and the Kronecker product with I_Q multiplies the rank by Q . Different sources involve disjoint sets of copy variables, so the stacked system is block diagonal after reordering. $\square$

Proposition 8.7 says the agreement constraints contain no redundancy. It says nothing about the full problem: full row rank of the agreement block does not imply LICQ for the complete nonlinear program, since the nonlinear rows are not accounted for.

8.5 The reformulated problem

For vehicle a , collect into a single block the trajectory it owns and the copies it holds,

$$z_a := \text{col}\left(x_{a,\text{free}}, (\tilde{x}_{a \leftarrow i})_{i \in I_a^{\text{copy}} \setminus \{a\}}\right) \in \mathbb{R}^{d_a}, \quad d_a = Q |I_a^{\text{copy}}|, \quad (63)$$

and define the selector $S_{a \leftarrow i}$ extracting the source- i part,

$$S_{a \leftarrow i} z_a = \begin{cases} x_{a,\text{free}}, & i = a, \\ \tilde{x}_{a \leftarrow i}, & i \neq a, \end{cases} \quad \chi_{a \leftarrow i}^{\text{full}}(z_a) := (p_i^0, S_{a \leftarrow i} z_a). \quad (64)$$

Write $Z = \text{col}(z_1, \dots, z_N)$. The reformulated problem is

$$\begin{array}{ll} \min_{(z_a)_{a \in V}} & \sum_{a \in V} f_a(\chi_{a \leftarrow a}^{\text{full}}(z_a)) \\ \text{s.t.} & g_{a,m,k}^{\mathcal{W}}(\chi_{a \leftarrow a}^{\text{full}}(z_a)) \leq 0, \quad a \in V, m = 1, \dots, M_{\mathcal{W}}, k = 1, \dots, K, \\ & g_{a,k}^{\text{spd}}(\chi_{a \leftarrow a}^{\text{full}}(z_a)) \leq 0, \quad a \in V, k = 0, \dots, K-1, \\ & g_{a,o,k}^{\text{obs,seg}}(\chi_{a \leftarrow a}^{\text{full}}(z_a)) \leq 0, \quad a \in V, o \in \mathcal{O}, k = 0, \dots, K-1, \\ & h_{i,j,k}^{\text{inter,coll,sq}}(\chi_{a \leftarrow i}^{\text{full}}(z_a), \chi_{a \leftarrow j}^{\text{full}}(z_a)) \leq 0, \quad \{i, j\} \in E^{\text{all}}, a = \gamma(\{i, j\}), k = 0, \dots, K-1, \\ & S_{a \leftarrow i} z_a - S_{b \leftarrow i} z_b = 0, \quad i \in V, \{a, b\} \in E_i^{\text{eq}}. \end{array} \quad (\text{P}')$$

Each source objective appears exactly once, attached to its own vehicle. Each pairwise constraint appears exactly once, attached to its guardian, and is evaluated entirely from data that the guardian holds. Every nonlinear row is local to one block. The only rows linking different blocks are the linear agreement constraints of the last line. This is precisely the structure a splitting method can exploit.

Denote by $\mathcal{F}_{\text{copy}}(t_0)$ the feasible set of (P'). The conditions under which the reformulation is exact are stated as assumptions and then, where they are consequences rather than hypotheses, proved.

Assumption 8.8 (Architecture conditions).

- E1.** (P) and (P') use the same data and the same objective.
- E2.** The anchor (58) holds.
- E3.** Each G_i^{copy} is connected.
- E4.** T_i^{eq} is a spanning tree of G_i^{copy} , and agreement is imposed on all of its edges.
- E5.** Each pair has exactly one guardian.
- E6.** $V_i^{\text{req}} \subseteq V_i^{\text{copy}}$ for every i .
- E7.** Every pair of E^{all} is covered, for every segment.

Theorem 8.9 (Exactness of the reformulation). *Under Assumption 8.8,*

$$\text{proj}_{x_{\text{free}}} \mathcal{F}_{\text{copy}}(t_0) = \mathcal{F}_{0\text{-D}}^{\text{free}}(t_0), \quad (65)$$

the optimal values coincide, $J_{\text{copy}}^ = F_{0\text{-D}}^*$, and the projection maps the global minimizers of (P') exactly onto those of (P).*

Proof. Projection is feasible. Let $Z \in \mathcal{F}_{\text{copy}}(t_0)$ and let x_{free} be its projection, i.e. $x_{a,\text{free}} = S_{a \leftarrow a} z_a$. The workspace, speed and obstacle rows of (P') are exactly those of (P) for each vehicle. For a pair $\{i, j\}$ with guardian a , the constraint imposed in (P') is evaluated at $\chi_{a \leftarrow i}^{\text{full}}(z_a)$ and $\chi_{a \leftarrow j}^{\text{full}}(z_a)$. By Theorem 8.2, applicable because of E2–E4, each of these equals the corresponding source trajectory. Hence the guardian’s inequality is literally $h_{i,j,k}^{\text{inter,coll,sq}}(x_i, x_j) \leq 0$. By E5 and E7 every pair and segment is covered exactly once, so all constraints of (33) hold and $x_{\text{free}} \in \mathcal{F}_{0-D}^{\text{free}}(t_0)$.

Every feasible point has a feasible representation. Let $x_{\text{free}} \in \mathcal{F}_{0-D}^{\text{free}}(t_0)$ and define $\tilde{x}_{a \leftarrow i} := x_{i,\text{free}}$ for every maintained copy. Then every agreement constraint holds with both sides equal; the owner-local rows hold because x_{free} is feasible for (P) ; and each guardian row reduces, as above, to the corresponding physical constraint, which holds. Hence $Z \in \mathcal{F}_{\text{copy}}(t_0)$ and projects to x_{free} .

Objective. By E1 and (64), the objective of (P') at any Z equals $\sum_a f_a(x_a)$ evaluated at the projection, since the selector applied to a vehicle’s own source returns that source and no copy contributes an objective term. The objective is therefore constant on the fibres of the projection and equal to F .

Values and minimizers. The two displayed inclusions show that the projection maps $\mathcal{F}_{\text{copy}}(t_0)$ onto $\mathcal{F}_{0-D}^{\text{free}}(t_0)$ and that every point of $\mathcal{F}_{0-D}^{\text{free}}(t_0)$ has a preimage. Combined with pointwise equality of the objectives, minimizing over $\mathcal{F}_{\text{copy}}(t_0)$ and over $\mathcal{F}_{0-D}^{\text{free}}(t_0)$ gives the same value, and Z^* is a global minimizer of (P') if and only if its projection is one of (P) . No uniqueness is needed anywhere in this argument. $\square$

Proposition 8.10 (The representation is unique). *For a fixed x_{free} and a fixed architecture, the values of all maintained copies are uniquely determined: $\tilde{x}_{a \leftarrow i} = x_{i,\text{free}}$.*

Proof. Immediate from Theorem 8.2, which used only E2–E4. $\square$

Remark 8.11 (What the reformulation does not do). Theorem 8.9 is a statement about sets and values. It does not convexify anything: $\mathcal{F}_{\text{copy}}(t_0)$ is nonconvex whenever $\mathcal{F}_{0-D}^{\text{free}}(t_0)$ is. It says nothing about what happens when agreement holds only approximately, which is the situation in any finite computation; that question is addressed quantitatively in Section 10.6, and the answer is not automatic. And it is a statement about a static reformulation, not about an algorithm: convergence, stale information, packet loss and intermediate-iterate safety are all outside its scope.

8.6 Choosing the copy topology

Two design questions remain, and both have exact answers.

Proposition 8.12 (Relay-free criterion). *A relay-free construction for source i , i.e. one with $V_i^{\text{copy}} = V_i^{\text{req}}$, exists if and only if the subgraph of G_0^{comm} induced by V_i^{req} is connected.*

Proof. E3 requires G_i^{copy} to be connected and E6 requires it to contain V_i^{req} . If no relays are added, the holder set is exactly V_i^{req} , and a connected subgraph on it exists precisely when the induced subgraph is connected. $\square$

When the criterion fails, relays are necessary, and the vehicles that must serve as relays are determined by the graph: an articulation vertex separating the source from a required guardian lies on every path between them and therefore on every admissible connected copy topology. Choosing a minimum-cost copy topology for positive additive edge costs is the graph Steiner tree problem with terminals V_i^{req} ; minimality is an optimization of the architecture, not a requirement for exactness, and is not pursued here.

For the agreement tree itself, a source-rooted shortest-path spanning tree is used. The reason is a bound that will matter in Section 10.6: if every tree edge has agreement residual at most τ_{eq} , then by summing along the unique path from the source,

$$\|\tilde{x}_{a \leftarrow i} - x_{i,\text{free}}\| \leq h_{i,a} \tau_{\text{eq}}, \quad (66)$$

where $h_{i,a}$ is the source-to-holder hop count. A shortest-path tree minimizes $h_{i,a}$ for every holder simultaneously, hence minimizes this bound. It does not minimize realized error, latency or congestion, and no such claim is made. Finally, every spanning tree on n_i holders has the same total incidence $\sum_a \deg(a) = 2(n_i - 1)$, so the choice of tree redistributes agreement work between vehicles but cannot reduce its total.

9 The Distributed Solver

Section 8 produced a problem with the structure a splitting method needs: nonconvex but *local* inequality constraints, and *linear* constraints linking blocks. This section selects an algorithm family matched to that structure, maps (P') onto it, and derives each update step. The assumptions the mapping requires are stated explicitly, and Section 10 reports which of them were verified for the instance actually computed.

9.1 Selecting an algorithm family

Three families were examined against four criteria: whether the reformulated problem can be expressed in the family’s problem class without modification; whether the family’s convergence hypotheses are plausible here; whether the whole algorithm, including its coordination step, is decentralized; and what a small residual actually certifies.

Augmented Lagrangian based alternating direction inexact Newton (ALADIN) methods, introduced by Houska, Frasch and Diehl [40], solve local nonlinear programs, build sensitivity information and solve a coordination quadratic program. The local theory is strong but it requires second-order regularity, and by Proposition 6.9 the exact safety functions are not globally C^2 ; the coordination step is also a coupled central computation. Retained as a reference, not adopted.

The *bi-level distributed* variant of ALADIN, due to Engelmann, Jiang, Houska and Faulwasser [41], decomposes that coordination step and admits an inner decentralized splitting method. Its pairwise structure matches the agreement architecture well. However, relay directions can lie in the active-constraint nullspace with zero exact local Lagrangian curvature, which obstructs the positive-curvature hypothesis of the stronger local result. Removing relays removes that one obstruction without establishing the remaining hypotheses. Retained as a fallback.

The family adopted is the *two-level method* of Sun and Sun [42], which embeds a specially structured three-block alternating direction method inside an augmented Lagrangian framework. Its problem class is exactly

$$\min_{Z \in X, U \in \bar{X}} F(Z) \quad \text{subject to} \quad AZ + BU = 0, \quad (67)$$

with X compact and possibly nonconvex, $\bar{X}$ compact and convex, and F continuously differentiable. The decisive property is that the complicated geometry is carried by the *set* X rather than by smooth constraint functions, so no global C^2 requirement is imposed on the safety model. The price is a different set of hypotheses, stated in Section 9.9, which are made explicit rather than assumed.

9.2 Compactification and the split

Two preparatory steps bring (P') into the form (67).

Compactification. The class requires a compact local set. A vehicle's own trajectory is already confined to $\mathcal{W}^K$ by its workspace rows, but a foreign copy carries no such row. Choose a compact convex box $\mathcal{B}$ with $\mathcal{W} \subset \text{int}(\mathcal{B})$ and put $\mathcal{B}_K := \mathcal{B}^K$. Define X_a as the set of blocks z_a satisfying: the workspace, speed and obstacle rows of source a ; every pairwise row guarded by a ; and, for each foreign source $i \in I_a^{\text{copy}} \setminus \{a\}$, the bound $S_{a \leftarrow i} z_a \in \mathcal{B}_K$. Set $X := \prod_a X_a$; each X_a is compact and in general nonconvex. The source block receives no duplicate box.

Lemma 9.1 (The added bounds exclude nothing). *For every $x_{\text{free}} \in \mathcal{F}_{0-D}^{\text{free}}(t_0)$, the representation of Proposition 8.10 satisfies the added bounds. Consequently the compactification changes neither $\text{proj}_{x_{\text{free}}}$ of the feasible set, nor the optimal value, nor the set of global minimizers.*

Proof. Feasibility of x_{free} places every free sample of every source in $\mathcal{W}$, so $x_{i,\text{free}} \in \mathcal{W}^K$. By Proposition 8.10 each foreign copy equals its source, so it lies in $\mathcal{W}^K \subset \text{int}(\mathcal{B}_K)$ and satisfies the bound strictly. The conclusion then follows from Theorem 8.9, whose second half constructs exactly this representation. $\square$

Splitting the agreement constraints. For each agreement edge $e = (a, b; i)$ introduce an auxiliary variable $u_e \in \mathbb{R}^Q$ and replace (59) by the pair

$$S_{a \leftarrow i} z_a - u_e = 0, \quad S_{b \leftarrow i} z_b - u_e = 0. \quad (68)$$

Eliminating u_e recovers the original constraint exactly, so nothing is lost. Stacking $U = \text{col}(u_e)$ over all $M_{\text{eq}} = \sum_i (n_i - 1)$ edges and writing the two rows of (68) for every edge gives a linear system

$$AZ + BU = 0, \quad B_e = \begin{bmatrix} -I_Q \\ -I_Q \end{bmatrix}, \quad B^\top B = 2I, \quad (69)$$

with $m_{\text{split}} = 2QM_{\text{eq}}$ scalar rows. Finally set $\bar{X} := \mathcal{B}_K^{M_{\text{eq}}}$, which is compact and convex; at agreement the auxiliary equals the common source trajectory, which lies in the interior of $\mathcal{B}_K$ by Lemma 9.1, so the split is exact and reversible.

The identity $B^\top B = 2I$ is not cosmetic. It is what makes the auxiliary update below a plain average, and it is one of the reasons this family fits the architecture.

9.3 Outer and inner levels

Introduce a slack $\xi \in \mathbb{R}^{m_{\text{split}}}$ and write the agreement system as $AZ + BU + \xi = 0$ with $\xi = 0$ required. The outer level drives ξ to zero by an augmented Lagrangian method; the inner level solves, for fixed outer parameters, the problem

$$\min_{Z \in X, U \in \bar{X}, \xi} F(Z) + \langle \Lambda^r, \xi \rangle + \frac{\beta^r}{2} \|\xi\|^2 \quad \text{s.t.} \quad AZ + BU + \xi = 0, \quad (70)$$

where Λ^r is the outer multiplier and $\beta^r > 0$ the outer penalty at outer iteration r . Problem (70) has three blocks, Z , U and ξ , and is solved by an alternating direction method with penalty ρ^r and multiplier y . The penalty is not free:

$$\rho^r = 2\beta^r. \quad (71)$$

This relation is required by the convergence analysis of the source method, not chosen for convenience, and it is monitored numerically in Section 10.

The augmented Lagrangian of the inner problem is

$$\mathcal{L}_{\rho^r}(Z, U, \xi, y) = F(Z) + \langle \Lambda^r, \xi \rangle + \frac{\beta^r}{2} \|\xi\|^2 + \langle y, AZ + BU + \xi \rangle + \frac{\rho^r}{2} \|AZ + BU + \xi\|^2, \quad (72)$$

understood as $+\infty$ outside $X \times \bar{X}$.

9.4 The local step, and why it is separable

Minimizing (72) over $Z \in X$ with U, ξ, y fixed at their previous values gives

$$\min_{Z \in X} F(Z) + \langle y^{t-1}, AZ + BU^{t-1} + \xi^{t-1} \rangle + \frac{\rho^r}{2} \|AZ + BU^{t-1} + \xi^{t-1}\|^2. \quad (73)$$

The crucial structural point is that this separates across vehicles. Each row of $AZ + BU + \xi$ is indexed by a half-edge (e, a) and, by (68), reads

$$q_{e,a}^{r,t}(z_a) := S_{a \leftarrow i} z_a - u_e^{r,t-1} + \xi_{e,a}^{r,t-1}, \quad (74)$$

which involves the single block z_a . Since $F(Z) = \sum_a f_a$ is also separable and $X = \prod_a X_a$, problem (73) decomposes into N independent problems, one per vehicle:

$$\min_{z_a \in X_a} \phi_a^{r,t}(z_a), \quad \phi_a^{r,t}(z_a) = f_a(\chi_{a \leftarrow a}^{\text{full}}(z_a)) + \sum_{(e,a) \in \mathcal{H}_a} \left[\langle y_{e,a}^{r,t-1}, q_{e,a}^{r,t}(z_a) \rangle + \frac{\rho^r}{2} \|q_{e,a}^{r,t}(z_a)\|^2 \right], \quad (75)$$

where $\mathcal{H}_a$ is the set of half-edges incident to a . Each vehicle solves (75) using only its own block, its own incident multipliers and the auxiliary values of its neighbours. This is where decentralization is realized.

The method requires from this step a *stationary point* of the local problem together with a decrease of the local objective, not a global minimizer of a nonconvex program — a distinction that matters, since (75) is nonconvex whenever X_a carries pairwise or obstacle rows. The implementation additionally enforces the sufficient decrease condition

$$\phi_a^{r,t}(z_a^{r,t}) \leq \phi_a^{r,t}(z_a^{r,t-1}) \quad \text{for every } a. \quad (76)$$

9.5 The auxiliary step is an average and a projection

Proposition 9.2 (Edge update). *Minimizing (72) over $U \in \bar{X}$ with $Z = Z^{r,t}$ and $\xi = \xi^{r,t-1}$, $y = y^{r,t-1}$ fixed decouples across edges, and for the edge $e = (a, b; i)$ the solution is*

$$u_e^{r,t} = \text{Proj}_{\mathcal{B}_K} \left(\frac{\psi_{e,a}^{r,t} + \psi_{e,b}^{r,t}}{2} \right), \quad \psi_{e,c}^{r,t} := w_{e,c}^{r,t} + \xi_{e,c}^{r,t-1} + \frac{1}{\rho^r} y_{e,c}^{r,t-1}, \quad (77)$$

where $w_{e,c}^{r,t} = S_{c \leftarrow i} z_c^{r,t}$ is the value selected at endpoint c .

Proof. Only the two half-edge rows of e involve u_e , and they involve no other auxiliary, so the objective decouples across edges. Dropping terms independent of u_e , the edge subproblem is

$$\min_{u_e \in \mathcal{B}_K} \sum_{c \in \{a,b\}} \left[\langle y_{e,c}, w_{e,c} - u_e + \xi_{e,c} \rangle + \frac{\rho^r}{2} \|w_{e,c} - u_e + \xi_{e,c}\|^2 \right].$$

Ignoring the constraint for a moment, the gradient with respect to u_e is

$$-\sum_c y_{e,c} - \rho^r \sum_c (w_{e,c} - u_e + \xi_{e,c}) = 2\rho^r u_e - \sum_c (y_{e,c} + \rho^r (w_{e,c} + \xi_{e,c})),$$

which vanishes at $u_e = \frac{1}{2}(\psi_{e,a} + \psi_{e,b})$. The objective is a quadratic in u_e with Hessian $2\rho^r I$, a positive multiple of the identity, so constrained minimization over the convex set $\mathcal{B}_K$ is the Euclidean projection of the unconstrained minimizer. Since $\mathcal{B}_K$ is a box, that projection is componentwise clipping. $\square$

Two properties of (77) matter for the architecture. Both endpoints compute the same value from the same two quantities, so there is no edge master and no tie-breaking rule. And the update requires one exchange with one neighbour, not a broadcast.

9.6 The slack and multiplier steps

Proposition 9.3 (Slack update and the exact identity). *Let $c^{r,t} := AZ^{r,t} + BU^{r,t}$. Minimizing (72) over ξ gives*

$$\xi^{r,t} = -\frac{\Lambda^r + y^{r,t-1} + \rho^r c^{r,t}}{\beta^r + \rho^r}, \quad (78)$$

and the multiplier update

$$y^{r,t} = y^{r,t-1} + \rho^r (c^{r,t} + \xi^{r,t}) \quad (79)$$

then yields, in exact arithmetic, the identity

$$\Lambda^r + \beta^r \xi^{r,t} + y^{r,t} = 0. \quad (80)$$

Proof. The slack is unconstrained, and the terms of (72) involving ξ are $\langle \Lambda^r, \xi \rangle + \frac{\beta^r}{2} \|\xi\|^2 + \langle y^{r,t-1}, c^{r,t} + \xi \rangle + \frac{\rho^r}{2} \|c^{r,t} + \xi\|^2$. Setting the gradient to zero gives $\Lambda^r + \beta^r \xi + y^{r,t-1} + \rho^r (c^{r,t} + \xi) = 0$, i.e. (78); the Hessian is $(\beta^r + \rho^r)I \succ 0$, so this is the minimizer. Substituting (79),

$$\Lambda^r + \beta^r \xi^{r,t} + y^{r,t} = \Lambda^r + \beta^r \xi^{r,t} + y^{r,t-1} + \rho^r c^{r,t} + \rho^r \xi^{r,t} = [\Lambda^r + y^{r,t-1} + \rho^r c^{r,t}] + (\beta^r + \rho^r) \xi^{r,t},$$

and the bracket equals $-(\beta^r + \rho^r) \xi^{r,t}$ by (78). $\square$

Identity (80) is exact in real arithmetic and therefore a usable integrity test: any drift in its residual indicates an implementation or arithmetic fault rather than slow convergence. Both it and the relation (71) are monitored.

9.7 Inner stopping criteria

Combining the optimality condition of the local step with the multiplier update (79) shows that the quantities

$$R_1^{r,t} = \left\| \rho^r A^\top [B(U^{r,t-1} - U^{r,t}) + \xi^{r,t-1} - \xi^{r,t}] \right\|, \quad (81)$$

$$R_2^{r,t} = \left\| \rho^r B^\top (\xi^{r,t-1} - \xi^{r,t}) \right\|, \quad (82)$$

$$R_3^{r,t} = \left\| AZ^{r,t} + BU^{r,t} + \xi^{r,t} \right\|, \quad (83)$$

measure how far the current iterate is from satisfying the stationarity conditions of (70). The inner loop stops when $R_j^{r,t} \leq \epsilon_j^r$ for $j = 1, 2, 3$, with positive tolerances satisfying $\epsilon_j^r \rightarrow 0$ as $r \rightarrow \infty$. Note that ρ^r does not appear in (83), so the three tolerances are naturally of different magnitudes; Section 10.4 returns to how they should be scaled relative to one another.

These are global quantities, but they need no coordinator. Fix a spanning tree T^{redux} of the communication graph and evaluate them by neighbour-to-neighbour reduction along it, with two accounting rules. For R_1 , each vehicle must sum its incident contributions *before* squaring, since $A^\top$ collects all half-edges incident to a block. For R_2 , each edge contributes $\|\Delta\xi_{e,a} + \Delta\xi_{e,b}\|^2$ once; this is obtained without double counting by having each endpoint contribute $\frac{1}{2} \|\Delta\xi_{e,a} + \Delta\xi_{e,b}\|^2$. A fixed reduction order also fixes the floating-point summation order, which is what makes the computation reproducible; it does not make the arithmetic order-independent.

9.8 Outer updates and initialization

At the end of an inner solve the outer multiplier is updated with a safeguard, and the penalty is increased only if the slack has not decreased sufficiently:

$$\Lambda^{r+1} = \text{Proj}_{[\underline{\Lambda}, \bar{\Lambda}]}(\Lambda^r + \beta^r \xi^r), \quad (84)$$

$$\beta^{r+1} = \begin{cases} \beta^r, & \|\xi^r\| \leq \omega \|\xi^{r-1}\|, \\ \vartheta \beta^r, & \text{otherwise,} \end{cases} \quad \rho^{r+1} = 2\beta^{r+1}, \quad (85)$$

with $\vartheta > 1$ and $\omega \in [0, 1)$. The projection in (84) is the standard safeguard that keeps the outer multiplier bounded; without it the method can degenerate into a pure penalty method requiring $\beta^r \rightarrow \infty$. The bounds must be chosen generously, since a multiplier pinned at a bound indicates that the safeguard, rather than the problem, is determining the iterate.

The parameter classes are

$$\beta^0 \geq \frac{1}{4}, \quad \beta^1 = \vartheta \beta^0, \quad \vartheta > 1, \quad \omega \in [0, 1), \quad \Lambda^1 \in [\underline{\Lambda}, \bar{\Lambda}]. \quad (86)$$

The lower bound on β^0 comes from the source analysis through (71).

Inner initialization must satisfy $Z^{r,0} \in X$, $U^{r,0} \in \bar{X}$ and the identity $\Lambda^r + \beta^r \xi^{r,0} + y^{r,0} = 0$. The initialization used here is the constant trajectory of (40), with every copy and every auxiliary set equal to its source, which is feasible by Theorem 6.1 and satisfies $AZ^{1,0} + BU^{1,0} = 0$. Taking $\xi^{1,0} = 0$ and $y^{1,0} = -\Lambda^1$ satisfies the identity. For $r \geq 2$ the previous outer output is reused and the inner multiplier is reset exactly by $y^{r,0} = -\Lambda^r - \beta^r \xi^{r,0}$.

One consequence of taking the outer reference $\xi^0 = 0$ should be noted, since it affects the first penalty decision: the test in (85) at $r = 1$ compares $\|\xi^1\|$ with $\omega \|\xi^0\| = 0$, so unless $\xi^1 = 0$ exactly the penalty increases once at the first outer step.

9.9 Assumptions required by the convergence theory

The convergence results of the adopted family rest on four hypotheses [42]. They are listed here with their status for the present problem, and they are referred to as Source Assumptions to distinguish them from Assumption 8.8.

Source Assumption 1. The problem is feasible and the set of stationary points is nonempty.

Status: feasibility is established by Theorem 6.1 together with Theorem 8.9; nonemptiness of the generalized stationary set is established only conditionally.

Source Assumption 2. The objective is continuously differentiable, X is compact, and $\bar{X}$ is compact and convex. *Status:* satisfied. F is a polynomial; compactness of X follows from the workspace rows and the added bounds of Lemma 9.1; $\bar{X}$ is a box. Note that only the *objective* is required to be smooth — the nonsmooth safety geometry lives in the set X , which is exactly why this family was selected.

Source Assumption 3. Each local step returns an *exact* stationary point of (75) together with exact nonincrease of the augmented Lagrangian. *Status:* this is the strongest hypothesis, and Section 10.7 examines it in detail for the computed instance.

Source Assumption 4. A uniform upper bound on the augmented Lagrangian at inner initialization, across outer iterations. *Status:* required only by the fixed-tolerance variant discussed in Appendix A2, which is not the variant used here.

The limit-point statements of the theory are expressed through the limiting normal cone. For a closed set C , the regular normal cone at x is

$$\hat{N}_C(x) := \left\{ v : \limsup_{\substack{z \rightarrow x, z \in C \\ z \neq x}} \frac{\langle v, z - x \rangle}{\|z - x\|} \leq 0 \right\}, \quad (87)$$

and the limiting normal cone is

$$N_C(x) := \{ v : \exists x_\nu \in C, x_\nu \rightarrow x, v_\nu \rightarrow v, v_\nu \in \hat{N}_C(x_\nu) \}. \quad (88)$$

The stationarity system for the split problem is then

$$Z \in X, \quad U \in \bar{X}, \quad AZ + BU = 0, \quad 0 \in \nabla F(Z) + A^\top y + N_X(Z), \quad 0 \in B^\top y + N_{\bar{X}}(U). \quad (89)$$

Reaching (89) in the limit additionally requires a limit point of the *actual* multiplier sequence $\{y^r\}$ along a subsequence converging to the primal limit. The source notes that this follows from a sequentially bounded constraint qualification or from a constant positive linear dependence condition. Those conditions supply multiplier sequences for the smooth program; they do not by themselves show that the specific sequence produced by (79) is a bounded one. The following result closes that step for the present structure.

Theorem 9.4 (Boundedness of the generated multiplier sequence). *Write $w = (Z, U)$ and $E = [A \ B]$. Along a subsequence indexed by ν , assume:*

- (i) $w^{r\nu} \rightarrow w^*$ with $Ew^* = 0$, and every active inequality is C^1 near w^* ;
- (ii) E has full row rank and there exists d with $Ed = 0$ and $\nabla g_\ell(w^*)^\top d < 0$ for every active ℓ ;
- (iii) the generated multipliers satisfy

$$\nabla \Phi(w^{r\nu}) + E^\top y^{r\nu} + G(w^{r\nu})^\top \mu^{r\nu} = e^{r\nu}, \quad \mu^{r\nu} \geq 0, \quad e^{r\nu} \rightarrow 0,$$

with zero multipliers on inactive inequalities.

Then $\{(y^{r\nu}, \mu^{r\nu})\}$ is bounded, and $\{y^{r\nu}\}$ has a convergent further subsequence matching the same primal limit.

Proof. Suppose not, and pass to a further subsequence with $\theta_\nu := \|(y^{r\nu}, \mu^{r\nu})\| \rightarrow \infty$. Normalizing, $(\hat{y}^\nu, \hat{\mu}^\nu) := (y^{r\nu}, \mu^{r\nu})/\theta_\nu$ lies on the unit sphere, so a further subsequence converges to some $(\bar{y}, \bar{\mu})$ with $\|(\bar{y}, \bar{\mu})\| = 1$ and $\bar{\mu} \geq 0$. A constraint strictly inactive at w^* remains inactive for large ν by continuity, so its multiplier is zero there and $\bar{\mu}$ is supported on $\mathcal{A}(w^*)$.

Divide the relation in (iii) by θ_ν . The primal sequence is bounded and $\nabla \Phi$ is continuous, so $\nabla \Phi(w^{r\nu})/\theta_\nu \rightarrow 0$; likewise $e^{r\nu}/\theta_\nu \rightarrow 0$ since $e^{r\nu} \rightarrow 0$. The active gradients converge to $\nabla g_\ell(w^*)$ by (i). Passing to the limit,

$$E^\top \bar{y} + \sum_{\ell \in \mathcal{A}(w^*)} \bar{\mu}_\ell \nabla g_\ell(w^*) = 0. \quad (90)$$

Take the inner product of (90) with the direction d of (ii). Since $Ed = 0$, the first term contributes $\langle \bar{y}, Ed \rangle = 0$, leaving $\sum_\ell \bar{\mu}_\ell \nabla g_\ell(w^*)^\top d = 0$. Every coefficient is nonnegative and every directional derivative is strictly negative, so each product is nonpositive and the sum is zero only if $\bar{\mu} = 0$.

Then (90) reduces to $E^\top \bar{y} = 0$. Full row rank of E makes $E^\top$ injective, so $\bar{y} = 0$, contradicting $\|(\bar{y}, \bar{\mu})\| = 1$. Hence the sequence is bounded, and finite-dimensional compactness supplies the convergent further subsequence. $\square$

Theorem 9.4 is a conditional statement and must be read as one. Its hypothesis (iii) is what Source Assumptions 1–3 deliver; it is not an independent fact about the algorithm. Section 10.7 examines whether hypothesis (iii) is available for the instance actually computed, and the answer there is negative, for a reason that turns out to be structural rather than particular to that instance.

9.10 Summary of the method and of its mapping onto this problem

The method has two levels: an outer augmented Lagrangian on a slack variable, and an inner three-block alternating direction method whose penalty is tied to the outer one by $\rho^r = 2\beta^r$. Nothing in that skeleton is new here; what this section contributes is the mapping — the demonstration that the reformulated problem lands in the source problem class without modification, that each step of the method becomes an operation the swarm can actually perform without a coordinator, and that the hypotheses the theory needs are either satisfied, conditional or absent, stated one by one. Table 9.1 records that mapping in full.

Table 9.1 Every step of the two-level method becomes an operation the swarm can perform without a central coordinator, and each hypothesis the convergence theory needs is recorded as satisfied, conditional or not required.

Element	What the method prescribes	What it becomes here	Central coordinator?
Problem class	$\min F(Z)$ over $Z \in X$, $U \in \bar{X}$ subject to $AZ + BU = 0$, with X compact and possibly nonconvex, $\bar{X}$ compact convex, F continuously differentiable (67).	Problem (P') after compactification and splitting. The nonconvex safety geometry is carried by the set X , never by smooth constraint functions, which is why no global C^2 requirement is placed on the safety model.	—
Outer level	Augmented Lagrangian on the slack, with multiplier Λ^r and penalty β^r .	Unchanged (70). The multiplier is safeguarded onto $[\underline{\Lambda}, \bar{\Lambda}]$ and the penalty grows by ϑ only when the slack fails to contract by ω , (84)–(85).	No: scalars only
Inner level	Three-block alternating direction method with $\rho^r = 2\beta^r$.	Unchanged (71). The tie between the two penalties is imposed by the convergence analysis and is not a tuning choice.	No
Block 1: the local step	Minimize the augmented Lagrangian over $Z \in X$ with the other blocks fixed.	Separates exactly across vehicles (73): vehicle a solves its own nonlinear program ϕ_a over its own block z_a (75). Every nonlinear row is local to one block.	No: fully local
Block 2: the auxiliary step	Minimize over $U \in \bar{X}$ with the other blocks fixed.	A closed form (77): the average of the two half-edge values, followed by a projection onto the box. It is an average precisely because $B^\top B = 2I$.	No: one exchange per edge
Block 3: slack and multiplier	Closed-form slack minimization followed by a dual ascent step.	Equations (78) and (79), which together preserve the identity $\Lambda^r + \beta^r \xi^{r,t} + y^{r,t} = 0$ exactly (80); that identity is used as an implementation check.	No
Inner stopping	Three residuals below three vanishing tolerances.	R_1, R_2, R_3 of (81)–(83), evaluated by neighbour-to-neighbour reduction along a fixed tree T^{redux} under two accounting rules. Fixing the reduction order is what makes the computation bit-reproducible.	No
Hypotheses	Source Assumptions 1–4.	Assumption 2 is satisfied outright. Assumption 1 holds conditionally. Assumption 3, exact local stationarity, is the strongest and is examined in Section 10.7. Assumption 4 is required only by the fixed-tolerance variant, which is not the variant used.	—

Element	What the method prescribes	What it becomes here	Central coordinator?
Multiplier boundedness	Referred to a sequentially bounded constraint qualification or a constant positive linear dependence condition, which supply multiplier sequences for the smooth program.	Those conditions do not by themselves bound the sequence the algorithm actually generates. Theorem 9.4 closes that step for this structure, under hypotheses stated explicitly and checked rather than assumed.	—

Two points in the table deserve to be carried forward rather than read once. The first is that the decentralization claim is structural and not incidental: it follows from where the constraints were placed in Section 8, since each nonlinear row was attached to a single block and only the linear agreement rows link blocks at all. The second is that the hypothesis column is not uniformly green, and the report does not present it as such. Section 10.7 shows that Source Assumption 3 cannot be met by any floating-point implementation of the local step, which locates the remaining gap precisely: it is not a gap in the mapping, but the absence of an inexact-oracle convergence result for this family.

10 Numerical Study

This section reports a numerical study of the formulation and the solver on a small instance chosen so that every quantity in it can be inspected. The instance is deliberately minimal: two vehicles, a planar workspace, an eight-interval horizon and no obstacles. It is not a demonstration of scale, and no scaling claim is made from it. It is a test of whether the objects defined in Sections 3–9 behave as the analysis says they should, and of what can be certified about the result.

10.1 Instance and parameters

The physical instance is a symmetric crossing: two vehicles whose straight goal-directed paths intersect, so that the pairwise coupling of (38) is genuinely engaged even though the obstacle register is empty. Table 10.1 lists the data in full.

Table 10.1 The numerical study uses a two-vehicle symmetric crossing instance with an empty obstacle register, so that the pairwise separation constraint is the only constraint linking the two vehicles.

Quantity	Value
Number of vehicles N	2
Spatial dimension q	2
Prediction intervals K	8
Sampling period Δt	0.5 s
Planning horizon $K\Delta t$	4 s
Workspace $\mathcal{W}$	$[-3, 3] \times [-3, 3]$ m
Vehicle radii r_1, r_2	0.10 m
Pairwise safety margin δ_{12}^{uav}	0.20 m
Required centre clearance $\kappa_{12}^{\text{uav}} = r_1 + r_2 + \delta_{12}^{\text{uav}}$	0.40 m
Maximum speeds $v_1^{\text{max}}, v_2^{\text{max}}$	1.5 m/s
Start of vehicle 1, p_1^0	$(-2, 0)$ m
Goal of vehicle 1, g_1	$(2, 0)$ m
Start of vehicle 2, p_2^0	$(0, -2)$ m
Goal of vehicle 2, g_2	$(0, 2)$ m
Terminal weights α_1, α_2	20
Step weights β_1, β_2	1
Obstacle register $\mathcal{O}$	empty
Free scalar decision variables $n = NqK$	32

The objective weights satisfy the admissibility requirement $\alpha_i, \beta_i \geq 0$ and the strict-convexity condition of Theorem 6.3 for $K \geq 2$, namely $\beta_i > 0$. Their particular values were not obtained from a weight-selection study, and no optimality is claimed for them; Section 10.8 reports a sensitivity check showing that the qualitative behaviour of the instance does not depend on this choice.

The architecture of Section 8 instantiates minimally at $N = 2$: the guardian map assigns the single pair to vehicle 1, so vehicle 1 holds one copy of source 2, there is one agreement edge and no relay. Hence $Q = qK = 16$, $M_{\text{eq}} = 1$, $d_1 = 32$, $d_2 = 16$ and $m_{\text{split}} = 32$. The counting results of Theorems 8.3 and 8.4 reduce to $\binom{2}{2} = 1$ required foreign copy and a maximum guardian load of $\lceil 1/2 \rceil = 1$, both attained. It should be said plainly that at $N = 2$ several of the structural results of Section 8 — optimal guardian balancing, the Steiner structure of the copy topology, mandatory relays at articulation vertices — are satisfied trivially, and exercising them requires $N \geq 4$.

Algorithm parameters are $\beta^0 = 1$, hence $\beta^1 = 2$ by (86); $\vartheta = 2$; $\omega = 0.5$; $\Lambda^1 = 0$ with bounds ± 100 ; and inner tolerances $\epsilon_1^r = \epsilon_2^r = 10^{-4}/r$, $\epsilon_3^r = 10^{-6}/r$, which are positive and vanish as required. Agreement acceptance thresholds are $\tau_{\text{eq}} = \tau_{\text{target}} = 10^{-6}$ m. The local problems (75) are solved with IPOPT (Interior Point OPTimizer) [43] through CasADi [44], with solver tolerance 10^{-9} and no bound relaxation. Execution is made deterministic by fixing the hash seed and restricting all numerical libraries to a single thread.

10.2 Design of the study

Three configurations were run, in a designed sequence in which exactly one element changes at a time, so that the effect of each change can be attributed.

Configuration 1 used an acceptance test for the local step that required the multipliers returned by the local solver to satisfy complementarity exactly on every row. The run stopped early, at the sixteenth inner iteration of the first outer iteration, because the local solver returns multipliers of magnitude around 10^{-5} on rows that are far from active, and complementarity as originally tested is not satisfied by such values.

Configuration 2 refined the acceptance test. Complementarity in the Karush–Kuhn–Tucker sense requires the multiplier of a *strictly inactive* inequality to be zero, so for rows verified strictly inactive by an independent evaluation the diagnostic multiplier is set to zero, while the raw value returned by the solver is retained in the record. This changes no constraint, no update and no trajectory; it changes only how the returned multipliers are interpreted. With this test the run proceeded, but it reached the inner iteration cap of 100 within the first outer iteration and stopped there. At that point agreement was far from attained — $R_{\text{TCC,eq}} = 0.1258$ m against a threshold of 10^{-6} m — and the source trajectories violated the required separation by 0.1139 m. Since the combined body radii are 0.20 m, this is a violation of the required clearance and not a demonstrated overlap of vehicles, but it is a violation of the model as posed.

Configuration 3 changed exactly one parameter relative to Configuration 2: the inner iteration cap, from 100 to 1000. Everything else — physical data, architecture, solver, tolerances, update formulas, outer cap, time limit — was held at its previous value. This isolates the question of whether the cap, rather than the method, produced the outcome of Configuration 2.

10.3 Results

Configuration 3 was executed twice in independent fresh processes. Both executions terminated at the same point, $(r, t) = (10, 3)$, after 853 complete inner iterations, and the two saved final states are *bit-for-bit identical*, with SHA-256 digest `d444e1d5df70c64639f5a3f0de1445bc82150ec9797a7b9533a76feebd58e46`. The first outer iteration required 211 inner iterations, which answers the question the configuration was designed to ask: the cap of 100 was binding for this instance.

The outer progression is recorded in Table 10.2.

Table 10.2 The agreement residuals contract by roughly a factor of five per outer iteration from $r = 1$ to $r = 9$, and the first outer iteration exceeds the inner iteration cap that this configuration was designed to test.

r	inner t	$\ \xi^r\ $	$R_{\text{TCC,eq}}$ (m)	R_{target} (m)
1	211	$9.028962253367044 \times 10^{-2}$	$1.2768898917888183 \times 10^{-1}$	$9.028975013124303 \times 10^{-2}$
2	202	$1.7461463312267986 \times 10^{-2}$	$2.4694004025418662 \times 10^{-2}$	$1.746129770102144 \times 10^{-2}$
3	164	$3.645464643064285 \times 10^{-3}$	$5.155309106016253 \times 10^{-3}$	$3.6453540279768504 \times 10^{-3}$
4	121	$7.715252249663599 \times 10^{-4}$	$1.0909810971517794 \times 10^{-3}$	$7.714401319423628 \times 10^{-4}$
5	79	$1.628939614843912 \times 10^{-4}$	$2.302719353329657 \times 10^{-4}$	$1.628268469908902 \times 10^{-4}$
6	42	$3.389893843838477 \times 10^{-5}$	$4.786206784085708 \times 10^{-5}$	$3.384359273188062 \times 10^{-5}$
7	18	$6.802221596989993 \times 10^{-6}$	$9.533951795113956 \times 10^{-6}$	$6.741521965830736 \times 10^{-6}$
8	8	$1.283381764762773 \times 10^{-6}$	$1.7006142757418199 \times 10^{-6}$	$1.2025158865596899 \times 10^{-6}$
9	5	$3.963968424180342 \times 10^{-7}$	$6.408605500628833 \times 10^{-7}$	$4.5315684074440574 \times 10^{-7}$
10	3	$5.181917114188811 \times 10^{-7}$	$7.850016699224997 \times 10^{-7}$	$5.550800040449635 \times 10^{-7}$

The agreement residuals contract by roughly a factor of five per outer iteration from $r = 1$ to $r = 9$. At the returned iterate,

$$\begin{aligned}
R_1 &= 9.780703313666428 \times 10^{-6}, & R_2 &= 2.6597130100645397 \times 10^{-15}, \\
R_3 &= 4.157895314285983 \times 10^{-8}, & \|\xi\| &= 5.181917114188811 \times 10^{-7}, \quad (91) \\
R_{\text{TCC,eq}} &= 7.850016699224997 \times 10^{-7} \text{ m}, & R_{\text{target}} &= 5.550800040449635 \times 10^{-7} \text{ m},
\end{aligned}$$

all within their thresholds, with algebraic integrity residuals $R_{\text{id}} = 2.99 \times 10^{-22}$ and $R_{\rho\beta} = 0$. The objective value is

$$F = 4.0157423760136215, \quad f_1 = 2.0078734359074213, \quad f_2 = 2.0078689401062, \quad (92)$$

with terminal goal errors 0.0230 m and 0.0266 m. This value is descriptive; on its own it certifies nothing about optimality, and Section 10.8 supplies the external comparison that gives it meaning.

Three features of this run must be stated alongside the result, because they bound how it should be read.

1. **The returned iterate is the later of two that satisfied the acceptance criteria, and it is the weaker one.** The stopping rule requires the acceptance predicate on two consecutive outer iterates and returns the last complete snapshot; it is therefore the later one by construction. Here every agreement residual is larger at $r = 10$ than at $r = 9$.
2. **The penalty-increase test of (85) fails at $r = 10$.** With $\omega = 0.5$ we have $\|\xi^{10}\| = 5.1819 \times 10^{-7}$ against $\omega \|\xi^9\| = 1.9820 \times 10^{-7}$, so the test fails and β would have doubled from 4 to 8 had the iteration continued. It passed at every earlier outer iteration, and β had been constant at 4 since $r = 2$. No statement about behaviour beyond $r = 10$ is supported.
3. **The thresholds were met with limited headroom:** 21.5% for $R_{\text{TCC,eq}}$, 44.5% for R_{target} and 2.2% for R_1 . The exact stopping point is therefore sensitive to constants that are inputs to the study rather than properties of the problem.

10.4 On the tolerance schedule

Feature 3 above has an identifiable cause, and recording it is more useful than recording the number alone. The source method suggests that ϵ_1 and ϵ_2 be of order $\mathcal{O}(\rho^r \epsilon_3^r)$, since ρ^r appears in (81)–(82) but not in (83). The schedule used here, $10^{-4}/r$ and $10^{-6}/r$, does not follow that scaling: at $r = 10$, where $\rho^{10} = 8$, the suggested magnitude is about 8×10^{-7} whereas 10^{-5} was used, a factor of roughly 12 looser. The consequence is visible in the inner iteration counts, which fall 211, 202, 164, 121, 79, 42, 18, 8, 5, 3: as the residuals contract geometrically while the tolerances relax only as $1/r$, the inner loop exits ever earlier, and by $r = 10$ it exits after three iterations on a tolerance well above the residual level it is meant to resolve. No hypothesis of the convergence theory is violated — the tolerances are positive and vanish — but the schedule is not well matched to the observed contraction. A schedule tied to the outer contraction, for instance $\epsilon_1^r = \epsilon_2^r = c \rho^r \epsilon_3^r$ with ϵ_3^r decaying at least geometrically at rate at most ω , is the indicated choice for a successor study.

10.5 Certifying the safety of the computed plan

The physical requirement is $\sigma \geq 0$ for every margin. A floating-point test of the form $\hat{\sigma} \geq -\tau$ cannot establish this when $\hat{\sigma}$ is close to zero: it admits margins that are truly negative. For a certificate, the signs are therefore decided in exact arithmetic.

The saved coordinates are binary64 numbers, hence exact dyadic rationals, and the model constants are exact decimal literals; in particular $\kappa_{12}^{\text{uav}} = 2/5$ exactly. Every branch decision in (25) is a comparison of rational numbers and is therefore exact, and the sign test is performed on the *squared* margin $M - \kappa^2$, which requires no square root.

Evaluating all required margins on the source trajectories — eight initial and workspace checks, sixty-four sampled workspace margins, sixteen speed margins, eight pairwise segment margins, and an obstacle family verified empty — gives no failures. The binding quantity is the pairwise margin on segment $k = 3$, whose exact value is

$$M_{1,2,3}^{\text{uav}} - (\kappa_{12}^{\text{uav}})^2 = \frac{P}{Q} > 0, \quad (93)$$

where, with digits grouped in threes so that the value can be checked by hand,

$$P = 2\,359\,246\,339\,737\,244\,210\,484\,930\,680\,683\,636\,361\,274\,901\,726\,413\,405\,985\,560\,497,$$

$$Q = 5\,368\,394\,241\,616\,423\,559\,259\,674\,901\,405\,253\,777\,730\,294\,016\,954\,048\,679\,647\,169\,740\,800.$$

approximately $4.3946965024440439 \times 10^{-7} \text{ m}^2$, attained in the interior regime at the exact closest-point parameter

$$\hat{\lambda} = \frac{2\,447\,760\,472\,664\,976\,541\,094\,683\,039\,794\,479}{2\,646\,822\,713\,140\,499\,409\,576\,550\,887\,611\,213} \approx 0.92479200080638113. \quad (94)$$

The minimum exact workspace margin and the minimum exact squared speed margin are, respectively,

$$\begin{aligned} \sigma_{\min}^W &= \frac{4\,607\,004\,478\,089\,813}{4\,503\,599\,627\,370\,496} \approx 1.02296 \text{ m}, \\ (\sigma^{\text{spd}})_{\min}^2 &= \frac{1\,431\,476\,901\,226\,393\,583\,963\,385\,819\,543\,727}{5\,192\,296\,858\,534\,827\,628\,530\,496\,329\,220\,096} \approx 0.27569 \text{ m}^2. \end{aligned}$$

An independent implementation in directed-rounding interval arithmetic at one hundred decimal digits encloses (93) with a strictly positive lower bound, agreeing with the exact value.

Remark 10.1 (Why the squared margin, and not the distance). The unsquared margin $D - \kappa$ is badly conditioned at this solution. Here $M = D^2 \approx 0.1600004$ while $\kappa^2 = 0.16$, so forming $\sqrt{M} - \kappa$ cancels about six significant digits: a one-unit perturbation in the last place of the square root moves the margin by around 10^{-16} in absolute terms, which is 2×10^{-10} relative to a margin of 5.49×10^{-7} . Three independent floating-point evaluations of the same quantity agree only to about eight significant figures. The exact rational certificate is therefore stated for the squared margin, and the unsquared value $\approx 5.4933669 \times 10^{-7}$ m is reported for readability only.

10.6 How large is that margin, and what does it depend on?

The certificate of Section 10.5 is exact, and it should be read together with its scale. The certified separation margin is 5.49×10^{-7} m on a required clearance of 0.4 m. Three further quantities are needed to interpret it, and Table 10.3 collects them together with the certified value itself:

Table 10.3 The certified source-side separation margin exceeds the guardian-side margin by about four orders of magnitude, and both are bounded in magnitude by the copy-versus-source discrepancy.

Quantity at the returned iterate	Value
Guardian-side separation margin, segment $k = 3$	1.435×10^{-10} m
Guardian-side separation margin, segment $k = 4$	7.324×10^{-11} m
Source-side separation margin, segment $k = 3$ (certified)	5.4934×10^{-7} m
Source-side separation margin, segment $k = 4$	5.5677×10^{-7} m
Largest per-sample copy-versus-source discrepancy	6.265×10^{-7} m
$R_{\text{TCC,eq}}$	7.850×10^{-7} m

The constraint the solver actually enforces is the guardian-side one, evaluated on vehicle 1's *copy* of source 2, and it is active to within 1.4×10^{-10} m — that is, the optimizer drives the pair to the separation boundary. The certified quantity is the source-side margin, and it is of the same order as the agreement error. That is not a coincidence.

Lemma 10.2 (Agreement error bounds the discrepancy between the two margins). *Fix a pair $\{i, j\}$ and a segment k , put $a = p_i^k - p_j^k$, $b = p_i^{k+1} - p_j^{k+1}$ and $D = \min_{\lambda \in [0,1]} \|(1 - \lambda)a + \lambda b\|$. Let the guardian evaluate the same expression with p_j replaced by its copy $\tilde{x}_{a \leftarrow j}$, obtaining $\tilde{D}$, and set*

$$\delta^k := \tilde{x}_{a \leftarrow j}^k - p_j^k, \quad \Delta_k := \max\{\|\delta^k\|, \|\delta^{k+1}\|\}.$$

Then $|D - \tilde{D}| \leq \Delta_k$. Consequently, with source and guardian margins $\sigma = D - \kappa_{ij}^{\text{uav}}$ and $\tilde{\sigma} = \tilde{D} - \kappa_{ij}^{\text{uav}}$,

$$\sigma \geq \tilde{\sigma} - \Delta_k, \tag{95}$$

so enforcing $\tilde{\sigma} \geq 0$ yields only $\sigma \geq -\Delta_k$.

Proof. For each $\lambda \in [0, 1]$ the two relative positions differ by $(1 - \lambda)\delta^k + \lambda\delta^{k+1}$, whose norm is at most $(1 - \lambda)\|\delta^k\| + \lambda\|\delta^{k+1}\| \leq \Delta_k$ by the triangle inequality and convexity. By the reverse

triangle inequality, the two norms differ by at most Δ_k pointwise in λ , hence $\|(1 - \lambda)\tilde{a} + \lambda\tilde{b}\| \geq D - \Delta_k$ for every λ , and taking the minimum gives $\tilde{D} \geq D - \Delta_k$. Exchanging the roles of the two configurations gives the reverse inequality, and subtracting the common constant κ_{ij}^{uav} gives (95). $\square$

Lemma 10.2 is nearly tight on this run: the two margins differ by 5.492×10^{-7} m against the bound $\Delta_3 = 6.265 \times 10^{-7}$ m, so the inequality is attained to within 88%. The consequence must be stated plainly.

The exact certificate of Section 10.5 establishes the safety of *this computed plan*. It does not establish that the method produces safe plans. When the guardian constraint is active, (95) bounds the magnitude of the source-side margin by the agreement error but does not control its *sign*. Configuration 2 realized the same mechanism with the opposite sign: agreement error 0.1258 m and a source-side margin of -0.1139 m.

This is a property of the formulation as it currently stands, and it has an exact remedy that is stated here and left for the continuation of the work. Suppose the guardian enforces the tightened clearance $\kappa_{ij}^{\text{uav}} + \bar{\Delta}$ with $\bar{\Delta} \geq \max_k \Delta_k$. Then (95) gives $D \geq \tilde{D} - \Delta_k \geq \kappa_{ij}^{\text{uav}} + \bar{\Delta} - \Delta_k \geq \kappa_{ij}^{\text{uav}}$, so the source trajectories are safe unconditionally. Moreover $\Delta_k \leq R_{\text{TCC,eq}}$, and by (66) the agreement error is bounded by $h_{\text{max}}\tau_{\text{eq}}$, so $\bar{\Delta} = h_{\text{max}}\tau_{\text{eq}}$ suffices — one micrometre for the present architecture. The cost of this remedy must be stated with it: the tightened problem is a conservative inner approximation of (P), so Theorem 8.9 no longer applies unchanged, and exactness of the reformulation is traded for an a-priori safety guarantee. That trade is a modelling decision, and it is discussed again in Section 15.

10.7 The local step and the exactness hypothesis

Source Assumption 3 requires each local step to return an *exact* stationary point of (75). What a numerical solver returns is a point at which the stationarity residual is below its tolerance. Whether the distinction matters can be settled for this run, and it is settled here rather than assumed away.

Consider the very first local step, at outer iteration $r = 1$, inner iteration $t = 1$, for vehicle 2. At that point $\rho^1 = 4$, $y^{1,0} = 0$ and $\xi^{1,0} = 0$, so by (75) the local objective is

$$\phi_2(z_2) = f_2(z_2) + 2\|z_2 - u^0\|^2, \quad (96)$$

with u^0 the constant initialization. Vehicle 2 guards no pair and the obstacle register is empty, so its local set carries only workspace and speed rows.

Consider the coordinate $p_{2,y}^1$, the y -component of vehicle 2 at the first free sample. Differentiating (96) with respect to it, using $p_{2,y}^0 = -2$, $\alpha_2 = 20$, $\beta_2 = 1$ and $u_y^0 = -2$: the terminal term does not involve this coordinate; the step terms of segments 0 and 1 contribute $2(p_{2,y}^1 + 2)$ and $2p_{2,y}^1 - 2p_{2,y}^2$; and the proximal term contributes $4(p_{2,y}^1 + 2)$. Hence

$$\frac{\partial \phi_2}{\partial p_{2,y}^1} = 8p_{2,y}^1 - 2p_{2,y}^2 + 12. \quad (97)$$

Reconstructing the two coordinates from the recorded values by exact rounding-cell enumeration — that is, determining every binary64 number consistent with the recorded data — and evaluating (97) in exact rational arithmetic over all consistent pairs gives

$$\frac{558383}{2^{50}} \leq \frac{\partial \phi_2}{\partial p_{2,y}^1} \leq \frac{558393}{2^{50}}, \quad (98)$$

that is, between 4.9594×10^{-10} and 4.9595×10^{-10} : strictly positive.

Now examine the normal cone. The two workspace rows involving $p_{2,y}^1$ and the two speed rows of segments 0 and 1 are all verified strictly inactive at that point, and every other row of vehicle 2's local set is independent of that coordinate. Hence, in a neighbourhood, the local set is a cylinder in the $p_{2,y}^1$ direction, so every element of $N_{X_2}(z_2^{1,1})$ has zero component there — a conclusion that requires no constraint qualification. Combining with (98),

$$0 \notin \nabla \phi_2(z_2^{1,1}) + N_{X_2}(z_2^{1,1}). \quad (99)$$

The decrease condition, by contrast, holds: an exact outward enclosure gives $\phi_2(z_2^{1,1}) < 55.15977310579323 < 320 = \phi_2(z_2^{1,0})$, a certified decrease of more than 264.84.

So the local step satisfies the decrease requirement and fails exact stationarity, and one failure at a required step is enough: the hypothesis of Theorem 9.4 is not available for this run, and neither is the stationary-limit conclusion of the source theory.

It would be a mistake to read this as a defect of this particular execution. A local solve terminating at a finite tolerance is not an exact stationary point except by coincidence, so Source Assumption 3 is unattainable by *any* floating-point realization, independently of the instance. The observed gradient of 4.96×10^{-10} is simply the solver stopping at its tolerance of 10^{-9} . What this analysis identifies is therefore not a computational error but a genuine gap between the available theory and any implementation of it: what is needed is an inexact-oracle version of the convergence result, in which the local step is allowed a quantified error and that error is propagated. The elements of such an accounting are already visible. If $n_Z \in N_X(Z^{r,t})$ is a justified normal vector, define

$$\begin{aligned} e_Z^{r,t} &:= \nabla F(Z^{r,t}) + A^\top y^{r,t-1} + n_Z^{r,t} + \rho^r A^\top (AZ^{r,t} + BU^{r,t-1} + \xi^{r,t-1}), \\ d_Z^{r,t} &:= \nabla F(Z^{r,t}) + A^\top y^{r,t} + n_Z^{r,t}, \end{aligned} \quad (100)$$

so that $d_Z^{r,t} \in \nabla F(Z^{r,t}) + A^\top y^{r,t} + N_X(Z^{r,t})$ and, subtracting and using (79) together with (81),

$$\|d_Z^{r,t}\| \leq \|e_Z^{r,t}\| + R_1^{r,t}. \quad (101)$$

A small R_1 therefore certifies target stationarity only to the extent that the local error $\|e_Z\|$ is itself controlled, and controlling it requires a construction of n_Z that justifies feasibility, the active gradients, and nonnegativity and zero-value of the multipliers on inactive rows. Establishing such a bound is future work, and is listed as such in Section 15.

10.8 An independent centralized reference

Every quantity reported so far is internal to the algorithm. The residuals R_1, R_2, R_3 measure the splitting method against itself; $R_{\text{TCC},\text{eq}}$ measures agreement between copies; $\|\xi\|$ measures

the outer slack. None of them can distinguish *solving* (P) from *converging to a stationary point of the agreement-violation problem*, an outcome the convergence theory of the adopted family explicitly permits. An instrument external to the method is needed, and it also supplies the first numerical check of Theorem 8.9, which until now has been proved but never tested. That instrument is a direct centralized solve of (P) on the same instance. It is a validation oracle, not a proposed architecture: it requires a central computation, it does not scale, and it is precisely the design this work sets out to replace. Its role here is the one an analytical solution plays for a discretization scheme.

The comparison is made fair by construction: the physical data are those of Section 10.1; the exact segment-safety functions are the *same* implementation used by the distributed solver, so both models evaluate identical geometry; and the backend is the same, IPOPT (Interior Point OPTimizer) through CasADi, with tolerance 10^{-10} . Twenty starts were used: the constant initialization of (40) and nineteen workspace-clipped Gaussian perturbations of it. Table 10.4 reports the outcome.

Table 10.4 An independent centralized solve of the same instance reproduces the distributed objective value to seven significant figures and the distributed trajectory to thirty-three micrometres, and finds two distinct global minimizers.

Quantity	Value
Starts converged	20/20
Centralized optimal value F^*	4.015742261564965
Distributed value F	4.0157423760136215
Absolute objective gap	$1.1444865677390226 \times 10^{-7}$
Relative objective gap	$2.85 \times 10^{-6} \%$
Distinct global minimizers found	2
Separation between them	0.9609102855881685 m
$\ x_{\text{distributed}} - x_{\text{nearest}}\ _2$	$3.261077006949802 \times 10^{-5}$ m
Largest coordinate difference	$8.875261982552174 \times 10^{-6}$ m
Separation margin at that minimizer	$1.0649314763355733 \times 10^{-10}$ m

Two conclusions follow, and a third observation.

The reformulation is numerically corroborated. The distributed solution of (P') and a direct centralized solution of (P) agree to seven significant figures in objective value and to thirty-three micrometres in stacked ℓ_2 norm over the whole thirty-two-coordinate trajectory. This is evidence for Theorem 8.9 — feasible set, value and optimizer set preserved — on one instance.

Non-uniqueness occurs on this instance. All twenty starts converge to the same objective value but to *two* distinct trajectories, separated by almost a metre. They are the mirror pair of the symmetric crossing: vehicle 1 yielding to one side of the meeting point or to the other. This is a concrete realization of Theorem 6.4 on the instance actually solved, and it illustrates the consequence drawn in Section 6.4: the natural stability object here is the solution correspondence, and small changes in the data may switch which equal-cost route is returned.

The coupling constraint is active for structural reasons. Sweeping the terminal weight with $\beta_i = 1$ fixed, the separation constraint is active at the centralized optimum for every

value tested, as Table 10.5 records:

Table 10.5 The pairwise separation constraint is active at the centralized optimum for every terminal weight tested, so its activity follows from the crossing geometry rather than from the weights chosen for the study.

α	F^*	minimum separation margin (m)
0.5	3.232602115	1.095×10^{-11}
1	3.595555556	4.543×10^{-12}
2	3.805996205	4.398×10^{-12}
5	3.943255351	4.433×10^{-12}
10	3.991167318	5.366×10^{-12}
20	4.015742262	1.061×10^{-11}
50	4.030702504	3.116×10^{-11}

Activity therefore follows from the symmetric crossing geometry, not from the particular weights chosen in Section 10.1. This matters for Section 10.6: because the optimum itself sits on the separation boundary, *any* method solving this instance to finite accuracy terminates within the agreement error of that boundary, and the scale effect identified there is structural rather than accidental.

Finally, the limits of this comparison. Multistart interior-point optimization on a nonconvex feasible set is not a proof of global optimality, twenty starts is not exhaustive, and Theorem 6.4 guarantees there is no uniqueness to appeal to. The centralized value is numerical evidence on one instance, and is reported as such.

11 A Physics-Based Simulation Environment

The numerical study of Section 10 is a two-dimensional computation: it answers questions about the formulation and the solver, and it says nothing about how a plan looks when it is executed by vehicles in a three-dimensional world with rendering, physics and message passing. A separate physics-based simulation environment was therefore built for that purpose, in ROS 2 and Gazebo, and this section presents it. Every scene, every vehicle model and both demonstrations reported below run in ROS 2 and Gazebo; nothing in this section is a plot or an abstract animation.

Its status should be clear from the outset. The environment is the validation track on which later versions of this work will be tested. In its present form it does not run the solver of Section 9, and the two tracks are kept apart deliberately: mixing them would make it impossible to say which component a given observation is about.

11.1 Environment

The scene is a UM6P-inspired campus built for this project: a paved courtyard, a set of campus buildings, landscaping, a pergola and a landmark, together with two landing pads and two vehicle models. All geometry is generated by a deterministic script from a written scene specification, so the environment can be rebuilt from source rather than distributed as

opaque assets. The installed scene comprises seven models in total: the five static models just listed, and exactly two vehicles. Figure 11.1 shows it as it appears when launched.



Figure 11.1 The parked campus scene shows the courtyard, the surrounding buildings, the landscaping and the two landing pads, with both vehicles at rest and no simulation stepping yet performed.

The software stack is Ubuntu 24.04 with ROS 2 Jazzy [45] and Gazebo Sim [46], connected by the standard ROS–Gazebo bridge. The delivery is organized as five packages: one for the campus description, one for the bringup, one shared implementation package, and one package for each demonstration. This separation exists so that a scenario can be added without touching the environment, and so that the environment can be inspected without running any scenario.

A specific design decision is worth recording because it is easy to get wrong. The environment can be launched in a *paused* state, with zero simulation time and no stepping. This gives a scene that can be inspected, photographed and checked for correct entity counts and poses before any motion occurs, which separates errors in the scene from errors in the motion. Figure 11.2 shows the scene inspected in this state from two further viewpoints.

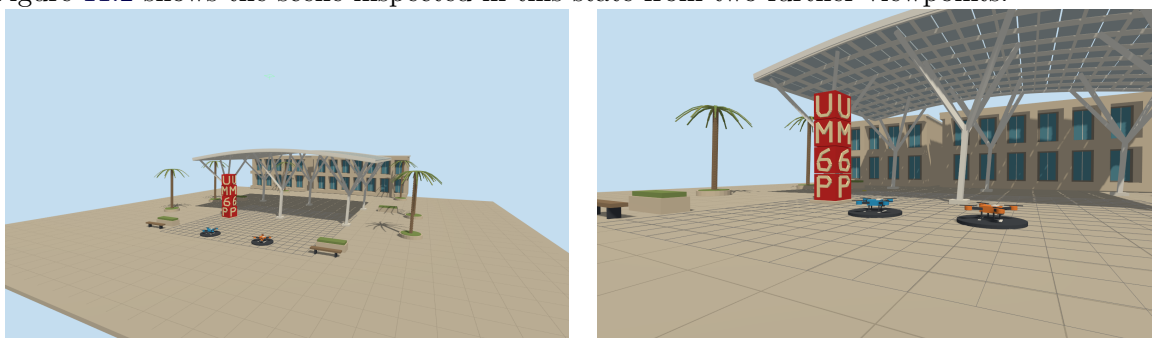


Figure 11.2 A wide framing of the campus (left) and a closer view of the pergola and the landmark (right) allow the seven static models to be checked against the scene specification before any motion is commanded.

11.2 Two demonstrations

Two scenarios were implemented on this environment. Both are *kinematic*: motion is prescribed by geometric position commands together with a yielding rule that holds one vehicle when a predicted crossing conflict is detected. There is no rotor model, no attitude dynamics and no flight controller, and the yielding rule is not the solver of Section 9. Table 11.1 states what each scenario does and what it records.

Table 11.1 Three scenarios were implemented on the campus environment, and each one records an endpoint measured in simulation time rather than in wall-clock time.

Scenario	Behaviour	Recorded endpoint
Parked campus	Two static vehicles in the paused scene; no stepping.	Zero simulation time
Demonstration 1	Two vehicles take off together, travel, yield at a crossing conflict and land on opposite pads. No added obstacle.	Mission complete at 51.92 s of simulation time
Demonstration 2	The same structure with one stationary sphere in the environment, giving a bypass followed by a later crossing and yield.	Mission complete at 54.22 s of simulation time

The recorded endpoints are *simulation* time, not wall-clock time and not a measure of computation.

Figures 11.3 to 11.6 show both demonstrations as they run in ROS 2 and Gazebo. Demonstration 1 is displayed with a head-up display reporting the live inter-vehicle clearance, so that the separation actually being maintained is legible in the recording itself rather than having to be reconstructed afterwards.



Figure 11.3 Demonstration 1 runs with a head-up display that reports the live inter-vehicle clearance, which makes the separation being maintained visible in the recording itself.

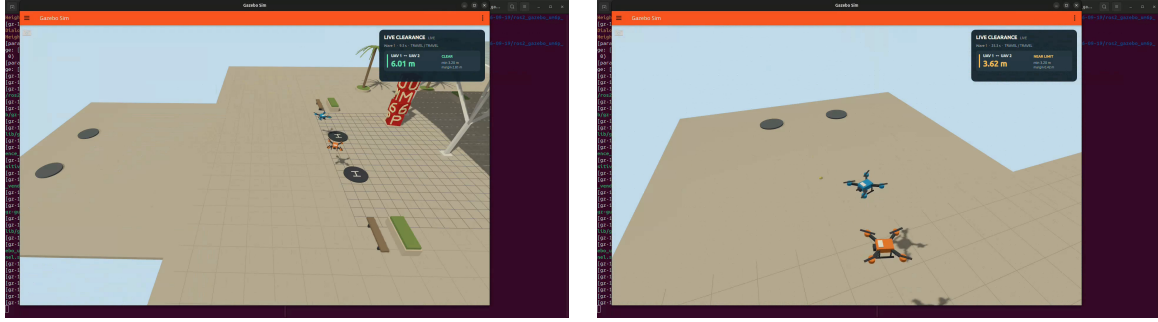


Figure 11.4 Demonstration 1 at ten seconds (left) and at eighteen seconds (right) of the recording: the two vehicles travel towards opposite pads, and the crossing conflict is resolved by one vehicle yielding.

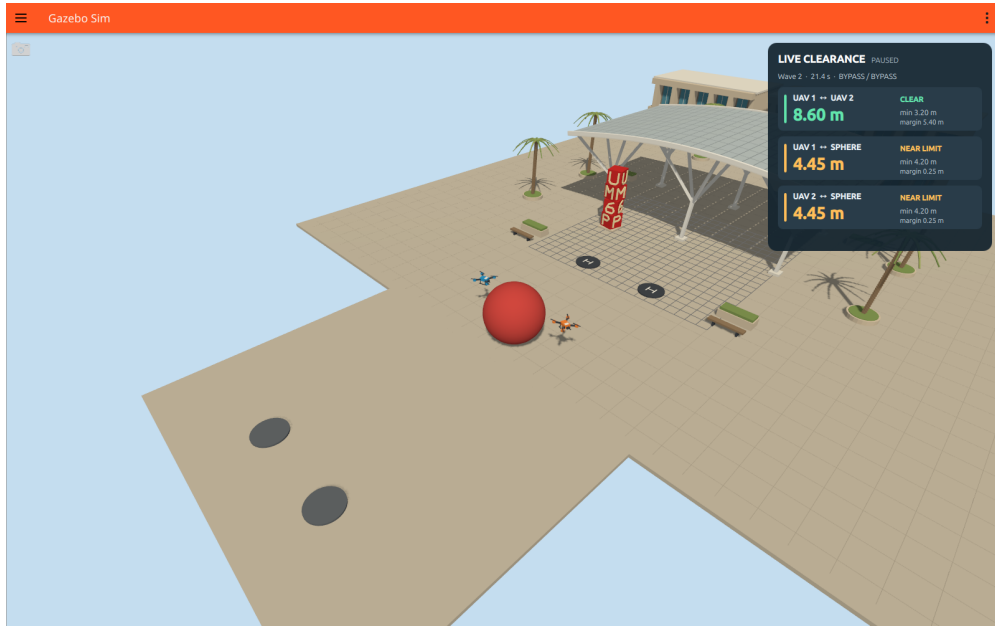


Figure 11.5 Demonstration 2 adds one stationary sphere to the same environment, which the vehicles must bypass before the crossing conflict arises.

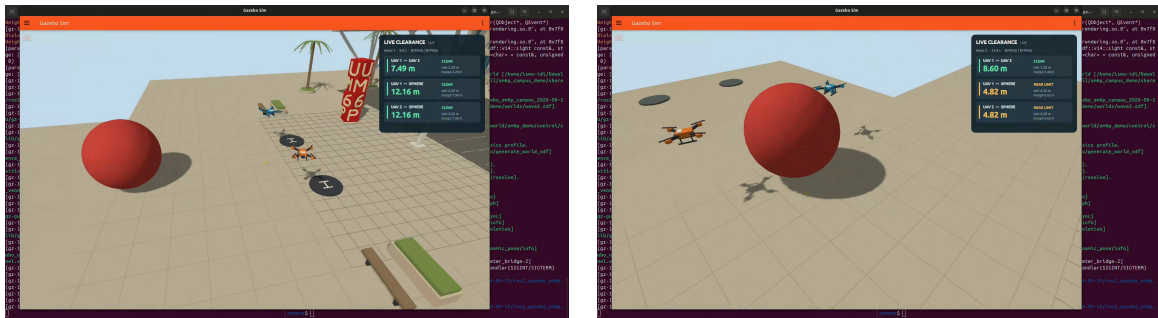


Figure 11.6 Demonstration 2 at ten seconds (left) and at eighteen seconds (right) of the recording: the approach to the sphere, and the bypass that precedes the later crossing and yield.

The times quoted in Figures 11.4 and 11.6 are positions in the supplied recordings and not simulation time; the mission endpoints in Table 11.1 are the simulation-time quantities. Complete recordings of both demonstrations, together with the campus description, the launch files and the scenario packages, are available in the project repository [55].

11.3 What was verified, and what determinism means here

Each scenario was run both with the graphical interface and in a fresh rebuild from a new directory, and the following were checked in both: the entity count and initial poses of the scene; simultaneous take-off; the yielding behaviour at the crossing; continuous geometric clearance between the two vehicles throughout the run; correct pause and resume; the telemetry published on the ROS side, including the clock; the poses reported by the renderer rather than only by the physics engine; the landings; and an orderly shutdown of every child process. Test runs were given a unique namespace and simulation partition, so that a verification run cannot be contaminated by another session — an isolation failure observed early in development and fixed.

Determinism deserves a precise statement, since a loose one would be misleading. The recorded trajectory is a deterministic function of the simulation tick: every tick common to two runs carries identical poses and phases. The tick at which an external pause request lands is *not* deterministic, so two runs paused mid-flight will report their mid-flight snapshots one tick apart. The completed states are identical.

11.4 What these demonstrations do and do not show

They show that the environment is real, that two vehicles can be commanded through a coordinated manoeuvre in it, that geometric clearance can be monitored continuously during execution, and that the whole thing is reproducible from a fresh build. That is the intended contribution: a working, inspectable validation track.

They do not execute the formulation of this report. No optimization problem is solved, no agreement constraint is imposed, and the saved plan of Section 10 is not replayed. The clearances used by the yielding rule belong to the demonstration and are not the model constants of Section 10.1; in particular, the presence of a sphere in Demonstration 2 does not constitute a test of the obstacle constraint family (37), which, as Section 10.3 records, was not exercised by the numerical study. Nothing here validates flight dynamics, and nothing here is evidence about real vehicles.

12 Extension to Dynamic and Uncertain Environments

The model of Section 3 is deterministic by construction. This section sets out how the extension to dynamic environments and uncertainty should be approached, what has to be decided before any guarantee can be stated, and which of the results of this report would survive. It is a scoping study: it fixes no uncertainty model, states no theorem and reports no computation. That restraint is deliberate, and the reason is given below.

12.1 Which simplifications were made so that they could be revisited

Several choices in Section 3 were made in a form that makes them removable one at a time: the workspace is static; obstacle centres are static and exactly known; the initial positions are treated as exact data; communication and information edge sets are given rather than derived from state; and separation is an all-pairs physical requirement that is deliberately

not identified with communication availability, by (5). Each of these is a place where time dependence or uncertainty can enter without redesigning the model around it.

12.2 Dynamic environments and receding-horizon planning

The first extension reintroduces time dependence in the environment:

$$\mathcal{W} \rightarrow \mathcal{W}(t), \quad c_o \rightarrow c_o(t), \quad C_i(t_0) \rightarrow C_i(t), \quad h_{ij}(x_i, x_j) \rightarrow h_{ij}(x_i, x_j; t). \quad (102)$$

One point must be settled before anything else: the closed-form expressions of Section 4 were derived for a *static* obstacle centre. With a moving centre, the relative position along a segment is no longer $a + \lambda d$ with constant a, d , and (25) does not apply. Either the obstacle motion is modelled as piecewise linear on the same grid, in which case the relative motion is again affine in λ and the same derivation goes through with modified a and d , or a new inter-sample analysis is required. Reusing the static formula without that check would reintroduce exactly the failure of Example 4.3 in a different guise.

Because the environment changes while the vehicles fly, the natural framework is receding-horizon planning: estimate the current state, predict over a finite horizon, solve, execute only the first portion, and repeat. The relevant literature is that of Section 2, in particular [17, 18, 19, 24]. This dynamic layer should be built and understood *before* stochastic uncertainty is introduced, so that the two effects are not confounded.

12.3 Sensing and communication uncertainty are different problems

Uncertainty in what a vehicle *measures* and uncertainty in what it *receives* are separate, and conflating them is a common source of confusion.

On the sensing side, the first task is to identify which quantities are actually uncertain — own position, neighbour states, obstacle positions or motion, disturbances — and only then to choose a representation: a bounded uncertainty set, a random variable with a justified distribution, or an empirical model. The order matters. A distribution introduced in order to make a chance constraint writable is not a model of anything.

On the communication side, the effects are link availability, distance or state dependence, packet loss, delay and stale information, and temporary disconnection. Two reserved symbols exist for this purpose, an edge-availability indicator and the latency function $\zeta(e, t_k)$, and the graph may become time-dependent or random. Throughout, the separation (5) must be preserved: *loss of a communication link must never silently delete a physical safety requirement*. It changes who can evaluate a constraint, not whether the constraint holds.

This is also where the reformulation of Section 8 needs genuine re-examination rather than reuse. Theorem 8.2 assumes agreement holds exactly on a connected tree. Under delay, loss or asynchrony, copies are stale and that hypothesis fails. Since Section 10.6 has already shown that the source-side safety margin is controlled by the agreement error even in the deterministic case, the question of what agreement error can be guaranteed under unreliable communication is not a refinement but a central one.

12.4 Probabilistic and robust safety

Once the uncertainty is defined, safety can be restated probabilistically, in the generic pairwise form

$$\mathbb{P}(h_{ij}(x_i, x_j; t, \zeta) \leq 0) \geq 1 - \varepsilon_{ij}, \quad (103)$$

with ζ the uncertainty and ε_{ij} an admissible risk level. Before (103) means anything, six things must be fixed: the random variables; the probability space or data model; the safety event itself; the interpretation and justification of ε ; whether risk is allocated per pair or jointly across the swarm; and whether the guarantee is asserted at one instant, at the samples, or over the whole horizon — which, given Example 4.3, is not a detail. Robust counterparts in the sense of [16] are the alternative when a bounded set is more honest than a distribution.

The deterministic margins δ_i^{obs} and δ_{ij}^{uav} also need reconsideration. They may remain as they are, or they may be tightened as a function of the uncertainty. Either choice is a modelling decision and must be made explicitly — which connects directly to the tightening already identified for a different reason in Section 10.6.

12.5 Which results would have to be re-established

The extension does not inherit the results of this report automatically. Once one precise uncertain model is fixed, the following must be re-examined: inter-sample safety under moving obstacles; nonemptiness and existence, since the constant trajectory may no longer be feasible if the environment moves towards a stationary vehicle; stability with respect to the new data; regularity of the new constraint functions; the constraint-qualification analysis and therefore Theorem 7.8; the exactness of the reformulation under the new communication model, that is Theorem 8.9; the hypotheses required by the solver; and the interpretation of every safety and convergence statement. The goal is not to repeat the whole analysis but to identify precisely which conclusions survive and which must be weakened.

A minimal first numerical study would proceed in four steps: the deterministic case of Section 10 as reference; a known moving obstacle; bounded position uncertainty; and finally a probabilistic formulation — each step changing one thing.

13 Higher-Order Trajectory Representations

The trajectory model of Section 4 represents a path by sampled positions joined linearly. A natural question, raised during the supervision of this work, is whether a smoother parameterization would be preferable. This section reports a scoping study of that question. Its conclusion is that the direction is viable and clearly defined, and that it is not a drop-in replacement, for a specific reason.

13.1 The current model already is a spline

It is worth stating first that the present representation is not a discrete approximation awaiting continuous treatment. The reconstruction (18) is a continuous function of time determined by finitely many parameters, and it is exactly a degree-one spline on the planning grid. The

natural progression is therefore not from discrete to continuous trajectories, but

$$\text{continuous piecewise-linear} \longrightarrow \text{higher-order finite-dimensional representation}, \quad (104)$$

which is a change of regularity, not a change of category. The present model's regularity is deliberately limited: the path is continuous, the velocity may jump at the knots, and no acceleration, curvature or jerk claim is made.

13.2 A candidate formulation

The natural first candidate is a B-spline,

$$p_i(t; c_i) = \sum_{\ell=0}^{n_c-1} c_{i\ell} \mathcal{B}_{\ell, \varrho}(t), \quad (105)$$

with degree ϱ , a fixed knot vector, and vector control points $c_{i\ell} \in \mathbb{R}^q$ as decision variables. The problem stays finite-dimensional. B-splines are the natural first object because they combine a polynomial representation with local support, an explicit derivative structure and the convex-hull property [47].

For a controlled comparison, a first variant should change *only* the parameterization, keeping the workspace, obstacle model, safety radii, horizon, initial data, architecture and — initially — the objective unchanged. Otherwise an observed difference cannot be attributed. The assumptions to fix in advance are: a fixed horizon; a fixed degree and knot vector, with knot positions not optimized initially; the same basis for all copies of one source trajectory; a clamped construction so that the known initial position is imposed by parameterization rather than by an added equality; sufficient degree for whichever derivatives are constrained; a bounded coefficient domain or another route to compactness; and continuous-time rather than knot-only constraints.

13.3 How each ingredient would transfer

Table 13.1 takes the ingredients of the present formulation one by one and states how each would carry over.

Table 13.1 Each ingredient of the present formulation transfers to a spline parameterization in a different way, and only some of the transfers are direct.

Ingredient	Transfer	Status
Fixed initial position	Impose $p_i(t_0) = \hat{p}_i(t_0)$ by a clamped construction rather than by an added equality.	Depends on basis and knots
Terminal goal	Replace p_i^K by $p_i(t_K; c_i)$.	Direct
Step regularizer	Keep the evaluation grid and use $\sum_k \ p_i(t_{k+1}) - p_i(t_k)\ ^2$ for comparability.	Direct
Workspace	Control points in the convex $\mathcal{W}$ give containment for the whole curve by the convex-hull property.	Sufficient, possibly conservative
Speed bound	Differentiate the spline; bound $\dot{p}_i(t)$ directly, or bound the derivative control points.	New formulation needed
Obstacle and pairwise safety	Must hold along the whole curve, not at knots or control points.	New analysis required
Agreement	Copy coefficient vectors and impose agreement on them; with a common basis, equal coefficients means equal trajectories.	Plausible, needs proof
Solver mapping	Same broad architecture may apply.	All assumptions to re-establish

13.4 The decisive difficulty

The row marked in bold is the obstruction, and it is worth being explicit about it. The present model enjoys an exact, closed-form, branch-complete minimum-distance expression for every segment, derived in Section 4, and that expression is what makes the safety constraints exactly enforceable and cheap. That property does not survive raising the degree. For a curve of degree ϱ , the squared distance to a point is a polynomial of degree 2ϱ in the curve parameter, and its minimum over a span is no longer available in closed form.

Four routes exist: exact or certified minimization over each span, for instance by real root isolation; conservative separation certificates from the convex hull of the control points; subdivision and Bernstein bounds; or sampling accompanied by a proved error bound. What is *not* admissible is sampling without such a bound, since that would reintroduce precisely the failure of Example 4.3.

13.5 Verdict, and the Sobolev direction

As a research direction the answer is affirmative: a spline trajectory remains finitely parameterized, fits naturally into the optimization framework, and brings useful structure. As a replacement for the present model the answer is that no conclusion is available: the current model has proved exact segment safety and a completed analysis, and a higher-order representation would require renewed safety, existence, regularity, constraint-qualification and solver-assumption work. The recommended next step is one tightly controlled variant — plau-

sibly a clamped cubic B-spline with the same horizon and data — with the continuous-time safety formulation derived first and the comparison made only afterwards.

A related question is whether to regard a trajectory as an element of a Sobolev space $H^s([t_0, t_K]; \mathbb{R}^q)$ and the spline space as a finite-dimensional subspace, so that approximation theory quantifies the error as the knot spacing is refined [48]. That viewpoint would be useful later for deciding what functional regularity is physically meaningful for a vehicle trajectory, which derivative norms should be controlled, and how discretization error interacts with continuous-time safety. It is not required for the present model, and nothing in this report depends on it. Applied work using optimized B-spline control points for aerial vehicles [49] and piecewise polynomial trajectories for quadrotors [50] indicates that the computational side is well trodden; the safety side is what would need new work here.

14 What Is Established, and What Is Not

A formulation is only as useful as the precision of the boundary around it. This section collects, in one place, what the preceding sections establish and what they do not. It is intended to be read as a whole, and every entry is traceable to the result that supports it.

14.1 Established

1. **Exact inter-sample safety.** Under the reconstruction (18), workspace membership and the speed bound transfer exactly from the samples to the continuum (Propositions 4.1 and 4.2), and obstacle and pairwise clearance are enforced over complete segments through the closed-form expressions (25), with no auxiliary variable and no smoothing. Example 4.3 shows this is necessary.
2. **Existence.** The feasible set is nonempty and compact and the problem attains its minimum (Theorem 6.1), by a constructive argument that relies on the goal being soft.
3. **A complete convexity criterion, and the failure of uniqueness.** The objective is convex (Proposition 6.2) and strictly convex exactly under (41) (Theorem 6.3); nevertheless an admissible instance exists with a strictly convex objective and two distinct global minimizers (Theorem 6.4), and the same phenomenon is observed numerically on the instance actually solved (Section 10.8).
4. **Set-valued stability.** Outer stability of the feasible correspondence and lower semi-continuity of the value hold generally (Proposition 6.6); full continuity holds when only objective data move (Proposition 6.7); and what constraint perturbation additionally requires is stated (Proposition 6.8). Communication changes have no effect on the problem at all.
5. **Regularity of the safety functions.** The squared segment value functions are locally Lipschitz, continuously differentiable where the segment is nondegenerate — with the gradient formula (48), valid across the regime switches — and generally neither twice differentiable there nor differentiable at a collapsed segment (Proposition 6.9).

6. **A first-order theory that locates its own failures.** Three explicit mechanisms break the standard constraint qualifications (Proposition 7.2), so there is no general guarantee of either linear independence (Corollary 7.4) or the Mangasarian–Fromovitz condition (Corollary 7.6); what survives unconditionally is the workspace subsystem (Proposition 7.7); and first-order necessity is stated conditionally on exactly those hypotheses (Theorem 7.8).
7. **Exactness of the reformulation.** Agreement on physical trajectories is infeasible (Proposition 8.1); agreement on copies of the same source is realized on a spanning tree (Theorem 8.2); and the resulting problem has the same feasible-set projection, the same optimal value and the same global minimizers as the original (Theorem 8.9), with the added compactification excluding nothing (Lemma 9.1). The architecture’s cost and realizability are characterized exactly (Theorems 8.3, 8.4, 8.6, Propositions 8.5, 8.7, 8.12).
8. **A derived, decentralized algorithm.** Each update step is derived from the optimality conditions of its block (Section 9.4, Propositions 9.2 and 9.3); the local step separates across vehicles; the auxiliary update is a two-neighbour average with no edge master; and the exact identity (80) provides an arithmetic integrity test.
9. **A reproducible computation with an exact safety certificate.** On the instance of Section 10.1, two independent executions produce bit-identical results; all agreement and stopping tolerances are met; and the safety of the computed plan is certified in exact rational arithmetic (93), independently confirmed by interval arithmetic.
10. **External corroboration of the reformulation.** A centralized solve of the same instance reproduces the objective to seven significant figures and the trajectory to 3.26×10^{-5} m (Section 10.8).

14.2 Not established, and not claimed

1. **Global optimality.** The feasible set is nonconvex, first-order conditions are necessary and not sufficient, and no global optimality is claimed for any computed point. Multistart agreement is evidence, not proof.
2. **Universal constraint qualification.** None holds in general; this is proved, not merely unproved.
3. **Convergence of the distributed method on this problem class.** The available theory requires an exact local stationary point at every step, and Section 10.7 shows that requirement is not met — and, being unattainable in floating-point arithmetic, cannot be met by any implementation. What is needed is an inexact-oracle result, which does not currently exist.
4. **Safety as a property of the method.** The certificate of Section 10.5 applies to one computed plan. Lemma 10.2 shows the source-side margin is bounded in magnitude by the agreement error and *its sign is not controlled*; Configuration 2 realized the opposite sign. An a-priori guarantee requires the tightening identified in Section 10.6, which is not implemented here.

5. **Behaviour at scale.** The study is $N = 2$, $K = 8$, planar, with an empty obstacle register. Several structural results of Section 8 are satisfied trivially at $N = 2$. The obstacle constraint family, though specified and analysed, was not exercised end-to-end.
6. **Probabilistic safety.** No uncertainty model is part of this formulation, so no probabilistic statement is available. Section 12 scopes the extension; it does not perform it.
7. **Real flight.** A certified plan is not a certified flown trajectory. No vehicle dynamics, tracking controller, actuator model or disturbance is included, and the demonstrations of Section 11 run no solver.
8. **Asynchronous or unreliable communication.** The reformulation assumes a fixed connected graph and exact agreement. Nothing here covers delay, packet loss, staleness or topology change.

14.3 Statements that should survive into any presentation of this work

Five points are easy to overstate, and are therefore recorded explicitly. The reformulation does not convexify the problem. Full row rank of the agreement constraints does not imply a constraint qualification for the full program. First-order conditions do not certify optimality on a nonconvex feasible set. A small algorithmic residual establishes neither agreement nor physical safety — the two are separate quantities, related only through Lemma 10.2. And the word *global* in the convergence literature for methods of this family refers to convergence from arbitrary initialization, not convergence to a global optimizer.

15 Future Work

This section sets out the continuation of the work. It separates what is committed from what is intended and from what is longer-term, because those are different kinds of statement.

15.1 Immediate continuation

The first priority is to complete the directed-research task as specified. The present report is not an incomplete piece of work, but the ambition for it is higher than its current state, and the following are the immediate items.

Reproduce and improve the numerical study. The computation of Section 10 should be repeated and extended in several controlled configurations rather than one. Three specific improvements are already identified by the analysis in this report:

- the inner tolerance schedule should be tied to the outer contraction, as Section 10.4 makes precise, which is expected to remove the stagnation observed between the ninth and tenth outer iterations;
- the obstacle scenario, which is fully specified but was not run, should be executed, so that the constraint family (37) is exercised end-to-end;
- the study should be repeated with $N \geq 4$, where the guardian balancing of Theorem 8.4, the relay structure and the storage identity of Proposition 8.5 become non-trivial.

Close the safety gap identified in Section 10.6. The tightening $\kappa_{ij}^{\text{uav}} \rightarrow \kappa_{ij}^{\text{uav}} + h_{\max}\tau_{\text{eq}}$ converts a per-run certificate into an a-priori guarantee, at the cost of making the reformulated problem a conservative inner approximation. The trade should be analysed and decided rather than left implicit.

Establish an inexact-oracle statement. Section 10.7 shows that the exactness hypothesis of the available theory cannot be met by any floating-point implementation. The bound (101) indicates the shape of the required result: a convergence statement in which the local step is allowed a quantified error and that error is propagated to the limit. This is, in the author’s view, the most interesting open theoretical question raised by this work.

15.2 Model extensions

The uncertainty extension scoped in Section 12 is the principal modelling direction: dynamic environments, receding-horizon planning, sensing and communication uncertainty, and probabilistic or robust safety, together with the re-audit of which results survive. The higher-order trajectory representation of Section 13 is the second, with the continuous-time safety formulation as its first obstacle. Both are defined precisely enough to be started.

The simulation environment of Section 11 should be developed until it can execute the formulation rather than a prescribed motion, which requires connecting the solver to the environment and deciding how a plan computed over a horizon is handed to an execution layer.

15.3 A scientific article

After validation by the supervisors, the intention is to write a scientific article on this work. The author’s own target for a first complete draft is before 31 December 2026. This is a personal working deadline; it has not been agreed with the supervisors and carries no commitment on their part.

15.4 Longer-term directions

Full vehicle aerodynamics. The present model deliberately excludes rotor allocation, attitude dynamics, aerodynamic forces and plant identification. Building a full quadrotor aerodynamic model on top of the planning baseline established here is a substantial piece of work and an attractive one. It is also work that another doctoral student could take up directly, precisely because the planning layer is now specified and its interface is clear; that would save duplicated effort within the group.

Mean-field and continuum limits. The present work is finite-agent and graph-based. Mean-field, continuum and partial-differential-equation-constrained formulations of swarm behaviour were suggested as a possible doctoral direction and remain of interest, in particular as a way of asking what the coordination structure studied here becomes as the swarm size grows.

Physical experiments. The author’s intention is that this research should not remain purely theoretical and simulated. The concrete plan is to build a small physical swarm: a Bambu Lab A1 fused-deposition printer (3D printer), for airframes and fixtures, together with Crazyflie micro aerial vehicles as the flying platform. Figures 15.1 and 15.2 show what each of the

two is and at what scale it works. The printer is intended to be bought from the Moroccan supplier 3D Ware [53], which lists it in stock at a delivered cost of approximately 5 000 MAD. Crazyflie is chosen deliberately rather than opportunistically: it is supported in ROS 2 and in Gazebo, so the same environment used in Section 11 can host both the simulated and the physical experiments, and the transition does not require abandoning the tooling built here. A separate preparatory note on this has been started. It should be said clearly that moving from a certified plan to a flown trajectory raises the whole question of tracking, disturbance and actuation that Section 14 excludes, and that this is exactly why the step is worth taking carefully.



Figure 15.1 The Bambu Lab A1 is a fused-deposition 3D printer of the bed-slinger type: the toolhead traverses a fixed gantry while the heated bed carries the part front to back, giving a build volume of 256 mm on each side. That volume is large enough for the airframes, mounts and fixtures a small indoor swarm needs, which is what would make fabrication on site practical.



Figure 15.2 A Crazyflie is a nano quadrotor weighing about 27 g and measuring 92 mm between opposite motors, which is why it fits in one hand. Two properties matter for this work: it is small enough that several can be flown together indoors, and light enough that a collision damages neither the vehicle nor the room, so separation constraints can be tested rather than merely trusted.

15.5 A Niger and Sahel perspective

The direction of this work is not purely mathematical, and the author's intention is that it should in time serve the Republic of Niger and the wider Sahel. The region is at present the setting of a severe and protracted security and humanitarian crisis. For the first quarter of 2026 alone, the United Nations Office for the Coordination of Humanitarian Affairs records 1 179 security incidents and 2.8 million internally displaced persons across Burkina Faso, Mali and western Niger, and reports that insecurity and access constraints significantly limit the coverage of the humanitarian response, particularly in hard-to-reach and newly affected zones [51]. Figure 15.3 locates the three states concerned.

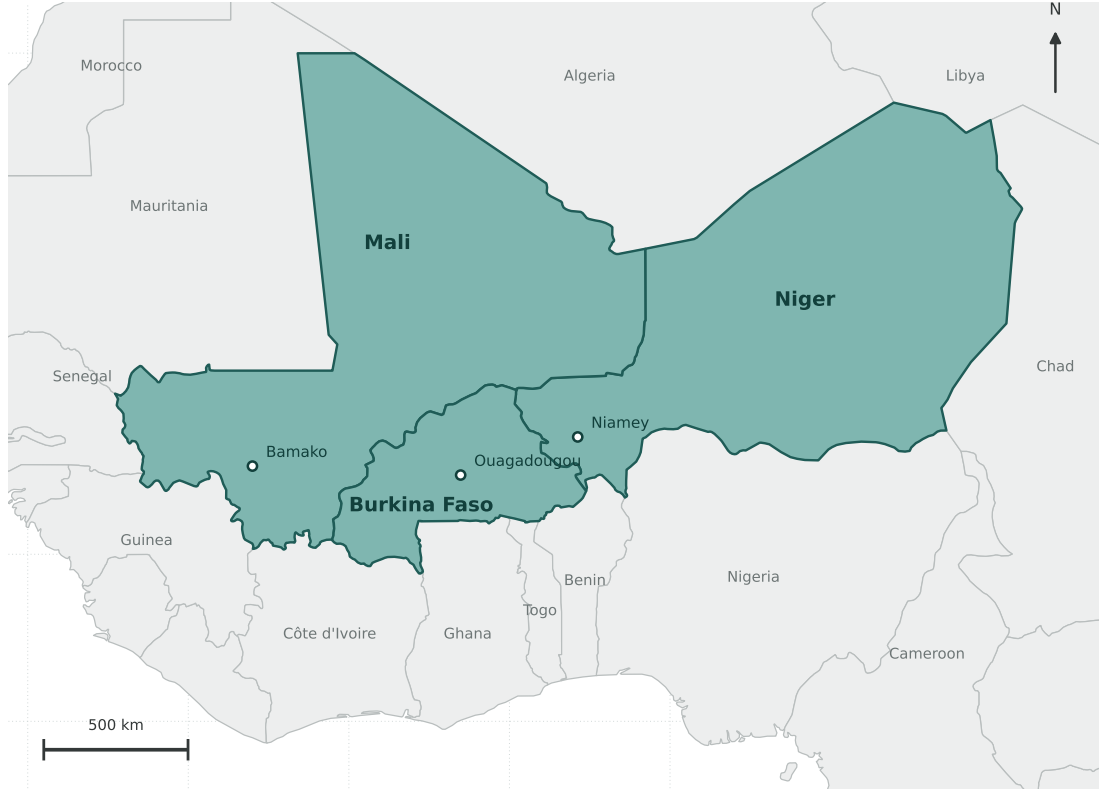


Figure 15.3 Mali, Burkina Faso and Niger — the three states of the Alliance of Sahel States — shown with their capitals. The three together span about 3 050 km from west to east, and all three are landlocked, which is one reason distances, terrain and communication coverage dominate the operational problem there. Boundaries are drawn from the public-domain Natural Earth 1:50 m dataset [52] in an equirectangular projection with standard parallel 16° N, with Morocco shown in its full territorial extent.

Unmanned aerial vehicles are already used in that setting and will continue to be: for surveillance and reconnaissance, for logistics over terrain on which ground movement is slow or unsafe, and for the defence and security tasks the situation imposes. What is hard there is not flight itself. It is coordination. Missions extend over long distances; communication is intermittent, degraded or actively denied; the environment is known only partially in advance and changes during the mission; and a vehicle must frequently continue to act sensibly on information that is already stale rather than halt and wait to be told what to do.

Those are precisely the conditions that the uncertainty extension scoped in Section 12 is written for, and they are the reason that extension is the priority direction of this work rather than one option among several. Three properties of the present formulation matter for it specifically, and all three were settled in the deterministic model rather than deferred.

- **Safety does not depend on connectivity.** Separation is imposed on every pair in E^{all} , whether or not that pair can communicate, by the separation of the three notions in (5). Sparse or degraded communication changes how a constraint is evaluated and enforced; it never deletes the constraint. In an environment where links fail, that distinction is the difference between a method that degrades and one that becomes unsafe.
- **Agreement is between copies, not between vehicles.** Because Trajectory-Copy Consensus imposes agreement on computational copies of a single source trajectory (Section 8),

a vehicle that loses contact still holds a well-defined object: its own plan, together with its copies of what it last knew of its neighbours. It does not hold an average of incompatible intentions, and it does not have to abandon its plan to remain consistent.

- **The guarantees are stated with their conditions.** Section 14 states what is established and what is not, and every result in this report carries the hypotheses it needs. That is the minimum a method must offer before it can responsibly be proposed for a mission in which failure has a human cost.

The longer-term goal, then, is to carry this line of work back to Niger and the Sahel: to develop swarm coordination methods that remain trustworthy in communication-difficult and unpredictable environments, and to contribute the mathematical and aerial-robotics competence built here to the institutions of the region. This is stated as a direction of work and not as a claim about the present report, which concerns a deterministic model of two vehicles in a fully known environment.

15.6 Explicitly outside the present scope

The following are relevant but are not part of this line of work unless the scope changes, and they are listed so that their absence is understood as a decision: high-fidelity environmental physics such as wind fields, turbulence and inter-vehicle aerodynamic interaction; a full perception stack including simultaneous localization and mapping, vision-based detection and sensor fusion; learning-based black-box planning as the central method, as opposed to the constrained-optimization direction taken here; global optimality guarantees for the complete nonconvex problem; and regulatory or airworthiness certification, which is a different activity from mathematical certification and is not addressed by anything in this report.

16 Conclusion

This report has developed a decentralized path-planning formulation for UAV swarms built on constrained consensus-based dynamics, and has carried it through in full for a deterministic instance.

The starting observation was that the word *consensus*, taken at face value, is not merely inefficient for this problem but infeasible: requiring two vehicles to agree on their physical trajectories contradicts the requirement that they stay apart, and Proposition 8.1 makes that precise. The resolution adopted here is to make the agreement variable a *computational copy* of a source trajectory rather than the trajectory itself. Theorem 8.9 shows this costs nothing: the reformulated problem has the same feasible set, the same optimal value and the same global minimizers as the original. The result is a problem in which every nonlinear constraint is local to a single vehicle and the only inter-vehicle coupling is linear, which is exactly the structure a splitting method needs, and Section 9 derives such a method step by step.

Alongside this, the report has tried to be as clear about limits as about results. The safety model is exact between samples, and Example 4.3 shows why that is not optional. A global minimizer exists, and Theorem 6.4 shows that a second one may exist alongside it even when the objective is strictly convex — a fact confirmed numerically on the very instance solved

here. The first-order theory is developed together with three explicit mechanisms that break the standard constraint qualifications, so the conditions under which it may be applied are visible. And the safety of the computed plan is certified in exact rational arithmetic, while Lemma 10.2 shows precisely what that certificate does and does not extend to: it is a statement about one plan, because the sign of the physical margin is governed by the agreement error and is not controlled a priori. Identifying that, and identifying its remedy, is one of the more useful outcomes of the study.

The analysis of the local step in Section 10.7 leads to a similar conclusion about the algorithm. The available convergence theory requires an exactly stationary local solution at every step, and no floating-point implementation can supply one. The gap is not between this implementation and the theory; it is between *any* implementation and the theory, and closing it requires an inexact-oracle result that does not yet exist. Stating this plainly seems more valuable than reporting a convergence claim that the evidence does not support.

What has been contributed is therefore an integration with a boundary. The components — constrained consensus, coupled-constraint distributed optimization, chance-constrained planning, distributed predictive control, safety filtering — exist and are mature, and the report does not claim otherwise. What did not exist, and what has been assembled here, is a single constrained consensus formulation for UAV swarm path planning that carries nonconvex local feasibility, exact pairwise geometric coupling enforced between samples, an agreement mechanism compatible with physical separation, and an explicit account of where its own guarantees end. The deterministic instance is complete, checkable and reproducible. The extensions to dynamic environments, uncertainty, higher-order trajectories, larger swarms and eventually physical vehicles are defined precisely enough to be started, and that is the purpose a baseline is meant to serve.

Appendix

A1 Correspondence with the source algorithm

The distributed method of Section 9 is a mapping of (P') onto the two-level scheme of [42]. The correspondence is recorded item by item in Table A1.1 so that the translation can be checked against the source.

Table A1.1 Each item of the source method maps onto an object of this report, and the status column records whether the translation is exact.

Source item	In this report	Status
Problem class, compact X , compact convex $\bar{X}$, affine coupling	(67), with $X = \prod_a X_a$, $\bar{X} = \mathcal{B}_K^{Meq}$	Exact; compactness by Lemma 9.1
Slack reformulation and outer penalty on $\xi = 0$	(70)	Exact structural translation
Augmented Lagrangian and the identity $\Lambda + \beta\xi + y = 0$	(72), (80)	Exact in real arithmetic; monitored numerically
Inner stopping criteria and the relation $\rho = 2\beta$	(81)–(83), (71)	Exact; the factor two is required, not chosen
First-block subproblem	(75)	Exact, and separable across vehicles
Outer safeguarded multiplier and adaptive penalty	(84), (85)	The variant used here
Source Assumption 1	feasibility and stationary-set nonemptiness	Feasibility established; nonemptiness conditional
Source Assumption 2	smooth objective, compact sets	Satisfied
Source Assumption 3	exact local stationarity and decrease	Decrease holds; exact stationarity fails — Section 10.7
Limit-point characterization	(89)	Conditional on the above; not global optimality
Matching multiplier limit	Theorem 9.4	Proved under its hypotheses; hypotheses unmet here

A2 The fixed-tolerance variant

The source method admits a second variant in which the inner tolerances are held at a fixed target $\varepsilon > 0$ rather than vanishing,

$$\epsilon_1^r = \epsilon_2^r = 2\epsilon_3^r = \varepsilon, \quad \beta^{r+1} = \vartheta\beta^r \text{ unconditionally,} \quad (106)$$

so that the penalty grows at every outer iteration regardless of progress. This variant carries an additional hypothesis, Source Assumption 4: there exists $\bar{L} \in \mathbb{R}$ with $\mathcal{L}_{\rho^r}(Z^{r,0}, U^{r,0}, \xi^{r,0}, y^{r,0}) \leq \bar{L}$ for every r , that is, a bound on the augmented Lagrangian at inner initialization that is uniform over outer iterations. Under Source Assumptions 1–4 the variant admits a finite- ε complexity statement.

That variant is not the one used in this report, its complexity statement is not transferred to the variant that is used, and no verification of Source Assumption 4 is claimed. It is recorded because it is the natural fallback should a fixed accuracy target be preferred to a vanishing one, and because its penalty growth is unconditional, which changes the behaviour discussed in Section 10.3 substantially.

A3 Cardinalities and the diagnostic quantities

For an instance with N vehicles, K intervals, $M_{\mathcal{W}}$ workspace rows and $|\mathcal{O}|$ obstacles, the constraint families of Section 5.2 have exactly

$$|\mathcal{I}_{\mathcal{W}}| = NM_{\mathcal{W}}K, \quad |\mathcal{I}_{\text{spd}}| = NK, \quad |\mathcal{I}_{\text{obs}}| = N|\mathcal{O}|K, \quad |\mathcal{I}_{\text{uav}}| = \binom{N}{2}K \quad (107)$$

members. Checking these counts before interpreting any aggregate is what distinguishes a family that is genuinely empty from one whose evaluation failed. For the instance of Section 10.1 they evaluate to 64, 16, 0 and 8 respectively.

For each nonempty family, the reported summaries are the largest violation and the smallest margin,

$$R_{\mathcal{F}} = \max_{\ell \in \mathcal{I}_{\mathcal{F}}} [-\sigma_{\ell}]_+, \quad \sigma_{\mathcal{F}, \min} = \min_{\ell \in \mathcal{I}_{\mathcal{F}}} \sigma_{\ell}, \quad (108)$$

with the raw margins retained. A family that is verified empty has zero aggregate violation and no minimum margin, and is recorded as such rather than with a placeholder number; missing rows, failed evaluations or unavailable data are *not* an empty family, and the distinction is categorical rather than numerical.

The signed margins themselves are

$$\sigma_{i,m,k}^W = \frac{b_m - a_m^{\top} p_i^k}{\|a_m\|}, \quad i \in V, \quad m = 1, \dots, M_{\mathcal{W}}, \quad k = 1, \dots, K, \quad (109)$$

$$\sigma_{i,k}^{\text{spd}} = v_i^{\max} \Delta t - \|p_i^{k+1} - p_i^k\|, \quad i \in V, \quad k = 0, \dots, K-1, \quad (110)$$

$$\sigma_{i,o,k}^{\text{obs}} = D_{i,o,k}^{\text{obs}} - \kappa_{io}^{\text{obs}}, \quad i \in V, \quad o \in \mathcal{O}, \quad k = 0, \dots, K-1, \quad (111)$$

$$\sigma_{i,j,k}^{\text{uav}} = D_{i,j,k} - \kappa_{ij}^{\text{uav}}, \quad 1 \leq i < j \leq N, \quad k = 0, \dots, K-1. \quad (112)$$

The normalization by $\|a_m\|$ in the workspace margin is what makes it a distance to the facet, and it is the reason the representation must be validated by (39) first. Margins must always be evaluated on the *source* trajectories, never on a guardian's copy; Section 10.6 quantifies exactly how much the two can differ.

A4 Two residuals that are not the same quantity

The inner residual R_3 of (83) includes the slack, whereas agreement concerns $AZ + BU$ alone. Writing

$$R_{\text{target}}(Z, U) := \|AZ + BU\|, \quad \text{so that} \quad R_{\text{target}} \leq R_3 + \|\xi\|, \quad (113)$$

shows that a small R_3 is compatible with substantial agreement error when ξ compensates for it. The direct measure on the unsplit constraints is

$$R_{\text{TCC,eq}}(Z) = \max_{i \in V} \max_{\{a,b\} \in E_i^{\text{eq}}} \|S_{a \leftarrow i} z_a - S_{b \leftarrow i} z_b\|_2, \quad (114)$$

and this is the quantity that enters Lemma 10.2 and therefore the safety discussion. Both are reported in metres as stacked ℓ_2 norms over all samples and coordinates; neither is a single point-to-point distance.

A5 Reproducibility of the numerical study

The computation of Section 10 was made reproducible by fixing every source of variation that is under the experimenter’s control: the hash seed; single-threaded execution of all numerical libraries; a recorded solver version and option set; a fixed reduction order for the global sums of Section 9.7, which fixes the floating-point summation order; and a recorded digest of the source, the configuration and the resulting state. Two independent executions in fresh processes produced bit-identical saved states.

Reproducibility of this kind establishes that the computation is deterministic. It does not establish that it is correct, and the two are often confused: a deterministic implementation reproduces its own errors exactly. That is why the results of Section 10 are accompanied by an exact-arithmetic certificate, by an independent recomputation of the constraint values, and by the external centralized comparison of Section 10.8, none of which would be satisfied by an implementation that merely repeated itself.

A6 Image credits

The figures in this report come from three different places, and each is identified here.

Figures 11.1 to 11.6 were recorded from the simulation environment described in Section 11. They are the author’s own work, and the recordings they were taken from are available in the project repository [55].

Figure 15.3 was produced for this report from the public-domain Natural Earth 1:50 m cultural vectors [52].

Figure 15.1 is the manufacturer’s product image of the Bambu Lab A1, as published on the supplier listing cited in [53]. Figure 15.2 is the manufacturer’s product photograph of the Crazyflie 2.1, as published by Bitcraze AB [54]. Both are reproduced here for the sole purpose of identifying the hardware discussed in Section 15; neither is the author’s work, no endorsement by either manufacturer is claimed or implied, and all rights in those two images remain with their respective owners.

References

- [1] R. Olfati-Saber and R. M. Murray, “Consensus problems in networks of agents with switching topology and time-delays,” *IEEE Transactions on Automatic Control*, vol. 49, no. 9, pp. 1520–1533, 2004. doi:10.1109/TAC.2004.834113.
- [2] R. Olfati-Saber, J. A. Fax, and R. M. Murray, “Consensus and cooperation in networked multi-agent systems,” *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, 2007. doi:10.1109/JPROC.2006.887293.
- [3] W. Ren, R. W. Beard, and E. M. Atkins, “Information consensus in multivehicle cooperative control,” *IEEE Control Systems Magazine*, vol. 27, no. 2, pp. 71–82, 2007. doi:10.1109/MCS.2007.338264.
- [4] J. Cortés, “Distributed algorithms for reaching consensus on general functions,” *Automatica*, vol. 44, no. 3, pp. 726–737, 2008. doi:10.1016/j.automatica.2007.07.022.
- [5] A. Nedić and A. Ozdaglar, “Distributed subgradient methods for multi-agent optimization,” *IEEE Transactions on Automatic Control*, vol. 54, no. 1, pp. 48–61, 2009. doi:10.1109/TAC.2008.2009515.
- [6] A. Nedić, A. Ozdaglar, and P. A. Parrilo, “Constrained consensus and optimization in multi-agent networks,” *IEEE Transactions on Automatic Control*, vol. 55, no. 4, pp. 922–938, 2010. doi:10.1109/TAC.2010.2041686.
- [7] A. Nedić, A. Olshevsky, and M. G. Rabbat, “Network topology and communication-computation tradeoffs in decentralized optimization,” *Proceedings of the IEEE*, vol. 106, no. 5, pp. 953–976, 2018. doi:10.1109/JPROC.2018.2817461.
- [8] A. Falsone, K. Margellos, S. Garatti, and M. Prandini, “Dual decomposition for multi-agent distributed optimization with coupling constraints,” *Automatica*, vol. 84, pp. 149–158, 2017. doi:10.1016/j.automatica.2017.07.003.
- [9] X. Li, G. Feng, and L. Xie, “Distributed proximal algorithms for multi-agent optimization with coupled inequality constraints,” *IEEE Transactions on Automatic Control*, vol. 66, no. 3, pp. 1223–1230, 2021. doi:10.1109/TAC.2020.2989282.
- [10] X. Li, X. Yi, and L. Xie, “Distributed online optimization for multi-agent networks with coupled inequality constraints,” *IEEE Transactions on Automatic Control*, vol. 66, no. 8, pp. 3575–3591, 2021. doi:10.1109/TAC.2020.3021011.
- [11] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, “Distributed optimization and statistical learning via the alternating direction method of multipliers,” *Foundations and Trends in Machine Learning*, vol. 3, no. 1, pp. 1–122, 2011. doi:10.1561/22000000016.
- [12] L. Blackmore, M. Ono, and B. C. Williams, “Chance-constrained optimal path planning with obstacles,” *IEEE Transactions on Robotics*, vol. 27, no. 6, pp. 1080–1094, 2011. doi:10.1109/TRO.2011.2161160.

- [13] M. Ono and B. C. Williams, “Decentralized chance-constrained finite-horizon optimal control for multi-agent systems,” in *Proc. IEEE Conference on Decision and Control*, 2010, pp. 138–145. doi:10.1109/CDC.2010.5718144.
- [14] A. Prékopa, *Stochastic Programming*. Dordrecht: Springer, 1995. doi:10.1007/978-94-017-3087-7.
- [15] A. Shapiro, D. Dentcheva, and A. Ruszczyński, *Lectures on Stochastic Programming: Modeling and Theory*. Philadelphia, PA: SIAM, 2009. doi:10.1137/1.9780898718751.
- [16] A. Ben-Tal, L. El Ghaoui, and A. Nemirovski, *Robust Optimization*. Princeton, NJ: Princeton University Press, 2009. doi:10.1515/9781400831050.
- [17] W. B. Dunbar and R. M. Murray, “Distributed receding horizon control for multi-vehicle formation stabilization,” *Automatica*, vol. 42, no. 4, pp. 549–558, 2006. doi:10.1016/j.automatica.2005.12.008.
- [18] T. Keviczky, F. Borrelli, and G. J. Balas, “Decentralized receding horizon control for large scale dynamically decoupled systems,” *Automatica*, vol. 42, no. 12, pp. 2105–2115, 2006. doi:10.1016/j.automatica.2006.07.008.
- [19] L. Dai, Q. Cao, Y. Xia, and Y. Gao, “Distributed MPC for formation of multi-agent systems with collision avoidance and obstacle avoidance,” *Journal of the Franklin Institute*, vol. 354, no. 4, pp. 2068–2085, 2017. doi:10.1016/j.jfranklin.2016.12.021.
- [20] P. Wang and B. Ding, “A synthesis approach of distributed model predictive control for homogeneous multi-agent system with collision avoidance,” *International Journal of Control*, vol. 87, no. 1, pp. 52–63, 2014. doi:10.1080/00207179.2013.822100.
- [21] I. Ravanshadi, E. Amini Boroujeni, and M. Pourgholi, “Centralized and distributed model predictive control for consensus of non-linear multi-agent systems with time-varying obstacle avoidance,” *ISA Transactions*, vol. 133, pp. 75–90, 2023. doi:10.1016/j.isatra.2022.06.043.
- [22] R. R. Negenborn and J. M. Maestre, “Distributed model predictive control: An overview and roadmap of future research opportunities,” *IEEE Control Systems Magazine*, vol. 34, no. 4, pp. 87–97, 2014. doi:10.1109/MCS.2014.2320397.
- [23] J. M. Maestre and R. R. Negenborn, Eds., *Distributed Model Predictive Control Made Easy*. Dordrecht: Springer, 2014. doi:10.1007/978-94-007-7006-5.
- [24] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Santa Barbara, CA: Nob Hill Publishing, 2017.
- [25] B. Lindqvist, P. Sopasakis, and G. Nikolakopoulos, “A scalable distributed collision avoidance scheme for multi-agent UAV systems,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 9212–9218. doi:10.1109/IROS51168.2021.9636293.

- [26] S. S. Mansouri, C. Kanellakis, B. Lindqvist, F. Pourkamali-Anaraki, A.-A. Agha-Mohammadi, J. Burdick, and G. Nikolakopoulos, “A unified NMPC scheme for MAVs navigation with 3D collision avoidance under position uncertainty,” *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5740–5747, 2020. doi:10.1109/LRA.2020.3010485.
- [27] A. Papadimitriou, H. Jafari, S. S. Mansouri, and G. Nikolakopoulos, “Multi-stage NMPC for a MAV based collision free navigation under varying communication delays,” *Journal of Intelligent & Robotic Systems*, vol. 107, art. 33, 2023. doi:10.1007/s10846-023-01818-1.
- [28] A. D. Ames, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs with application to adaptive cruise control,” in *Proc. IEEE Conference on Decision and Control*, 2014, pp. 6271–6278. doi:10.1109/CDC.2014.7040372.
- [29] L. Wang, A. D. Ames, and M. Egerstedt, “Safety barrier certificates for collisions-free multirobot systems,” *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 661–674, 2017. doi:10.1109/TRO.2017.2659727.
- [30] A. Richards and J. How, “Aircraft trajectory planning with collision avoidance using mixed integer linear programming,” in *Proc. American Control Conference*, 2002, pp. 1936–1941. doi:10.1109/ACC.2002.1023918.
- [31] S. Mastellone, D. M. Stipanović, C. R. Graunke, K. A. Intlekofer, and M. W. Spong, “Formation control and collision avoidance for multi-agent non-holonomic systems: Theory and experiments,” *International Journal of Robotics Research*, vol. 27, no. 1, pp. 107–126, 2008. doi:10.1177/0278364907084441.
- [32] O. Khatib, “Real-time obstacle avoidance for manipulators and mobile robots,” *International Journal of Robotics Research*, vol. 5, no. 1, pp. 90–98, 1986. doi:10.1177/027836498600500106.
- [33] R. Pinnau, C. Totzeck, O. Tse, and S. Martin, “A consensus-based model for global optimization and its mean-field limit,” *Mathematical Models and Methods in Applied Sciences*, vol. 27, no. 1, pp. 183–204, 2017. doi:10.1142/S0218202517400061.
- [34] J. A. Carrillo, Y.-P. Choi, C. Totzeck, and O. Tse, “An analytical framework for consensus-based global optimization method,” *Mathematical Models and Methods in Applied Sciences*, vol. 28, no. 6, pp. 1037–1066, 2018. doi:10.1142/S0218202518500276.
- [35] C. Berge, *Topological Spaces: Including a Treatment of Multi-Valued Functions, Vector Spaces and Convexity*. Edinburgh: Oliver and Boyd, 1963.
- [36] J. F. Bonnans and A. Shapiro, *Perturbation Analysis of Optimization Problems*. New York, NY: Springer, 2000. doi:10.1007/978-1-4612-1394-9.
- [37] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*. Berlin: Springer, 1998. doi:10.1007/978-3-642-02431-3.
- [38] D. P. Bertsekas, *Nonlinear Programming*, 2nd ed. Belmont, MA: Athena Scientific, 1999.

- [39] J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. New York, NY: Springer, 2006. doi:10.1007/978-0-387-40065-5.
- [40] B. Houska, J. Frasch, and M. Diehl, “An augmented Lagrangian based algorithm for distributed nonconvex optimization,” *SIAM Journal on Optimization*, vol. 26, no. 2, pp. 1101–1127, 2016. doi:10.1137/140975991.
- [41] A. Engelmann, Y. Jiang, B. Houska, and T. Faulwasser, “Decomposition of nonconvex optimization via bi-level distributed ALADIN,” *IEEE Transactions on Control of Network Systems*, vol. 7, no. 4, pp. 1848–1858, 2020. doi:10.1109/TCNS.2020.3005079.
- [42] K. Sun and X. A. Sun, “A two-level distributed algorithm for nonconvex constrained optimization,” *Computational Optimization and Applications*, vol. 84, pp. 609–649, 2023. doi:10.1007/s10589-022-00433-4.
- [43] A. Wächter and L. T. Biegler, “On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming,” *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006. doi:10.1007/s10107-004-0559-y.
- [44] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, “CasADi: A software framework for nonlinear optimization and optimal control,” *Mathematical Programming Computation*, vol. 11, no. 1, pp. 1–36, 2019. doi:10.1007/s12532-018-0139-4.
- [45] Open Robotics, *ROS 2 Documentation: Jazzy Jalisco*. [Online]. Available: <https://docs.ros.org/en/jazzy/>
- [46] Open Robotics, *Gazebo Sim Documentation*. [Online]. Available: <https://gazebo.org/libs/sim/>
- [47] M. S. Floater, *An Introduction to Spline Theory*. Lecture notes, University of Oslo. [Online]. Available: https://www.uio.no/studier/emner/matnat/math/MAT4170/v25/undervisningsmateriale/spline_notes.pdf
- [48] Y. Bazilevs, L. Beirão da Veiga, J. A. Cottrell, T. J. R. Hughes, and G. Sangalli, “Iso-geometric analysis: Approximation, stability and error estimates for h -refined meshes,” *Mathematical Models and Methods in Applied Sciences*, vol. 16, no. 7, pp. 1031–1090, 2006. doi:10.1142/S0218202506001455.
- [49] V. Usenko, L. von Stumberg, A. Pangercic, and D. Cremers, “Real-time trajectory re-planning for MAVs using uniform B-splines and a 3D circular buffer,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 215–222. doi:10.1109/IROS.2017.8202160.
- [50] C. Richter, A. Bry, and N. Roy, “Polynomial trajectory planning for aggressive quadrotor flight in dense indoor environments,” in *Robotics Research*, Springer Tracts in Advanced Robotics, vol. 114, 2016, pp. 649–666. doi:10.1007/978-3-319-28872-7_37.
- [51] United Nations Office for the Coordination of Humanitarian Affairs, *Burkina Faso, Mali and Western Niger — Humanitarian Overview (As of 29 April 2026)*. Published 6 May

2026. [Online]. Available: <https://www.unocha.org/publications/report/burkina-faso/burkina-faso-mali-and-western-niger-humanitarian-overview-29-april-2026>
- [52] Natural Earth, *Admin 0 — Countries, 1:50m cultural vectors*. Public domain. [Online]. Available: <https://www.naturalearthdata.com/downloads/50m-cultural-vectors/>
- [53] 3D Ware, *Bambu Lab A1 — FDM/FFF 3D printer*. Product listing, Morocco. [Online]. Available: <https://pro.3dware.ma/produit/bambu-lab-a1/>
- [54] Bitcraze AB, *Crazyflie 2.1*. Product page and datasheet. [Online]. Available: <https://store.bitcraze.io/products/crazyflie-2-1>
- [55] I. Assoumane Idi, *Constrained Consensus-Based Dynamics for Decentralized Path Planning in UAV Swarms — supporting code, simulation environments and numerical results*. Public repository. [Online]. Available: <https://github.com/ismo-idi/dr1-ccbd-model0D>



Project repository — <https://github.com/ismo-idi/dr1-ccbd-model0D>